CIS Apple macOS 10.15 Catalina Benchmark

v3.0.0 - 11-21-2022

Terms of Use

Please see the below link for our current terms of use:

https://www.cisecurity.org/cis-securesuite/cis-securesuite-membership-terms-of-use/

Table of Contents

Terms of Use *1*
Table of Contents *2*
Overview *6*
Intended Audience 7
Consensus Guidance 8
Typographical Conventions 9
Recommendation Definitions *10*
Title 10
Assessment Status 10
Automated 10
Manual 10
Profile 10
Description 10
Rationale Statement 10
Impact Statement 11
Audit Procedure 11
Remediation Procedure 11
Default Value 11
References 11
CIS Critical Security Controls® (CIS Controls®) 11
Additional Information 11
Profile Definitions 12
Acknowledgements 13
Recommendations *14*
1 Install Updates, Patches and Additional Security Software 14
1.1 Ensure All Apple-provided Software Is Current (Automated) 15
1.2 Ensure Auto Update Is Enabled (Automated) 18
1.3 Ensure Download New Updates When Available Is Enabled (Automated) 20
1.4 Ensure Installation of App Update Is Enabled (Automated) 22
1.5 Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled (Automated) 25
1.6 Ensure Install of macOS Updates Is Enabled (Automated) 28
1.7 Ensure Software Update Deferment Is Less Than or Equal to 30 Days (Automated) 31
1.8 Ensure Computer Name Does Not Contain PII or Protected Organizational Information (Manual) 33
2 System Preferences 36
2.1 Bluetooth 37

2.1.1 Ensure Show Bluetooth Status in Menu Bar Is Enabled (Automated) 38
2.2 Date & Time 41
2.2.1 Ensure "Set time and date automatically" Is Enabled (Automated) 42
2.2.2 Ensure Time Is Set Within Appropriate Limits (Automated) 46
2.3 Desktop & Screen Saver 49
2.3.1 Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled (Automated) .. 50
2.3.2 Ensure Screen Saver Corners Are Secure (Automated) 55
2.4 Sharing 59
2.4.1 Ensure Remote Apple Events Is Disabled (Automated) 60
2.4.2 Ensure Internet Sharing Is Disabled (Automated) 62
2.4.3 Ensure Screen Sharing Is Disabled (Automated) 65
2.4.4 Ensure Printer Sharing Is Disabled (Automated) 67
2.4.5 Ensure Remote Login Is Disabled (Automated) 69
2.4.6 Ensure DVD or CD Sharing Is Disabled (Automated) 72
2.4.7 Ensure Bluetooth Sharing Is Disabled (Automated) 74
2.4.8 Ensure File Sharing Is Disabled (Automated) 76
2.4.9 Ensure Remote Management Is Disabled (Automated) 79
2.4.10 Ensure Content Caching Is Disabled (Automated) 82
2.4.11 Ensure AirDrop Is Disabled (Automated) 85
2.4.12 Ensure Media Sharing Is Disabled (Automated) 90
2.5 Security & Privacy 94
2.5.1 Encryption 95
2.5.1.1 Ensure FileVault Is Enabled (Automated) 96
2.5.1.2 Ensure all user storage APFS volumes are encrypted (Manual) 99
2.5.1.3 Ensure all user storage CoreStorage volumes are encrypted (Manual) 103
2.5.2 Firewall 107
2.5.2.1 Ensure Firewall Is Enabled (Automated) 108
2.5.2.2 Ensure Firewall Stealth Mode Is Enabled (Automated) 113
2.5.3 Ensure Location Services Is Enabled (Automated) 116
2.5.4 Audit Location Services Access (Manual) 119
2.5.5 Ensure Sending Diagnostic and Usage Data to Apple Is Disabled (Automated) 122
2.5.6 Ensure Limit Ad Tracking Is Enabled (Automated) 127
2.5.7 Ensure Gatekeeper Is Enabled (Automated) 130
2.5.8 Ensure a Custom Message for the Login Screen Is Enabled (Automated) 133
2.5.9 Ensure an Administrator Password Is Required to Access System-Wide Preferences (Automated) 136
2.5.10 Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled (Automated) 139
2.6 Apple ID 142
2.6.1 iCloud 143
2.6.1.1 Audit iCloud Keychain (Manual) 144
2.6.1.2 Audit iCloud Drive (Manual) 147
2.6.1.3 Ensure iCloud Drive Document and Desktop Sync Is Disabled (Automated) 150
2.6.2 Audit App Store Password Settings (Manual) 154
2.7 Time Machine 156
2.7.1 Ensure Backup Automatically is Enabled If Time Machine Is Enabled (Automated) 157
2.7.2 Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled (Automated) 161
2.8 Energy Saver 164
2.8.1 Ensure Wake for Network Access Is Disabled (Automated) 165
2.8.2 Ensure Power Nap Is Disabled for Intel Macs (Automated) 169
2.8.3 Ensure the OS is not Activate When Resuming from Sleep (Automated) 172
2.9 Ensure Legacy EFI Is Valid and Updating (Automated) 176

2.10 Audit Siri Settings (Manual) ... 179
2.11 Audit Sidecar Settings (Manual) ... 185
2.12 Audit Touch ID and Wallet & Apple Pay Settings (Manual) ... 189

3 Logging and Auditing ... 191
3.1 Ensure Security Auditing Is Enabled (Automated) ... 192
3.2 Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements (Automated) ... 194
3.3 Ensure install.log Is Retained for 365 or More Days and No Maximum Size (Automated) ... 196
3.4 Ensure Security Auditing Retention Is Enabled (Automated) ... 198
3.5 Ensure Access to Audit Records Is Controlled (Automated) ... 200
3.6 Ensure Firewall Logging Is Enabled and Configured (Automated) ... 203
3.7 Audit Software Inventory (Manual) ... 206

4 Network Configurations ... 208
4.1 Ensure Bonjour Advertising Services Is Disabled (Automated) ... 209
4.2 Ensure Show Wi-Fi status in Menu Bar Is Enabled (Automated) ... 212
4.3 Ensure HTTP Server Is Disabled (Automated) ... 215
4.4 Ensure NFS Server Is Disabled (Automated) ... 217

5 System Access, Authentication and Authorization ... 219
5.1 File System Permissions and Access Controls ... 220
5.1.1 Ensure Home Folders Are Secure (Automated) ... 221
5.1.2 Ensure System Integrity Protection Status (SIP) Is Enabled (Automated) ... 223
5.1.3 Ensure Apple Mobile File Integrity (AMFI) Is Enabled (Automated) ... 226
5.1.4 Ensure Appropriate Permissions Are Enabled for System Wide Applications (Automated) ... 228
5.1.5 Ensure No World Writable Files Exist in the System Folder (Automated) ... 230
5.1.6 Ensure No World Writable Files Exist in the Library Folder (Automated) ... 232
5.2 Password Management ... 234
5.2.1 Ensure Password Account Lockout Threshold Is Configured (Automated) ... 236
5.2.2 Ensure Password Minimum Length Is Configured (Automated) ... 239
5.2.3 Ensure Complex Password Must Contain Alphabetic Characters Is Configured (Manual) ... 241
5.2.4 Ensure Complex Password Must Contain Numeric Character Is Configured (Manual) ... 244
5.2.5 Ensure Complex Password Must Contain Special Character Is Configured (Manual) ... 247
5.2.6 Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured (Manual) ... 250
5.2.7 Ensure Password Age Is Configured (Automated) ... 252
5.2.8 Ensure Password History Is Configured (Automated) ... 254
5.3 Ensure the Sudo Timeout Period Is Set to Zero (Automated) ... 256
5.4 Ensure a Separate Timestamp Is Enabled for Each User/tty Combo (Automated) ... 259
5.5 Ensure the "root" Account Is Disabled (Automated) ... 261
5.6 Ensure Automatic Login Is Disabled (Automated) ... 263
5.7 Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session (Automated) ... 266
5.8 Ensure a Login Window Banner Exists (Automated) ... 268
5.9 Ensure Users' Accounts Do Not Have a Password Hint (Automated) ... 271
5.10 Ensure Secure Keyboard Entry Terminal.app Is Enabled (Automated) ... 273

6 User Accounts and Environment ... 276
6.1 Accounts Preferences Action Items ... 277
6.1.1 Ensure Login Window Displays as Name and Password Is Enabled (Automated) ... 278
6.1.2 Ensure Show Password Hints Is Disabled (Automated) ... 280
6.1.3 Ensure Guest Account Is Disabled (Automated) ... 282
6.1.4 Ensure Guest Access to Shared Folders Is Disabled (Automated) ... 285
6.1.5 Ensure the Guest Home Folder Does Not Exist (Automated) ... 287

6.2 Ensure Show All Filename Extensions Setting is Enabled (Automated) 289
7 Applications 292
7.1 Mail 293
7.1.1 Ensure Protect Mail Activity in Mail Is Enabled (Manual) 294
7.2 Safari 296
7.2.1 Ensure Automatic Opening of Safe Files in Safari Is Disabled (Automated) 297
7.2.2 Audit History and Remove History Items (Manual) 301
7.2.3 Audit Passwords System Preference Setting (Manual) 306
7.2.4 Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled (Automated) 309
7.2.5 Ensure Prevent Cross-site Tracking in Safari Is Enabled (Automated) 313
7.2.6 Audit Hide IP Address in Safari Setting (Manual) 318
7.2.7 Ensure Advertising Privacy Protection in Safari Is Enabled (Automated) 321
7.2.8 Ensure Show Full Website Address in Safari Is Enabled (Automated) 324
Appendix: Summary Table 327
Appendix: CIS Controls v7 IG 1 Mapped Recommendations 335
Appendix: CIS Controls v7 IG 2 Mapped Recommendations 339
Appendix: CIS Controls v7 IG 3 Mapped Recommendations 344
Appendix: CIS Controls v7 Unmapped Recommendations 349
Appendix: CIS Controls v8 IG 1 Mapped Recommendations 350
Appendix: CIS Controls v8 IG 2 Mapped Recommendations 354
Appendix: CIS Controls v8 IG 3 Mapped Recommendations 359
Appendix: CIS Controls v8 Unmapped Recommendations 364
Appendix: Change History 365

Overview

All CIS Benchmarks focus on technical configuration settings used to maintain and/or increase the security of the addressed technology, and they should be used in **conjunction** with other essential cyber hygiene tasks like:

- Monitoring the base operating system for vulnerabilities and quickly updating with the latest security patches
- Monitoring applications and libraries for vulnerabilities and quickly updating with the latest security patches

In the end, the CIS Benchmarks are designed as a key **component** of a comprehensive cybersecurity program.

This document, CIS Apple macOS 10.15 Catalina Benchmark, provides prescriptive guidance for establishing a secure configuration posture for Apple macOS 10.15 Catalina. This guide was tested against Apple macOS 10.15 Catalina. To obtain the latest version of this guide, please visit http://benchmarks.cisecurity.org. If you have questions, comments, or have identified ways to improve this guide, please write us at feedback@cisecurity.org.
This Benchmark includes instructions for auditing and remediation containing three different methods: Graphical User Interface (GUI), Command Line Interface using Terminal (CLI), and Configuration Profiles. These may be used to evaluate current configuration status and make changes as desired. In most cases, all methods are supported by the Operating System and it is up to organizational implementation personnel on how best to implement. There are some recommendations that can only be managed through one of these methods. Each organization must decide if control management outside their standard process is required if no solution is possible through their organization's specific choice of implementation. It is best practice at this time for Enterprise-managed devices to use profiles for management. A mix of both profile device management and command line hardening scripts will be the most comprehensive solution.
More profile information
https://developer.apple.com/documentation/devicemanagement
https://developer.apple.com/documentation/devicemanagement/configuring_multiple_devices_using_profiles
Organizations that are using profiles should remember that a profile can limit what, if any, settings can be changed based on the profile payload. Even authorized organization technical personnel may not be able to change a setting with a profile in place. If technical personnel are expected to make changes that are contrary to profile settings, the profile may need to be reviewed in order to verify which accounts and what conditions apply, or a process to temporarily remove the profile should be in place.

Intended Audience

This document is intended for system and application administrators, security specialists, auditors, help desk, and platform deployment personnel who plan to develop, deploy, assess, or secure solutions that incorporate Apple macOS 10.15 Catalina.

Consensus Guidance

This CIS Benchmark was created using a consensus review process comprised of a global community of subject matter experts. The process combines real world experience with data-based information to create technology specific guidance to assist users to secure their environments. Consensus participants provide perspective from a diverse set of backgrounds including consulting, software development, audit and compliance, security research, operations, government, and legal.

Each CIS Benchmark undergoes two phases of consensus review. The first phase occurs during initial Benchmark development. During this phase, subject matter experts convene to discuss, create, and test working drafts of the Benchmark. This discussion occurs until consensus has been reached on Benchmark recommendations. The second phase begins after the Benchmark has been published. During this phase, all feedback provided by the Internet community is reviewed by the consensus team for incorporation in the Benchmark. If you are interested in participating in the consensus process, please visit https://workbench.cisecurity.org/.

Typographical Conventions

The following typographical conventions are used throughout this guide:

Convention	Meaning
`Stylized Monospace font`	Used for blocks of code, command, and script examples. Text should be interpreted exactly as presented.
`Monospace font`	Used for inline code, commands, or examples. Text should be interpreted exactly as presented.
<italic font in brackets>	Italic texts set in angle brackets denote a variable requiring substitution for a real value.
Italic font	Used to denote the title of a book, article, or other publication.
Note	Additional information or caveats

Recommendation Definitions

The following defines the various components included in a CIS recommendation as applicable. If any of the components are not applicable it will be noted or the component will not be included in the recommendation.

Title

Concise description for the recommendation's intended configuration.

Assessment Status

An assessment status is included for every recommendation. The assessment status indicates whether the given recommendation can be automated or requires manual steps to implement. Both statuses are equally important and are determined and supported as defined below:

Automated

Represents recommendations for which assessment of a technical control can be fully automated and validated to a pass/fail state. Recommendations will include the necessary information to implement automation.

Manual

Represents recommendations for which assessment of a technical control cannot be fully automated and requires all or some manual steps to validate that the configured state is set as expected. The expected state can vary depending on the environment.

Profile

A collection of recommendations for securing a technology or a supporting platform. Most benchmarks include at least a Level 1 and Level 2 Profile. Level 2 extends Level 1 recommendations and is not a standalone profile. The Profile Definitions section in the benchmark provides the definitions as they pertain to the recommendations included for the technology.

Description

Detailed information pertaining to the setting with which the recommendation is concerned. In some cases, the description will include the recommended value.

Rationale Statement

Detailed reasoning for the recommendation to provide the user a clear and concise understanding on the importance of the recommendation.

Impact Statement

Any security, functionality, or operational consequences that can result from following the recommendation.

Audit Procedure

Systematic instructions for determining if the target system complies with the recommendation

Remediation Procedure

Systematic instructions for applying recommendations to the target system to bring it into compliance according to the recommendation.

Default Value

Default value for the given setting in this recommendation, if known. If not known, either not configured or not defined will be applied.

References

Additional documentation relative to the recommendation.

CIS Critical Security Controls® (CIS Controls®)

The mapping between a recommendation and the CIS Controls is organized by CIS Controls version, Safeguard, and Implementation Group (IG). The Benchmark in its entirety addresses the CIS Controls safeguards of (v7) “5.1 - Establish Secure Configurations” and (v8) '4.1 - Establish and Maintain a Secure Configuration Process” so individual recommendations will not be mapped to these safeguards.

Additional Information

Supplementary information that does not correspond to any other field but may be useful to the user.

Profile Definitions

The following configuration profiles are defined by this Benchmark:

- **Level 1**

  Items in this profile intend to:

  - be practical and prudent;
  - provide a clear security benefit; and
  - not inhibit the utility of the technology beyond acceptable means.

- **Level 2**

  This profile extends the "Level 1" profile. Items in this profile exhibit one or more of the following characteristics:

  - are intended for environments or use cases where security is paramount
  - acts as defense in depth measure
  - may negatively inhibit the utility or performance of the technology.

Acknowledgements

This Benchmark exemplifies the great things a community of users, vendors, and subject matter experts can accomplish through consensus collaboration. The CIS community thanks the entire consensus team with special recognition to the following individuals who contributed greatly to the creation of this guide:

Author
Ron Colvin

Contributor
William Harrison
Mark Andersen CISSP, GMOB
Ben Montour
Sara Archacki
Hao Shu
Jorge Escobar
Jeffrey Compton
Tim Harrison CISSP, ICP, KMP, Center for Internet Security, New York
Laura Gardner
Michael Scarborough
Mauro Faccenda
Mischa van der Bent
Jason Olsen BSCS, ACSP 10.9, ACMT 2015, Comerica Bank
Bob Gendler
Dan Brodjieski
Allen Golbig
Jason Blake
Kari Byrd
John Mahlman

Editor
Eric Pinnell
Edward Byrd

Recommendations

1 Install Updates, Patches and Additional Security Software

Install Updates, Patches and Additional Security Software

1.1 Ensure All Apple-provided Software Is Current (Automated)

Profile Applicability:

- Level 1

Description:

Software vendors release security patches and software updates for their products when security vulnerabilities are discovered. There is no simple way to complete this action without a network connection to an Apple software repository. Please ensure appropriate access for this control. This check is only for what Apple provides through software update.

Software updates should be run at minimum every 30 days. Run the following command to verify when software update was previously run: `$ /usr/bin/sudo /usr/bin/defaults read /Library/Preferences/com.apple.SoftwareUpdate | grep -e LastFullSuccessfulDate`. The response should be in the last 30 days (*Example*): `LastFullSuccessfulDate = "2020-07-30 12:45:25 +0000";`

Rationale:

It is important that these updates be applied in a timely manner to prevent unauthorized persons from exploiting the identified vulnerabilities.

Impact:

Missing patches can lead to more exploit opportunities.

Audit:

Graphical Method:
Perform the following steps to ensure there are no available software updates:

1. Open `System Preferences`
2. Select `Software Update`
3. Select `Automatically keep my Mac up to date` to allow Software Update to check with Apple's servers for any outstanding updates
4. Select `Show Updates` to verify that there are no software updates available

Terminal Method:
Run the following command to verify there are no software updates:

```
$ /usr/bin/sudo /usr/sbin/softwareupdate -l

Software Update Tool

Finding available software
No new software available.
```

Computers that have installed pre-release software in the past will fail this check if there are pre-release software updates available when audited.

Remediation:

Graphical Method:
Perform the following steps to install all available software updates:

1. Open `System Preferences`
2. Select `Software Update`
3. Select `Show Updates`
4. Select `Update All`

Terminal Method:
Run the following command to verify what packages need to be installed:

```
$ /usr/bin/sudo /usr/sbin/softwareupdate -l
```

The output will include the following:

```
Software Update found the following new or updated software:
```

Run the following command to install all the packages that need to be updated:

```
$ /usr/bin/sudo /usr/sbin/softwareupdate -i -a -R
```

Or run the following command to install individual packages:

```
$ /usr/bin/sudo /usr/sbin/softwareupdate -i '<package name>'
```

example:

```
$ /usr/bin/sudo /usr/sbin/softwareupdate -l
Software Update Tool

Finding available software
Software Update found the following new or updated software:
   * iTunesX-12.8.2
        iTunes (12.8.2), 273614K [recommended]

$ sudo /usr/sbin/softwareupdate -i 'iTunesX-12.8.2'
Software Update Tool


Downloaded iTunes
Installing iTunes
Done with iTunes
Done.
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 Perform Automated Operating System Patch Management Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.4 Perform Automated Application Patch Management Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v7	3.4 Deploy Automated Operating System Patch Management Tools Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.	●	●	●
v7	3.5 Deploy Automated Software Patch Management Tools Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.	●	●	●

1.2 Ensure Auto Update Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Auto Update verifies that your system has the newest security patches and software updates. If "Automatically check for updates" is not selected, background updates for new malware definition files from Apple for XProtect and Gatekeeper will not occur.

http://macops.ca/os-x-admins-your-clients-are-not-getting-background-security-updates/

https://derflounder.wordpress.com/2014/12/17/forcing-xprotect-blacklist-updates-on-mavericks-and-yosemite/

Rationale:

It is important that a system has the newest updates applied so as to prevent unauthorized persons from exploiting identified vulnerabilities.

Impact:

Without automatic update, updates may not be made in a timely manner and the system will be exposed to additional risk.

Audit:

Graphical Method:
Perform the following steps to ensure the system is automatically checking for updates:

1. Open `System Preferences`
2. Select `Software Update`
3. Select `Advanced`
4. Verify that `Check for updates` is enabled

Terminal Method:
Run the following command to verify that software updates are automatically checked:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.SoftwareUpdate')\
.objectForKey('AutomaticCheckEnabled').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable the system to automatically check for updates:

1. Open `System Preferences`
2. Select `Software Update`
3. Select `Advanced`
4. Set `Check for updates` to enabled

Terminal Method:
Run the following command to enable auto update:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.SoftwareUpdate AutomaticCheckEnabled -bool
true
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.SoftwareUpdate`
2. The key to include is `AutomaticCheckEnabled`
3. The key must be set to `<true/>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 Perform Automated Operating System Patch Management Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.4 Perform Automated Application Patch Management Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v7	3.4 Deploy Automated Operating System Patch Management Tools Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.	●	●	●
v7	3.5 Deploy Automated Software Patch Management Tools Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.	●	●	●

1.3 Ensure Download New Updates When Available Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

In the GUI, both "Install macOS updates" and "Install app updates from the App Store" are dependent on whether "Download new updates when available" is selected.

Rationale:

It is important that a system has the newest updates downloaded so that they can be applied.

Impact:

If "Download new updates when available" is not selected, updates may not be made in a timely manner and the system will be exposed to additional risk.

Audit:

Graphical Method:
Perform the following steps to ensure the system is automatically checking for updates:

1. Open `System Preferences`
2. Select `Software Update`
3. Select `Advanced`
4. Verify that `Download new updates when available` is enabled

Terminal Method:
Run the following command to verify that software updates are automatically checked:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.SoftwareUpdate')\
.objectForKey('AutomaticDownload').js
EOS

true
```

Note: If automatic updates were selected during system setup, this setting may not have left an auditable artifact. Please turn off the check and re-enable when the GUI does not reflect the audited results.

Remediation:

Graphical Method:
Perform the following steps to enable the system to automatically check for updates:

1. Open `System Preferences`
2. Select `Software Update`
3. Select `Advanced`
4. Set `Download new updates when available` to enabled

Terminal Method:
Run the following command to enable auto update:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.SoftwareUpdate AutomaticDownload -bool true
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.SoftwareUpdate`
2. The key to include is `AutomaticDownload`
3. The key must be set to `<true/>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 Perform Automated Operating System Patch Management Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.4 Perform Automated Application Patch Management Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v7	3.4 Deploy Automated Operating System Patch Management Tools Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.	●	●	●
v7	3.5 Deploy Automated Software Patch Management Tools Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.	●	●	●

1.4 Ensure Installation of App Update Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that application updates are installed after they are available from Apple. These updates do not require reboots or administrator privileges for end users.

Rationale:

Patches need to be applied in a timely manner to reduce the risk of vulnerabilities being exploited.

Impact:

Unpatched software may be exploited.

Audit:

Graphical Method:
Perform the following steps to ensure that App Store updates install automatically:

1. Open `System Preferences`
2. Select `Software Updates`
3. Select `Advanced`
4. Verify that `Install app updates from the App Store` is enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Automatically Install App Updates` set to `True`

Terminal Method:
Run the following command to verify that macOS updates are automatically checked and installed:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.commerce')\
  .objectForKey('AutoUpdate'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.SoftwareUpdat
e')\
  .objectForKey('AutomaticallyInstallAppUpdates'))
  if ( pref1 == 1 || pref2 == 1 ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable App Store updates to install automatically:

1. Open `System Preferences`
2. Select `Software Updates`
3. Select `Advanced`
4. Set `Install app updates from the App Store` to enabled

Terminal Method:
Run the following command to to enable automatic checking and installing of macOS updates:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.commerce AutoUpdate -bool TRUE
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.SoftwareUpdate`
2. The key to include is `AutomaticallyInstallAppUpdates`
3. The key must be set to `<true/>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 Perform Automated Operating System Patch Management Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.4 Perform Automated Application Patch Management Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v7	3.4 Deploy Automated Operating System Patch Management Tools Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.	●	●	●
v7	3.5 Deploy Automated Software Patch Management Tools Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.	●	●	●

1.5 Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that system and security updates are installed after they are available from Apple. This setting enables definition updates for XProtect and Gatekeeper. With this setting in place, new malware and adware that Apple has added to the list of malware or untrusted software will not execute. These updates do not require reboots or end user admin rights.

Silently updated security data files in Monterey

https://support.apple.com/en-us/HT202491

XProtect is Apple's built-in, signature-based security tool for detection and removal of malware.

Protecting against malware in macOS

Rationale:

Patches need to be applied in a timely manner to reduce the risk of vulnerabilities being exploited.

Impact:

Unpatched software may be exploited.

Audit:

Graphical Method:
Perform the following steps to ensure that system data files and security updates install automatically:

1. Open `System Preferences`
2. Select `Software Updates`
3. Select `Advanced`
4. Verify that `Install system data files and security updates` is enabled

Terminal Method:
Run the following commands to verify that system data files and security updates are automatically checked:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.SoftwareUpdat
e')\
  .objectForKey('ConfigDataInstall'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.SoftwareUpdat
e')\
  .objectForKey('CriticalUpdateInstall'))
  if ( pref1 == 1 && pref2 == 1 ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Note: If automatic updates were selected during system setup, this setting may not have left an auditable artifact. Please turn off the check and re-enable when the GUI does not reflect the audited results.

Remediation:

Graphical Method:
Perform the following steps to enable system data files and security updates to install automatically:

1. Open `System Preferences`
2. Select `Software Updates`
3. Select `Advanced`
4. Set `Install system data files and security updates` to enabled

Terminal Method:
Run the following commands to enable automatic checking of system data files and security updates:

```
$ /usr/bin/sudo  /usr/bin/defaults write
/Library/Preferences/com.apple.SoftwareUpdate ConfigDataInstall -bool true

$ /usr/bin/sudo  /usr/bin/defaults write
/Library/Preferences/com.apple.SoftwareUpdate CriticalUpdateInstall -bool
true
```

Note: This remediation requires a log out and log in to show in the GUI.

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.SoftwareUpdate`
2. The key to include is `ConfigDataInstall`
3. The key must be set to `<true/>`
4. The key to also include is `CriticalUpdateInstall`
5. The key must be set to `<true/>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 Perform Automated Operating System Patch Management Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.4 Perform Automated Application Patch Management Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.7 Remediate Detected Vulnerabilities Remediate detected vulnerabilities in software through processes and tooling on a monthly, or more frequent, basis, based on the remediation process.		●	●
v7	3.4 Deploy Automated Operating System Patch Management Tools Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.	●	●	●
v7	3.5 Deploy Automated Software Patch Management Tools Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.	●	●	●

1.6 Ensure Install of macOS Updates Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that macOS updates are installed after they are available from Apple. This setting enables macOS updates to be automatically installed. Some environments will want to approve and test updates before they are delivered. It is best practice to test first where updates can and have caused disruptions to operations. Automatic updates should be turned off where changes are tightly controlled and there are mature testing and approval processes. Automatic updates should not be turned off simply to allow the administrator to contact users in order to verify installation. A dependable, repeatable process involving a patch agent or remote management tool should be in place before auto-updates are turned off.

Rationale:

Patches need to be applied in a timely manner to reduce the risk of vulnerabilities being exploited.

Impact:

Unpatched software may be exploited.

Audit:

Graphical Method:
Perform the following steps to ensure that macOS updates are set to auto update:

1. Open `System Preferences`
2. Select `Software Updates`
3. Select `Advanced`
4. Verify that `Install macOS updates` is enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Automatically Install macOS Updates` set to `True`

Terminal Method:
Run the following commands to verify that system data files and security updates are automatically checked:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.SoftwareUpdate')\
.objectForKey('AutomaticallyInstallMacOSUpdates').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable macOS updates to run automatically:

1. Open `System Preferences`
2. Select `Software Updates`
3. Select `Advanced`
4. Set `Install macOS updates` to enabled

Terminal Method:
Run the following commands to enable automatic checking of system data files and security updates:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.SoftwareUpdate
AutomaticallyInstallMacOSUpdates -bool TRUE
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.SoftwareUpdate`
2. The key to include is `AutomaticallyInstallMacOSUpdates`
3. The key must be set to `<true/>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 Perform Automated Operating System Patch Management Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.4 Perform Automated Application Patch Management Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v7	3.4 Deploy Automated Operating System Patch Management Tools Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.	●	●	●
v7	3.5 Deploy Automated Software Patch Management Tools Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.	●	●	●

1.7 Ensure Software Update Deferment Is Less Than or Equal to 30 Days (Automated)

Profile Applicability:

- Level 1

Description:

Apple provides the capability to manage software updates on Apple devices through mobile device management. Part of those capabilities permit organizations to defer software updates and allow for testing. Many organizations have specialized software and configurations that may be negatively impacted by Apple updates. If software updates are deferred, they should not be deferred for more than 30 days. This control only verifies that deferred software updates are not deferred for more than 30 days.

Manage software updates for Apple devices

Rationale:

Apple software updates almost always include security updates. Attackers evaluate updates to create exploit code in order to attack unpatched systems. The longer a system remains unpatched, the greater an exploit possibility exists in which there are publicly reported vulnerabilities.

Impact:

Some organizations may need more than 30 days to evaluate the impact of software updates.

Audit:

Graphical Method:
Perform the following steps to ensure that software updates are deferred at most 30 days:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that `Deferred Software Update Delays (Days)` is set to ≤ 30

Terminal Method:
Run the following command to verify that a profile is installed that defers software updates to at most 30 days:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('enforcedSoftwareUpdateDelay').js
EOS
```

If there is an output, it should be ≤ 30.
Note: If your organization does not use a software deferment mobile configuration, there will be no output and will pass the audit.

Remediation:

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `enforcedSoftwareUpdateDelay`
3. The key must be set to `<integer><1-30></integer>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 Perform Automated Operating System Patch Management Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v8	7.4 Perform Automated Application Patch Management Perform application updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.	●	●	●
v7	3.4 Deploy Automated Operating System Patch Management Tools Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.	●	●	●
v7	3.5 Deploy Automated Software Patch Management Tools Deploy automated software update tools in order to ensure that third-party software on all systems is running the most recent security updates provided by the software vendor.	●	●	●

1.8 Ensure Computer Name Does Not Contain PII or Protected Organizational Information (Manual)

Profile Applicability:

- Level 2

Description:

If the computer is used in an organization that assigns host names, it is a good idea to change the computer name to the host name. This is more of a best practice than a security measure. If the host name and the computer name are the same, computer support may be able to track problems down more easily.

For organizations or for users that self-administer their own computers, it is important to not use sensitive or personal information in computer names. The name of a computer that uses untrusted networks will be exposed at a minimum to the responsible network team of that network. While I may not care if the computer name of Ron Colvin's MacBook Pro is visible on a Hooters WiFi network, other uses may not feel the same way.

Examples of possibly inappropriate content in computer names include:

- User directory account names
- Computer directory account names where machine accounts exist
- Contact phone numbers
- Physical locations of offices or residences
- Personal information that can be augmented with Facebook data to assist social engineering attacks

Standard naming patterns avoid collisions and mitigate risk for computer users.

With mobile devices, using DHCP IP tracking has serious drawbacks; hostname or computer name tracking makes much more sense for those organizations that can implement it. If the computer is using different names for the "Computer Name" DNS and Directory environments, it can be difficult to manage Macs in an Enterprise asset inventory.

Rationale:

Part of IT security is having visibility into all of the devices for which an organization is responsible. Without a complete inventory, it is impossible to ensure all security controls are met on all organizational devices.

Default macOS naming deconfliction controls can create issues for appropriate management and tracking as well as privacy exposure. By default, the name of a macOS computer is derived from the first user created. If the user has multiple computers or an image is used without an appropriate name change, there will be multiple computers with names derived from the same user for discovery deconfliction. How many "Ron Colvin's MacBook Pro" should there be, and are any missing?

Local network auto renaming to avoid collisions also allows for the enumeration of local computer names. Computers should not be named after their users, especially on untrusted networks. For social engineering purposes, the computer name should not provide a full name of the user or an identifiable name that might be used to assist in targeted user attacks.

A documented plan to better enable a complete device inventory without exposing user or organizational information is part of mature security.

Audit:

Graphical Method:
Perform the following steps to verify the computer name:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Computer Name` is set to your organization's parameters

Remediation:

Graphical Method:
Perform the following steps to set the computer name:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Computer Name` to your organization's parameters

References:

1. https://support.apple.com/en-ca/guide/mac-help/mchlp1177/11.0/mac/11.0
2. https://uberagent.com/blog/choosing-macos-computer-names-wisely/
3. https://support.apple.com/en-ca/guide/mac-help/mchlp2322/11.0/mac/11.0

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	1.1 Establish and Maintain Detailed Enterprise Asset Inventory Establish and maintain an accurate, detailed, and up-to-date inventory of all enterprise assets with the potential to store or process data, to include: end-user devices (including portable and mobile), network devices, non-computing/IoT devices, and servers. Ensure the inventory records the network address (if static), hardware address, machine name, enterprise asset owner, department for each asset, and whether the asset has been approved to connect to the network. For mobile end-user devices, MDM type tools can support this process, where appropriate. This inventory includes assets connected to the infrastructure physically, virtually, remotely, and those within cloud environments. Additionally, it includes assets that are regularly connected to the enterprise's network infrastructure, even if they are not under control of the enterprise. Review and update the inventory of all enterprise assets bi-annually, or more frequently.	●	●	●
v7	9.1 Associate Active Ports, Services and Protocols to Asset Inventory Associate active ports, services and protocols to the hardware assets in the asset inventory.		●	●

2 System Preferences

This section contains recommendations related to configurable options in the *System Preferences* panel.

2.1 Bluetooth

Bluetooth is a short-range, low-power wireless technology commonly integrated into portable computing and communication devices and peripherals. Bluetooth is best used in a secure environment where unauthorized users have no physical access near the Mac. If Bluetooth is used, it should be secured properly (see below).

2.1.1 Ensure Show Bluetooth Status in Menu Bar Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

By showing the Bluetooth status in the menu bar, a small Bluetooth icon is placed in the menu bar. This icon quickly shows the status of Bluetooth, and can allow the user to quickly turn Bluetooth on or off.

Rationale:

Enabling "Show Bluetooth status in menu bar" is a security awareness method that helps understand the current state of Bluetooth, including whether it is enabled, discoverable, what paired devices exist, and what paired devices are currently active.

Impact:

Bluetooth is a useful wireless tool that has been widely exploited when configured improperly. The user should have insight into the Bluetooth status.

Audit:

Graphical Method:
Perform the following steps to ensure that Bluetooth status shows in the menu bar:

1. Open `System Preferences`
2. Select `Bluetooth`
3. Verify the `Show Bluetooth in menu bar` is selected

Terminal Method:
For each user, run the following command to verify that the Bluetooth status is enabled to show in the menu bar:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Preferences/com.apple.systemuiserver menuExtras |
/usr/bin/grep Bluetooth.menu

"/System/Library/CoreServices/Menu Extras/Bluetooth.menu"
```

Note: If the settings has not been changed from the default, then this audit will fail on the command line. Follow the remediation instructions to verify that it is set to a disabled status.
example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Preferences/com.apple.systemuiserver menuExtras |
/usr/bin/grep Bluetooth.menu

"/System/Library/CoreServices/Menu Extras/Bluetooth.menu"
```

Remediation:

Graphical Method:
Perform the following steps to enable Bluetooth status in the menu bar:

1. Open `System Preferences`
2. Select `Bluetooth`
3. Set `Show Bluetooth in menu bar` to enabled

Terminal Method:
For each user, run the following command to enable Bluetooth status in the menu bar:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Preferences/com.apple.systemuiserver menuExtras -
array-add "/System/Library/CoreServices/Menu Extras/Bluetooth.menu"
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Preferences/com.apple.systemuiserver menuExtras -
array-add "/System/Library/CoreServices/Menu Extras/Bluetooth.menu"
```

Note: If the remediation is run multiple times, multiple instances of the Bluetooth status will appear after rebooting the system. Command-click and drag the unwanted icons off the menu bar.

References:

1. http://osxdaily.com/2012/01/05/remove-icons-menu-bar-mac-os-x/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	13.9 Deploy Port-Level Access Control Deploy port-level access control. Port-level access control utilizes 802.1x, or similar network access control protocols, such as certificates, and may incorporate user and/or device authentication.			●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.2 Date & Time

This section contains recommendations related to the configurable items under the *Date & Time* panel.

2.2.1 Ensure "Set time and date automatically" Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Correct date and time settings are required for authentication protocols, file creation, modification dates, and log entries.

Note: If your organization has internal time servers, enter them here. Enterprise mobile devices may need to use a mix of internal and external time servers. If multiple servers are required, use the Date & Time System Preference with each server separated by a space.

Additional Note: The default Apple time server is time.apple.com. Variations include time.euro.apple.com. While it is certainly more efficient to use internal time servers, there is no reason to block access to global Apple time servers or to add a time.apple.com alias to internal DNS records. There are no reports that Apple gathers any information from NTP synchronization, as the computers already phone home to Apple for Apple services including iCloud use and software updates. Best practice is to allow DNS resolution to an authoritative time service for time.apple.com, preferably to connect to Apple servers, but local servers are acceptable as well.

Rationale:

Kerberos may not operate correctly if the time on the Mac is off by more than 5 minutes. This in turn can affect Apple's single sign-on feature, Active Directory logons, and other features.

Impact:

The `timed` service will periodically synchronize with named time servers and will make the computer time more accurate.

Audit:

Graphical Method:
Perform the following steps to ensure that the system's date and time are set automatically:

1. Open `System Preferences`
2. Select `Date & Time`
3. Verify that `Set date and time automatically` is enabled

Terminal Method:
Run the following command to ensure that date and time are automatically set:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -getusingnetworktime

Network Time: On
```

Remediation:

Graphical Method:
Perform the following steps to enable the date and time to be set automatically:

1. Open `System Preferences`
2. Select `Date & Time`
3. Set `Set date and time automatically` to enabled

Note: By default, the operating system will use `time.apple.com` as the time server. You can change to any time server that meets your organization's requirements.

Terminal Method:
Run the following commands to enable the date and time setting automatically:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -setnetworktimeserver
<your.time.server>

setNetworkTimeServer: <your.time.server>

$ /usr/bin/sudo /usr/sbin/systemsetup -setusingnetworktime on

setUsingNetworkTime: On
```

example:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -setnetworktimeserver time.apple.com

setNetworkTimeServer: time.apple.com

$ /usr/bin/sudo /usr/sbin/systemsetup -setusingnetworktime on

setUsingNetworkTime: On
```

Run the following commands if you have not set, or need to set, a new time zone:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -listtimezones

$ /usr/bin/sudo /usr/sbin/systemsetup -settimezone <selected time zone>
```

example:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -listtimezones

Time Zones:
 Africa/Abidjan
 ...
 Pacific/Wallis

$ /usr/bin/sudo /usr/sbin/systemsetup -settimezone America/New_York

Set TimeZone: America/New_York
```

Additional Information:

To learn more about `timed`, read: [Has anyone got the time? How High Sierra has changed time synchronisation]

Note: The profile configuration has been removed since it requires a specific time server to be set.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.4 Standardize Time Synchronization Standardize time synchronization. Configure at least two synchronized time sources across enterprise assets, where supported.		●	●
v7	6.1 Utilize Three Synchronized Time Sources Use at least three synchronized time sources from which all servers and network devices retrieve time information on a regular basis so that timestamps in logs are consistent.		●	●

2.2.2 Ensure Time Is Set Within Appropriate Limits (Automated)

Profile Applicability:

- Level 1

Description:

Correct date and time settings are required for authentication protocols, file creation, modification dates and log entries. Ensure that time on the computer is within acceptable limits. Truly accurate time is measured within milliseconds. For this audit, a drift under four and a half minutes passes the control check. Since Kerberos is one of the important features of macOS integration into Directory systems, the guidance here is to warn you before there could be an impact to operations. From the perspective of accurate time, this check is not strict, so it may be too great for your organization. Your organization can adjust to a smaller offset value as needed.

If there are consistent drift issues on the OS, some of the most common drift issues should be investigated:

- The chosen time server is not reachable based on network firewall rules on the current network
- The computer is offline often and the battery drains, and the network is not immediately available
- The chosen time server is a special internal or non-public time server that does not provide a reliable time source

Note: `ntpdate` has been deprecated with 10.14. `sntp` replaces that command.

Rationale:

Kerberos may not operate correctly if the time on the Mac is off by more than 5 minutes. This in turn can affect Apple's single sign-on feature, Active Directory logons, and other features. Audit check is for more than 4 minutes and 30 seconds ahead or behind.

Impact:

Accurate time is required for many computer functions.

Audit:

Terminal Method:
Run the following commands to verify the time is set within an appropriate limit:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -getnetworktimeserver
```

The output will include `Network Time Server:` and the name of your time server.
example: `Network Time Server: time.apple.com`

```
$ /usr/bin/sudo /usr/bin/sntp <your.time.server>
```

Ensure that the offset result(s) are between -270.x and 270.x seconds.
example:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -getnetworktimeserver

Network Time Server: time.apple.com

$ /usr/bin/sudo /usr/bin/sntp time.apple.com

+0.003399 +/- 0.058059 time.apple.com 17.253.24.253
```

Remediation:

Run the following commands to ensure your time is set within an appropriate limit:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -getnetworktimeserver
```

The output will include `Network Time Server:` and the name of your time server
example: `Network Time Server: time.apple.com`.

```
$ /usr/bin/sudo /usr/bin/sntp -sS <your.time.server>
```

example:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -getnetworktimeserver

Network Time Server: time.apple.com

$ /usr/bin/sudo /usr/bin/sntp -sS time.apple.com
```

Additional Information:

The associated check will fail if no network connection is available.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.4 Standardize Time Synchronization Standardize time synchronization. Configure at least two synchronized time sources across enterprise assets, where supported.		●	●
v7	6.1 Utilize Three Synchronized Time Sources Use at least three synchronized time sources from which all servers and network devices retrieve time information on a regular basis so that timestamps in logs are consistent.		●	●

2.3 Desktop & Screen Saver

This section contains recommendations related to the configurable items under the Desktop & Screen Saver panel.

It is vital that all users know how to quickly lock their screens if necessary. This capability in no way means that the user intends to hide unauthorized activity. The contents on the display screen may not be content authorized for someone that is visiting a user or their workspace. There are many simple ways to lock the screen, the obvious at this point is just selecting Lock Screen under the Apple menu or for a laptop to close the lid.

8 Ways to Lock Your Mac

2.3.1 Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

A locking screen saver is one of the standard security controls to limit access to a computer and the current user's session when the computer is temporarily unused or unattended. In macOS, the screen saver starts after a value is selected in the drop-down menu. 20 minutes or less is an acceptable value. Any value can be selected through the command line or script, but a number that is not reflected in the GUI can be problematic. 20 minutes is the default for new accounts.

Rationale:

Setting an inactivity interval for the screen saver prevents unauthorized persons from viewing a system left unattended for an extensive period of time.

Impact:

If the screen saver is not set, users may leave the computer available for an unauthorized person to access information.

Audit:

The preferred audit procedure for this control will evaluate every user account on the computer and will report on all users where the value has been set. If the default value of 20 minutes is used and the user has never changed the setting, there will not be an audit result on their compliant setting. The time is set in seconds so all outputs will be in that format.
Perform the following to ensure the system is set for the screen saver to activate in 20 minutes or less:
Run this script to verify the idle times for all users:

```
UUID=`ioreg -rd1 -c IOPlatformExpertDevice | grep "IOPlatformUUID" | sed -e
's/^.* "\(.*\)"$/\1/'`

for i in $(find /Users -type d -maxdepth 1)
do
  PREF=$i/Library/Preferences/ByHost/com.apple.screensaver.$UUID
  if [ -e $PREF.plist ]
  then
  echo -n "Checking User: '$i': "
  defaults read $PREF.plist idleTime 2>&1
  fi
done
```

Note: If the output of the script includes `The domain/default pair of (com.apple.screensaver, idleTime) does not exist` for any user, then the setting has not been changed from the default. Follow the remediation instructions to set the idle time to match your organization's policy.
For Macs with a single user:
Graphical Method:

1. Open `System Preferences`
2. Select `Desktop & Screen Saver`
3. Select `Screen Saver`
4. Verify that `Start after` is set for 20 minutes or less (≤1200)

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Idle Time` set to `≤1200`

Terminal Method:
Run the following command to verify that the screen saver idle time of individual users is set to less than or equal to 20 minutes:

```
$ /usr/bin/sudo -u <username> /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.screensaver')
\
  .objectForKey('idleTime'))
  if ( pref1 <= 1200 ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Note: If there is no output, then the setting has not been changed from the default and is considered not in compliance. Follow the remediation instructions to set the idle time to match your organization's policy.
or
Run the following command to verify that a profile is installed that enables a system-wide screensaver idle time of less than or equal to 20 minutes:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let timeout =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.screensaver')
\
.objectForKey('idleTime'))
  if ( timeout <= 1200 ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to set the screen saver to activate in 20 minutes or less:

1. Open `System Preferences`
2. Select `Desktop & Screen Saver`
3. Select `Screen Saver`
4. Select on option for `Start after` that is 20 minutes or less (≤1200)

Terminal Method:
Run the following command to set individual users to an idle time of the screen saver is set to 20 minutes or less (≤1200):

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults -currentHost write
com.apple.screensaver idleTime -int <value ≤1200>
```

example:

```
$ /usr/bin/sudo -u seconduser /usr/bin/defaults -currentHost write
com.apple.screensaver idleTime -int 600

$ /usr/bin/sudo -u seconduser /usr/bin/defaults -currentHost read
com.apple.screensaver idleTime

600
```

Note: Issues arise if the command line is used to make the setting something other than what is available in the GUI Menu. Choose either 1 (60), 2 (120), 5 (300), 10 (600), or 20 (1200) minutes to avoid any issues.
Profile Method:

1. The PayloadType string is `com.apple.screensaver`
2. The key to include is `idleTime`
3. The key must be set to `<integer><≤1200></integer>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.3 Configure Automatic Session Locking on Enterprise Assets Configure automatic session locking on enterprise assets after a defined period of inactivity. For general purpose operating systems, the period must not exceed 15 minutes. For mobile end-user devices, the period must not exceed 2 minutes.	●	●	●
v7	16.11 Lock Workstation Sessions After Inactivity Automatically lock workstation sessions after a standard period of inactivity.	●	●	●

2.3.2 Ensure Screen Saver Corners Are Secure (Automated)

Profile Applicability:

- Level 2

Description:

Hot Corners can be configured to disable the screen saver by moving the mouse cursor to a corner of the screen.

Rationale:

Setting a hot corner to disable the screen saver poses a potential security risk since an unauthorized person could use this to bypass the login screen and gain access to the system.

Audit:

Graphical Method:
Perform the following steps to ensure that a Hot Corner is not set to Disable Screen Saver:

1. Open `System Preferences`
2. Select `Desktop & Screen Saver`
3. Select `Screen Saver`
4. Select `Hot Corners...` and verify that `Disable Screen Saver` is not enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `<wvous-tl-corner>`, `<wvous-bl-corner>`, `<wvous-tr-corner>`, and `<wvous-br-corner>` not set to `6`

Terminal Method:
For all users, run the following commands to verify that Disable Screen Saver is not set as a Hot Corner:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read com.apple.dock wvous-tl-
corner

$ /usr/bin/sudo -u <username> /usr/bin/defaults read com.apple.dock wvous-bl-
corner

$ /usr/bin/sudo -u <username> /usr/bin/defaults read com.apple.dock wvous-tr-
corner

$ /usr/bin/sudo -u <username> /usr/bin/defaults read com.apple.dock wvous-br-
corner
```

Verify that the output does not have `6` as a key value. Any other number, or an output that includes `does not exist`, is compliant.
example:

```
$ /usr/bin/sudo -u seconduser /usr/bin/defaults read com.apple.dock wvous-tl-
corner

10
$ /usr/bin/sudo -u seconduser /usr/bin/defaults read com.apple.dock wvous-bl-
corner

2020-07-31 14:32:29.018 defaults[39521:1276494]
The domain/default pair of (com.apple.dock, wvous-bl-corner) does not exist

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read com.apple.dock wvous-tr-
corner

2020-07-31 14:32:32.403 defaults[39523:1276515]
The domain/default pair of (com.apple.dock, wvous-tr-corner) does not exist

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read com.apple.dock wvous-br-
corner

2020-07-31 14:32:36.045 defaults[39525:1276529]
The domain/default pair of (com.apple.dock, wvous-br-corner) does not exist
```

Run the following command to verify that a profile is installed secures screen saver corners:

```
$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep wvous-bl-
corner

$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep wvous-br-
corner

$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep wvous-tl-
corner

$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep wvous-tr-
corner
```

The output should include `wvous-bl-corner`, `wvous-br-corner`, `wvous-tl-corner`, and `wvous-tr-corner` are ≠ `6`;

Remediation:

Graphical Method:
Perform the following steps to disable a Hot Corner set to Disable Screen Saver:

1. Open `System Preferences`
2. Select `Desktop & Screen Saver`
3. Select `Screen Saver`
4. Select `Hot Corners...` and set any corner that is set to `Disable Screen Saver` to another setting that corresponds to your organization's requirements

Terminal Method:
Run the following command to turn off Disable Screen Saver for a Hot Corner:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write com.apple.dock <corner
that is set to '6'> -int 0
```

example:

```
$ /usr/bin/sudo -u seconduser /usr/bin/defaults write com.apple.dock wvous-
tl-corner -int 0

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read com.apple.dock wvous-tl-
corner

0
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.dock`
2. The key to include is `Forced`
3. The key must be set to the following:

```
<array>
        <dict>
                <key>mcx_preference_settings</key>
                        <dict>
                                <key>wvous-bl-corner</key>
                                <integer><≠6></integer>
                                <key>wvous-br-corner</key>
                                <integer><≠6></integer>
                                <key>wvous-tl-corner</key>
                                <integer><≠6></integer>
                                <key>wvous-tr-corner</key>
                                <integer><≠6></integer>
                        </dict>
                </dict>
</array>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.3 Configure Automatic Session Locking on Enterprise Assets Configure automatic session locking on enterprise assets after a defined period of inactivity. For general purpose operating systems, the period must not exceed 15 minutes. For mobile end-user devices, the period must not exceed 2 minutes.	●	●	●
v7	16.11 Lock Workstation Sessions After Inactivity Automatically lock workstation sessions after a standard period of inactivity.	●	●	●

2.4 Sharing

This section contains recommendations related to the configurable items under the *Sharing* panel.

2.4.1 Ensure Remote Apple Events Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Apple Events is a technology that allows one program to communicate with other programs. Remote Apple Events allows a program on one computer to communicate with a program on a different computer.

Rationale:

Disabling Remote Apple Events mitigates the risk of an unauthorized program gaining access to the system.

Impact:

With remote Apple events turned on, an AppleScript program running on another Mac can interact with the local computer.

Audit:

Graphical Method:
Perform the following steps to ensure that Remote Apple Events is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Remote Apple Events` is not enabled

Terminal Method:
Run the following commands to verify that Remote Apple Events is not set

```
$ /usr/bin/sudo /usr/sbin/systemsetup -getremoteappleevents

Remote Apple Events: Off
```

Remediation:

Graphical Method:
Perform the following steps to disable Remote Apple Events:

1. Open `System Preferences
2. Select `Sharing
3. Set `Remote Apple Events` to disabled

Terminal Method:
Run the following commands to set Remote Apple Events to Off:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -setremoteappleevents off

setremoteappleevents: Off
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.2 Ensure Internet Sharing Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Internet Sharing uses the open source `natd` process to share an internet connection with other computers and devices on a local network. This allows the Mac to function as a router and share the connection to other, possibly unauthorized, devices.

Rationale:

Disabling Internet Sharing reduces the remote attack surface of the system.

Impact:

Internet Sharing allows the computer to function as a router and other computers to use it for access. This can expose both the computer itself and the networks it is accessing to unacceptable access from unapproved devices.

Audit:

Graphical Method:
Perform the following steps to ensure Internet Sharing is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Internet Sharing` is not enabled

Terminal Method:
Run the following commands to verify that Internet Sharing is not set:

```
$ /usr/bin/sudo /usr/bin/defaults read
/Library/Preferences/SystemConfiguration/com.apple.nat >nul 2>&1 | grep -c
"Enabled = 1;"

0
```

or
Run the following command to verify that a profile is installed that automatically disables internet sharing:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.MCX')\
.objectForKey('forceInternetSharingOff').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to disable Internet Sharing:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Internet Sharing` to disabled

Terminal Method:
Run the following command to turn off Internet Sharing:

```
$ usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/SystemConfiguration/com.apple.nat NAT -dict Enabled -int
0
```

Note: Using the Terminal Method will not be reflected in the GUI, but will disable the underlying service.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.MCX`
2. The key to include is `forceInternetSharingOff`
3. The key must be set to `<true/>`

References:

1. STIGID AOSX-12-001270

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.3 Ensure Screen Sharing Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Screen Sharing allows a computer to connect to another computer on a network and display the computer's screen. While sharing the computer's screen, the user can control what happens on that computer, such as opening documents or applications, opening, moving, or closing windows, and even shutting down the computer.

While mature administration and management does not use graphical connections for standard maintenance, most help desks have capabilities to assist users in performing their work when they have technical issues and need support. Help desks use graphical remote tools to understand what the user sees and assist them so they can get back to work. For MacOS, some of these remote capabilities can use Apple's OS tools. Control is therefore not meant to prohibit the use of a just-in-time graphical view from authorized personnel with authentication controls. Sharing should not be enabled except in narrow windows when help desk support is required.

Rationale:

Disabling Screen Sharing mitigates the risk of remote connections being made without the user of the console knowing that they are sharing the computer.

Impact:

Help desks may require the periodic use of a graphical connection mechanism to assist users. Any support that relies on native MacOS components will not work unless a scripted solution to enable and disable sharing as neccessary.

Audit:

Graphical Method:
Perform the following steps to ensure Screen Sharing is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Screen Sharing` is not enabled

Terminal Method:
Run the following commands to verify that Screen Sharing is not set:

```
$ /usr/bin/sudo /bin/launchctl list | grep -c com.apple.screensharing

0
```

Remediation:

Graphical Method:
Perform the following steps to disable Screen Sharing:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Screen Sharing` to disabled

Terminal Method:
Run the following command to turn off Screen Sharing:

```
$ /usr/bin/sudo /bin/launchctl disable system/com.apple.screensharing
```

References:

1. https://support.apple.com/guide/mac-help/turn-screen-sharing-on-or-off-mh11848/mac

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.4 Ensure Printer Sharing Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

By enabling Printer Sharing, the computer is set up as a print server to accept print jobs from other computers. Dedicated print servers or direct IP printing should be used instead.

Rationale:

Disabling Printer Sharing mitigates the risk of attackers attempting to exploit the print server to gain access to the system.

Audit:

Graphical Method:
Perform the following steps to ensure that Printer Sharing is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Printer Sharing` is not enabled

Terminal Method:
Run the following command to verify that Printer Sharing is not enabled:

```
$ /usr/bin/sudo /usr/sbin/cupsctl | grep -c "_share_printers=0"

1
```

Remediation:

Graphical Method:
Perform the following steps to disable Printer Sharing:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Printer Sharing` to disabled

Terminal Method:
Run the following command to disable Printer Sharing:

```
$ /usr/bin/sudo /usr/sbin/cupsctl --no-share-printers
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.5 Ensure Remote Login Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Remote Login allows an interactive terminal connection to a computer.

Rationale:

Disabling Remote Login mitigates the risk of an unauthorized person gaining access to the system via Secure Shell (SSH). While SSH is an industry standard to connect to posix servers, the scope of the benchmark is for Apple macOS clients, not servers.

macOS does have an IP-based firewall available (pf, ipfw has been deprecated) that is not enabled or configured. There are more details and links in section 7.5. macOS no longer has TCP Wrappers support built in and does not have strong Brute-Force password guessing mitigations, or frequent patching of openssh by Apple. Since most macOS computers are mobile workstations, managing IP-based firewall rules on mobile devices can be very resource intensive. All of these factors can be parts of running a hardened SSH server.

Impact:

The SSH server built into macOS should not be enabled on a standard user computer, particularly one that changes locations and IP addresses. A standard user that runs local applications, including email, web browser, and productivity tools, should not use the same device as a server. There are Enterprise management toolsets that do utilize SSH. If they are in use, the computer should be locked down to only respond to known, trusted IP addresses and appropriate administrator service accounts.

For macOS computers that are being used for specialized functions, there are several options to harden the SSH server to protect against unauthorized access including brute force attacks. There are some basic criteria that need to be considered:

- Do not open an SSH server to the internet without controls in place to mitigate SSH brute force attacks. This is particularly important for systems bound to Directory environments. It is great to have controls in place to protect the system, but if they trigger after the user is already locked out of their account, they are not optimal. If authorization happens after authentication, directory accounts for users that don't even use the system can be locked out.
- Do not use SSH key pairs when there is no insight to the security on the client system that will authenticate into the server with a private key. If an attacker gets access to the remote system and can find the key, they may not need a password or a key logger to access the SSH server.
- Detailed instructions on hardening an SSH server, if needed, are available in the CIS Linux Benchmarks, but it is beyond the scope of this benchmark.

Audit:

Graphical Method:
Perform the following steps to ensure that Remote Login is disabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Remote Login` is not enabled

Terminal Method:
Run the following command to verify that Remote Login is disabled:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -getremotelogin

Remote Login: Off
```

Remediation:

Graphical Method:
Perform the following steps to disable Remote Login:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Remote Login` to disabled

Terminal Method:
Run the following command to disable Remote Login:

```
$ /usr/bin/sudo /usr/sbin/systemsetup -setremotelogin off

Do you really want to turn remote login off? If you do, you will lose this
connection and can only turn it back on locally at the server (yes/no)?
```

Entering yes will disable remote login.

Additional Information:

`man sshd_config`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.6 Ensure DVD or CD Sharing Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

DVD or CD Sharing allows users to remotely access the system's optical drive. While Apple does not ship Macs with built-in optical drives any longer, external optical drives are still recognized when they are connected. In testing, the sharing of an external optical drive persists when a drive is reconnected.

Rationale:

Disabling DVD or CD Sharing minimizes the risk of an attacker using the optical drive as a vector for attack and exposure of sensitive data.

Impact:

Many Apple devices are now sold without optical drives, however drive sharing may be needed for legacy optical media. The media should be explicitly re-shared as needed rather than using a persistent share. Optical drives should not be used for long-term storage. To store necessary data from an optical drive it should be copied to another form of external storage. Optionally, an image can be made of the optical drive so that it is stored in its original form on another form of external storage.

Audit:

Graphical Method:
Perform the following steps to ensure that DVD or CD Sharing is disabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `DVD or CD sharing` is not enabled

Terminal Method:
Run the following command to verify that DVD or CD Sharing is disabled

```
$ /usr/bin/sudo /bin/launchctl list | grep -c com.apple.ODSAgent

0
```

Remediation:

Graphical Method:
Perform the following steps to disable DVD or CD Sharing:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `DVD or CD sharing` to disabled

Terminal Method:
Run the following command to disable DVD or CD Sharing:

```
$ /usr/bin/sudo /bin/launchctl disable system/com.apple.ODSAgent
```

Note: If using the Terminal method, the GUI will still show the service checked until after a reboot.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.7 Ensure Bluetooth Sharing Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Bluetooth Sharing allows files to be exchanged with Bluetooth-enabled devices.

Rationale:

Disabling Bluetooth Sharing minimizes the risk of an attacker using Bluetooth to remotely attack the system.

Impact:

Control 2.1.1 discusses disabling Bluetooth if no paired devices exist. There is a general expectation that Bluetooth peripherals will be used by most users in Apple's ecosystem. It is possible that sharing is required and Bluetooth peripherals are not. Bluetooth must be enabled if sharing is an acceptable use case.

Audit:

Graphical Method:
Perform the following steps to verify that Bluetooth Sharing is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Bluetooth Sharing` is not enabled

Terminal Method:
Run the following command to verify that Bluetooth Sharing is disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults -currentHost read
com.apple.Bluetooth PrefKeyServicesEnabled

0
$ /usr/bin/sudo -u firstuser /usr/bin/defaults -currentHost read
com.apple.Bluetooth PrefKeyServicesEnabled

0
```

Remediation:

Graphical Method:
Perform the following steps to disable Bluetooth Sharing:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Bluetooth Sharing` to disabled

Terminal Method:
Run the following command to disable Bluetooth Sharing is disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults -currentHost write
com.apple.Bluetooth PrefKeyServicesEnabled -bool false
$ /usr/bin/sudo -u firstuser /usr/bin/defaults -currentHost write
com.apple.Bluetooth PrefKeyServicesEnabled -bool false
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	4.8 Log and Alert on Changes to Administrative Group Membership Configure systems to issue a log entry and alert when an account is added to or removed from any group assigned administrative privileges.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

2.4.8 Ensure File Sharing Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

File sharing from a user workstation creates additional risks, such as:

- Open ports are created that can be probed and attacked
- Passwords are attached to user accounts for access that may be exposed and endanger other parts of the organizational environment, including directory accounts
- Increased complexity makes security more difficult and may expose additional attack vectors

Apple's File Sharing uses a combination of SMB (Windows sharing) and AFP (Mac sharing)

Two common ways to share files using File Sharing are:

1. Apple File Protocol (AFP) AFP automatically uses encrypted logins, so this method of sharing files is fairly secure. The entire hard disk is shared to administrator user accounts. Individual home folders are shared to their respective user accounts. Users' "Public" folders (and the "Drop Box" folder inside) are shared to any user account that has sharing access to the computer (i.e. anyone in the "staff" group, including the guest account if it is enabled).
2. Server Message Block (SMB), Common Internet File System (CIFS) When Windows (or possibly Linux) computers need to access file shared on a Mac, SMB/CIFS file sharing is commonly used. Apple warns that SMB sharing stores passwords is a less secure fashion than AFP sharing and anyone with system access can gain access to the password for that account. When sharing with SMB, each user that will access the Mac must have SMB enabled.

Apple warns that SMB sharing stored passwords is less secure, and anyone with system access can gain access to the password for that account. When sharing with SMB, each user accessing the Mac must have SMB enabled. Storing passwords, especially copies of valid directory passwords, decrease security for the directory account and should not be used.

Rationale:

By disabling File Sharing, the remote attack surface and risk of unauthorized access to files stored on the system is reduced.

Impact:

File Sharing can be used to share documents with other users, but hardened servers should be used rather than user endpoints. Turning on File Sharing increases the visibility and attack surface of a system unnecessarily.

Audit:

Graphical Method:
Perform the following steps to ensure that File Sharing is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `File Sharing` is not enabled

Terminal Method:
Run the following command to verify that File Sharing is not enabled:

```
$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep -c AppleFileServer

0

$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep -c "com.apple.smbd"

0
```

Remediation:

Graphical Method:
Perform the following steps to disable File Sharing:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `File Sharing` to disabled

Terminal Method:
Run the following command to disable File Sharing:

```
$ /usr/bin/sudo /bin/launchctl disable system/com.apple.AppleFileServer

$ /usr/bin/sudo /bin/launchctl disable system/com.apple.smbd
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.9 Ensure Remote Management Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Remote Management is the client portion of Apple Remote Desktop (ARD). Remote Management can be used by remote administrators to view the current screen, install software, report on, and generally manage client Macs.

The screen sharing options in Remote Management are identical to those in the Screen Sharing section. In fact, only one of the two can be configured. If Remote Management is used, refer to the Screen Sharing section above on issues regard screen sharing.

Remote Management should only be enabled when a Directory is in place to manage the accounts with access. Computers will be available on port 5900 on a macOS System and could accept connections from untrusted hosts depending on the configuration, which is a major concern for mobile systems. As with other sharing options, an open port even for authorized management functions can be attacked, and both unauthorized access and Denial-of-Service vulnerabilities could be exploited. If remote management is required, the pf firewall should restrict access only to known, trusted management consoles. Remote management should not be used across the Internet without the use of a VPN tunnel.

Rationale:

Remote Management should only be enabled on trusted networks with strong user controls present in a Directory system. Mobile devices without strict controls are vulnerable to exploit and monitoring.

Impact:

Many organizations utilize ARD for client management.

Audit:

Graphical Mode:
Perform the following steps to verify that Remote Management is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Remote Management` is not enabled

Terminal Method:
Run the following command to verify that Remote Management is not enabled:

```
$ /usr/bin/sudo /bin/ps -ef | /usr/bin/grep -e ARDAgent

0  9233  8630   0  3:32pm ttys001    0:00.00 grep -e ARDAgent
```

Remediation:

Graphical Method:
Perform the following steps to disable Remote Management:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Remote Management` to disabled

Terminal Method:
Run the following command to disable Remote Management:

```
$ /usr/bin/sudo
/System/Library/CoreServices/RemoteManagement/ARDAgent.app/Contents/Resources
/kickstart -deactivate -stop

Starting...
Removed preference to start ARD after reboot.
Done.
```

Additional Information:

```
/System/Library/CoreServices/RemoteManagement/ARDAgent.app/Contents/Resources
/kickstart -help
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●
v7	14.3 Disable Workstation to Workstation Communication Disable all workstation to workstation communication to limit an attacker's ability to move laterally and compromise neighboring systems, through technologies such as Private VLANs or microsegmentation.		●	●

2.4.10 Ensure Content Caching Is Disabled (Automated)

Profile Applicability:

- Level 2

Description:

Starting with 10.13 (macOS High Sierra), Apple introduced a service to make it easier to deploy data from Apple, including software updates, where there are bandwidth constraints to the Internet and fewer constraints or greater bandwidth exist on the local subnet. This capability can be very valuable for organizations that have throttled and possibly metered Internet connections. In heterogeneous enterprise networks with multiple subnets, the effectiveness of this capability would be determined by how many Macs were on each subnet at the time new, large updates were made available upstream. This capability requires the use of mac OS clients as P2P nodes for updated Apple content. Unless there is a business requirement to manage operational Internet connectivity and bandwidth user endpoints should not store content and act as a cluster to provision data.

Content types supported by Content Caching in macOS

Rationale:

The main use case for Mac computers is as mobile user endpoints. P2P sharing services should not be enabled on laptops that are using untrusted networks. Content Caching can allow a computer to be a server for local nodes on an untrusted network. While there are certainly logical controls that could be used to mitigate risk, they add to the management complexity. Since the value of the service is in specific use cases organizations with the use case described above can accept risk as necessary.

Impact:

This setting will adversely affect bandwidth usage between local subnets and the Internet.

Audit:

Graphical Method:
Perform the following steps to ensure that Content Caching is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Content Caching` is not enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Allow Content Cachine` set to `False`

Terminal Method:
Run the following command to verify that Content Caching is not enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.AssetCache')\
  .objectForKey('Activated'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationac
cess')\
  .objectForKey('allowContentCaching'))
  if ( ( pref1 == 0 ) || ( pref2 == 0 ) ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to disable Content Caching:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Content Caching` to disabled

Terminal Method:
Run the following command to disable Content Caching:

```
$ /usr/bin/sudo /usr/bin/AssetCacheManagerUtil deactivate
```

The output will include `Content caching deactivated`
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowContentCaching`
3. The key must be set to `<false/>`

References:

1. https://support.apple.com/guide/mac-help/about-content-caching-mchl9388ba1b/
2. https://support.apple.com/guide/mac-help/set-up-content-caching-on-mac-mchl3b6c3720/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.4.11 Ensure AirDrop Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

AirDrop is Apple's built-in on demand ad hoc file exchange system that is compatible with both macOS and iOS. It uses Bluetooth LE for discovery that limits connectivity to Mac or iOS users that are in close proximity. Depending on the setting it allows everyone or only Contacts to share files when they are nearby to each other.

In many ways this technology is far superior to the alternatives. The file transfer is done over a TLS encrypted session, does not require any open ports that are required for file sharing, does not leave file copies on email servers or within cloud storage, and allows for the service to be mitigated so that only people already trusted and added to contacts can interact with you.

While there are positives to AirDrop, there are privacy concerns that could expose personal information. For that reason, AirDrop should be disabled, and should only be enabled when needed and disabled afterwards. The recommendation against enabling the sharing is not based on any known lack of security in the protocol but for specific user operational concerns.

- If AirDrop is enabled the Mac is advertising that a Mac is addressable on the local network and open to either unwanted AirDrop upload requests or for a negotiation on whether the remote user is in the user's contacts list Neither process is desirable.
- In most known use cases AirDrop use is ad hoc networking where AirDrop use is where Apple device users decide that a file should be exchanged and opt to use AirDrop which can be abled on the fly for that exchange.

For organizations concerned about any use of AirDrop because of Digital Loss Prevention (DLP) monitoring on other protocols JAMF has an article on reviewing AirDrop logs.

Detecting outbound AirDrop transfers and logging them

Rationale:

AirDrop can allow malicious files to be downloaded from unknown sources. Contacts Only limits may expose personal information to devices in the same area.

Impact:

Disabling AirDrop can limit the ability to move files quickly over the network without using file shares.

Audit:

Graphical Method:
Perform the following steps to ensure that AirDrop is disabled:

1. Open `System Settings` in the Menu Bar
2. Select `General`
3. Select `AirDrop & Handoff`
4. Verify that `AirDrop` is set to `No One`
5. Open `System Settings`
6. Select `Control Center`
7. Select `AirDrop`
8. Verify that `Don't show in Menu Bar` is not selected

or

1. Open `System Settings`
2. Select `Privacy & Security`
3. Select `Profiles`
4. Verify that an installed profile has `Allow AirDrop` set to `False`

Terminal Method:
For all users, run the following commands to verify whether AirDrop is disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read com.apple.NetworkBrowser
DisableAirDrop

1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read com.apple.NetworkBrowser
DisableAirDrop

1

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read com.apple.NetworkBrowser
DisableAirDrop

0

$ /usr/bin/sudo -u thirduser /usr/bin/defaults read com.apple.NetworkBrowser
DisableAirDrop

The domain/default pair of (com.apple.NetworkBrowser, DisableAirDrop) does
not exist
```

or
Run the following command to verify that a profile is installed that disabled AirDrop:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('allowAirDrop').js
EOS

false
```

Note: AirDrop can only be enabled or disabled through configuration profiles. Any additional settings need to be set through the GUI or CL

Remediation:

Graphical Method:
Perform the following steps to disable AirDrop:

1. Open `System Settings` in the Menu Bar
2. Select `General`
3. Select `AirDrop & Handoff`
4. Set `AirDrop` to `No One`
5. Open `System Settings`
6. Select `Control Center`
7. Set `AirDrop` to `Don't show in Menu Bar`

Terminal Method:
Run the following commands to disable AirDrop:

```
$ /usr/bin/sudo -u <username> defaults write com.apple.NetworkBrowser
DisableAirDrop -bool true
```

example:

```
$ /usr/bin/sudo -u seconduser defaults write com.apple.NetworkBrowser
DisableAirDrop -bool true
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowAirDrop`
3. The key must be set to `<false/>`

Note: AirDrop can only be enabled or disabled through configuration profiles. Any additional settings need to be set through the GUI or CL
Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://www.techrepublic.com/article/apple-airdrop-users-reportedly-vulnerable-to-security-flaw/
2. https://www.imore.com/how-apple-keeps-your-airdrop-files-private-and-secure
3. https://en.wikipedia.org/wiki/AirDrop

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	6.7 Centralize Access Control Centralize access control for all enterprise assets through a directory service or SSO provider, where supported.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	15.4 Disable Wireless Access on Devices if Not Required Disable wireless access on devices that do not have a business purpose for wireless access.			●

2.4.12 Ensure Media Sharing Is Disabled (Automated)

Profile Applicability:

- Level 2

Description:

Starting with macOS 10.15, Apple has provided a control which permits a user to share Apple downloaded content on all Apple devices that are signed in with the same Apple ID. This allows users to share downloaded Movies, Music, or TV shows with other controlled macOS, iOS and iPadOS devices, as well as photos with Apple TVs.

With this capability, guest users can also use media downloaded on the computer.

The recommended best practice is not to use the computer as a server, but to utilize Apple's cloud storage in order to download and use content stored there if content stored with Apple is used on multiple devices.

https://support.apple.com/guide/mac-help/set-up-media-sharing-on-mac-mchlp13371337/mac

Rationale:

Disabling Media Sharing reduces the remote attack surface of the system.

Impact:

Media Sharing allows for pre-downloaded content on a Mac to be available to other Apple devices on the same network. Leaving this disabled forces device users to stream or download content from each Apple authorized device. This sharing could even allow unauthorized devices on the same network media access.

Audit:

Graphical Method:
Perform the following steps to ensure that Media Sharing is not enabled:

1. Open `System Preferences`
2. Select `Sharing`
3. Verify that `Media Sharing` is not enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `homeSharingUIStatus` set to `0`
4. Verify that an installed profile has `legacySharingUIStatus` set to `0`
5. Verify that an installed profile has `mediaSharingUIStatus` set to `0`

Terminal Method:
Run the following command to verify that Media Sharing is not enabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
com.apple.amp.mediasharingd home-sharing-enabled

0
```

example:

```
$ /usr/bin/sudo -u test /usr/bin/defaults read com.apple.amp.mediasharingd
home-sharing-enabled

0

$ /usr/bin/sudo -u test2 /usr/bin/defaults read com.apple.amp.mediasharingd
home-sharing-enabled

1
```

or
Run the following command to verify that a profile is installed that disablesMedia Sharing:

```
$ /usr/bin/sudosudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.preferences.s
haring.SharingPrefsExtension')\
  .objectForKey('homeSharingUIStatus'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.preferences.s
haring.SharingPrefsExtension')\
  .objectForKey('legacySharingUIStatus'))
  let pref3 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.preferences.s
haring.SharingPrefsExtension')\
  .objectForKey('mediaSharingUIStatus'))
  if ( pref1 == 0 && pref2 == 0 && pref3 == 0 ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to disable Media Sharing:

1. Open `System Preferences`
2. Select `Sharing`
3. Set `Media Sharing` to disabled

Terminal Method:
Run the following command to disable Media Sharing:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
com.apple.amp.mediasharingd home-sharing-enabled -int 0
```

example:

```
$ sudo -u test2 /usr/bin/defaults write com.apple.amp.mediasharingd home-
sharing-enabled -int 0
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.preferences.sharing.SharingPrefsExtension`
2. The key to include is `homeSharingUIStatus`
3. The key must be set to `<integer>0</integer>`
4. The key to also include is legacySharingUIStatus
5. The key must be set to `<integer>0</integer>`
6. The key to also include is mediaSharingUIStatus
7. The key must be set to `<integer>0</integer>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.5 Security & Privacy

This section contains recommendations for configurable options under the *Security & Privacy* panel.

Additional privacy preference information from Apple

If the computer is present in an area where there are privacy concerns or sensitive activity is taking place, the Mac should be configured appropriately for the sensitive area.

Camera: If the computer is present in an area where there are privacy concerns or sensitive activity is taking place, the camera should be covered at those times. A permanent cover or alteration may be required when the computer is always located in a confidential area.

Microphone: If the computer is present in an area where there are privacy concerns or sensitive activity is taking place, the microphone input should be set to zero in the input tab of the Sound preference pane at those times. Individual management of applications with access to the microphone may be managed in the Security & Privacy Preference Pane under Microphone.

WiFi and Bluetooth Some organizations have comprehensive rules that cover the use of wireless technologies in order to implement operational security. There are often specific policies governing the use of both Bluetooth and Wi-Fi (802.11) that may include disabling the wireless capability in either software or hardware or both. Wireless access is part of the feature set required for mobile computers and is considered essential for most users.

Malware is continuously discovered that circumvents the privacy controls of the built-in Video, audio or network capabilities. No computer has perfect security, and even if all the drivers are disabled or removed, working drivers can be reintroduced by a determined attacker.

Additional info Apple Pays $100.5K Bug Bounty for Mac Webcam Hack

Mac users, update Zoom now — your microphone may be spying on you

Recommended settings for Wi-Fi routers and access points

Control access to the microphone on Mac

Bluetooth security

2.5.1 Encryption

Apple has created simple, easy-to-use encryption capabilities built into macOS. Those tools need to be utilized in order to protect information processed by macOS computers.

2.5.1.1 Ensure FileVault Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

FileVault secures a system's data by automatically encrypting its boot volume and requiring a password or recovery key to access it.

FileVault should be used with a saved escrow key to ensure that the owner can decrypt their data if the password is lost.

FileVault may also be enabled using command line using the `fdesetup` command. To use this functionality, consult the Der Flounder blog for more details (see link below under References).

Rationale:

Encrypting sensitive data minimizes the likelihood of unauthorized users gaining access to it.

Impact:

Mounting a FileVault encrypted volume from an alternate boot source will require a valid password to decrypt it.

Audit:

Graphical Method:
Perform the following steps to verify that FileVault is enabled:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `FileVault`
4. Verify that `FileVault` states `FileVault is turned on for the disk "<disk name>"`
5. Select `Show All`
6. Selection `Profile`
7. Verify that an installed profile has `FileVault Can't Disable` set to True

Terminal Method:
Run the following command to verify that FileVault is enabled and cannot be disabled:

```
$ /usr/bin/sudo /usr/bin/fdesetup status

FileVault is On

$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.MCX')\
.objectForKey('dontAllowFDEDisable').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable FileVault:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `FileVault`
4. Select `Turn On FileVault`

Note: This will allow you to create a recovery key for FileVault. Keep the key saved securely in case it is needed at a later date.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.MCX`
2. The key to include is `dontAllowFDEDisable`
3. The key must be set to `<true/>`

Note: This profile is required to pass the audit.

References:

1. https://derflounder.wordpress.com/2015/02/02/managing-yosemites-filevault-2-with-fdesetup/
2. https://derflounder.wordpress.com/2019/01/15/unlock-or-decrypt-your-filevault-encrypted-boot-drive-from-the-command-line-on-macos-mojave/
3. https://derflounder.wordpress.com/2021/10/29/use-of-filevault-institutional-recovery-keys-no-longer-recommended-by-apple/

Additional Information:

FileVault may not be desirable on a virtual OS. As long as the hypervisor and file storage are encrypted, the virtual OS does not need to be. Rather than checking if the OS is virtual and passing the control regardless of the encryption of the host system, the normal check will be run. Security officials can evaluate the comprehensive controls outside of the OS being tested.

Part of FileVault management in an Enterprise environment is to ensure key management if technical staff need to decrypt encrypted volumes. More information here: https://derflounder.wordpress.com/2021/10/29/use-of-filevault-institutional-recovery-keys-no-longer-recommended-by-apple/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.6 Encrypt Data on End-User Devices Encrypt data on end-user devices containing sensitive data. Example implementations can include: Windows BitLocker®, Apple FileVault®, Linux® dm-crypt.	●	●	●
v8	3.11 Encrypt Sensitive Data at Rest Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.		●	●
v7	13.6 Encrypt the Hard Drive of All Mobile Devices. Utilize approved whole disk encryption software to encrypt the hard drive of all mobile devices.	●	●	●
v7	14.8 Encrypt Sensitive Information at Rest Encrypt all sensitive information at rest using a tool that requires a secondary authentication mechanism not integrated into the operating system, in order to access the information.			●

2.5.1.2 Ensure all user storage APFS volumes are encrypted (Manual)

Profile Applicability:

- Level 1

Description:

Apple developed a new file system which was first made available in 10.12 and then became the default in 10.13. The file system is optimized for Flash and Solid-State storage and encryption. https://en.wikipedia.org/wiki/Apple_File_System macOS computers generally have several volumes created as part of APFS formatting, including Preboot, Recovery and Virtual Memory (VM), as well as traditional user disks.

All APFS volumes that do not have specific roles and do not require encryption should be encrypted. "Role" disks include Preboot, Recovery and VM. User disks are labelled with "(No specific role)" by default.

Rationale:

In order to protect user data from loss or tampering volumes, carrying data should be encrypted.

Impact:

While FileVault protects the boot volume, data may be copied to other attached storage and reduce the protection afforded by FileVault. Ensure all user volumes are encrypted to protect data.

Audit:

Terminal Method:
Run the following command to list the APFS Volumes:

```
$ /usr/bin/sudo /usr/sbin/diskutil ap list
```

Ensure all user data disks are encrypted.

example:

```
$ /usr/bin/sudo /usr/sbin/diskutil ap list

APFS Volume Disk (Role):   disk1s1 (No specific role)
Name:                      Macintosh HD (Case-insensitive)
Mount Point:               /
Capacity Consumed:         188514598912 B (188.5 GB)
FileVault:                 Yes (Unlocked)

APFS Containers (2 found)
|
+-- Container disk1 XXXX
|   ====================================================
|   APFS Container Reference:     disk1
|   Size (Capacity Ceiling):      249152200704 B (249.2 GB)
|   Minimum Size:                 249152200704 B (249.2 GB)
|   Capacity In Use By Volumes:   195635597312 B (195.6 GB) (78.5% used)
|   Capacity Not Allocated:       53516603392 B (53.5 GB) (21.5% free)
|   |
|   +-< Physical Store disk0s4 XXXXXY
|   |   ----------------------------------------------------------
|   |   APFS Physical Store Disk:   disk0s4
|   |   Size:                        249152200704 B (249.2 GB)
|   |
|   +-> Volume disk1s1 XXXXXZ
|   |   ---------------------------------------------------
|   |   APFS Volume Disk (Role):   disk1s1 (No specific role)
|   |   Name:                      HighSierra (Case-insensitive)
|   |   Mount Point:               /
|   |   Capacity Consumed:         188514598912 B (188.5 GB)
|   |   FileVault:                 Yes (Unlocked)
|   |
|   +-> Volume disk1s2 XXXXXZZ
|   |   ---------------------------------------------------
|   |   APFS Volume Disk (Role):   disk1s2 (Preboot)
|   |   Name:                      Preboot (Case-insensitive)
|   |   Mount Point:               Not Mounted
|   |   Capacity Consumed:         23961600 B (24.0 MB)
|   |   FileVault:                 No
|   |
|   +-> Volume disk1s3 XXXXXYY
|   |   ---------------------------------------------------
|   |   APFS Volume Disk (Role):   disk1s3 (Recovery)
|   |   Name:                      Recovery (Case-insensitive)
|   |   Mount Point:               Not Mounted
|   |   Capacity Consumed:         518127616 B (518.1 MB)
|   |   FileVault:                 No
|   |
```

```
|   +-> Volume disk1s4 XXXXXYYY
|       ----------------------------------------------------
|       APFS Volume Disk (Role):   disk1s4 (VM)
|       Name:                      VM (Case-insensitive)
|       Mount Point:               /private/var/vm
|       Capacity Consumed:         6442704896 B (6.4 GB)
|       FileVault:                 No
|
+-- Container disk4 XXXXXYYYY
    ====================================================
    APFS Container Reference:     disk4
    Size (Capacity Ceiling):      119824367616 B (119.8 GB)
    Minimum Size:                 143192064 B (143.2 MB)
    Capacity In Use By Volumes:   126492672 B (126.5 MB) (0.1% used)
    Capacity Not Allocated:       119697874944 B (119.7 GB) (99.9% free)
    |
    +-< Physical Store disk3s2 XXXXXYYYYYY
    |   ------------------------------------------------------------
    |   APFS Physical Store Disk:   disk3s2
    |   Size:                        119824371200 B (119.8 GB)
    |
    +-> Volume disk4s1 C4D99580-1FDA-43BF-BB62-B21BF7EE568C
        ----------------------------------------------------
        APFS Volume Disk (Role):   disk4s1 (No specific role)
        Name:                      Passport (Case-insensitive)
        Mount Point:               /Volumes/Passport
        Capacity Consumed:         839680 B (839.7 KB)
        FileVault:                 Yes (Unlocked)
```

Remediation:

Use Disk Utility to erase a user disk and format as APFS (Encrypted).
Note: APFS Encrypted disks will be described as "FileVault" whether they are the boot volume or not in the ap list.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.6 Encrypt Data on End-User Devices Encrypt data on end-user devices containing sensitive data. Example implementations can include: Windows BitLocker®, Apple FileVault®, Linux® dm-crypt.	●	●	●
v8	3.11 Encrypt Sensitive Data at Rest Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.		●	●
v7	13.6 Encrypt the Hard Drive of All Mobile Devices. Utilize approved whole disk encryption software to encrypt the hard drive of all mobile devices.	●	●	●
v7	14.8 Encrypt Sensitive Information at Rest Encrypt all sensitive information at rest using a tool that requires a secondary authentication mechanism not integrated into the operating system, in order to access the information.			●

2.5.1.3 Ensure all user storage CoreStorage volumes are encrypted (Manual)

Profile Applicability:

- Level 1

Description:

Apple introduced CoreStorage with 10.7. It is used as the default for formatting on macOS volumes prior to 10.13.

All HFS and CoreStorage Volumes should be encrypted.

Rationale:

In order to protect user data from loss or tampering, volumes carrying data should be encrypted.

Impact:

While FileVault protects the boot volume, data may be copied to other attached storage and reduce the protection afforded by FileVault. Ensure all user volumes are encrypted to protect data.

Audit:

Terminal Method:
Run the following command to list the CoreStorage Volumes:

```
$ /usr/bin/sudo /usr/sbin/diskutil cs list
```

Ensure all "Logical Volume Family" disks are encrypted

example:

```
$ /usr/bin/sudo /usr/sbin/diskutil cs list

CoreStorage logical volume groups (2 found)
|
+-- Logical Volume Group XXXXX
|   =========================================================
|   Name:         Macintosh HD
|   Status:       Online
|   Size:         250160967680 B (250.2 GB)
|   Free Space:   6516736 B (6.5 MB)
|   |
|   +-< Physical Volume XXXXXY
|   |   ----------------------------------------------------
|   |   Index:    0
|   |   Disk:     disk0s2
|   |   Status:   Online
|   |   Size:     250160967680 B (250.2 GB)
|   |
|   +-> Logical Volume Family XXXXXYY
|       ----------------------------------------------------------
|       Encryption Type:         AES-XTS
|       Encryption Status:       Unlocked
|       Conversion Status:       Complete
|       High Level Queries:      Fully Secure
|       |                        Passphrase Required
|       |                        Accepts New Users
|       |                        Has Visible Users
|       |                        Has Volume Key
|       |
|       +-> Logical Volume XXXXXYYY
|           ---------------------------------------------------
|           Disk:                  disk2
|           Status:                Online
|           Size (Total):          249802129408 B (249.8 GB)
|           Revertible:            Yes (unlock and decryption required)
|           LV Name:               Macintosh HD
|           Volume Name:           Macintosh HD
|           Content Hint:          Apple_HFS
|
```

```
+-- Logical Volume Group XXXXXYYYY
    =========================================================
    Name:         Passport
    Status:       Online
    Size:         119690149888 B (119.7 GB)
    Free Space:   1486848 B (1.5 MB)
    |
    +-< Physical Volume XXXXXYYY
    |   ----------------------------------------------------
    |   Index:    0
    |   Disk:     disk3s2
    |   Status:   Online
    |   Size:     119690149888 B (119.7 GB)
    |
    +-> Logical Volume Family XXXXXYYYYY
        ----------------------------------------------------------
        Encryption Type:         AES-XTS
        Encryption Status:       Unlocked
        Conversion Status:       Complete
        High Level Queries:      Fully Secure
        |                        Passphrase Required
        |                        Accepts New Users
        |                        Has Visible Users
        |                        Has Volume Key
        |
        +-> Logical Volume XXXXXYYYYYY
            ----------------------------------------------------
            Disk:                  disk4
            Status:                Online
            Size (Total):          119336337408 B (119.3 GB)
            Revertible:            No
            LV Name:               Passport
            Volume Name:           Passport
            Content Hint:          Apple_HFS
```

Remediation:

Use Disk Utility to erase a disk and format as macOS Extended (Journaled, Encrypted).

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.9 Encrypt Data on Removable Media Encrypt data on removable media.		●	●
v8	3.11 Encrypt Sensitive Data at Rest Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.		●	●
v7	13.6 Encrypt the Hard Drive of All Mobile Devices. Utilize approved whole disk encryption software to encrypt the hard drive of all mobile devices.	●	●	●
v7	14.8 Encrypt Sensitive Information at Rest Encrypt all sensitive information at rest using a tool that requires a secondary authentication mechanism not integrated into the operating system, in order to access the information.			●

2.5.2 Firewall

macOS has a built-in firewall that has two main configuration aspects. Both the Application Layer Firewall (ALF) and the Packet Filter Firewall (PF) can be used to secure running ports and services on a Mac. The Application Firewall is the one accessible in System Preferences under security. The PF firewall contains many more capabilities than ALF, but also requires a greater understanding of firewall recipes and rule configurations. For standard use cases on a Mac, the PF firewall is not necessary. macOS may expose server services that are reachable remotely, but that is not the primary use case or design. If custom use cases are required, the PF firewall can provide additional security. Macs that are used as mobile desktops do not need to use the PF firewall capabilities unless permanently open ports need to be protected with more granular IP access controls.

Additional information

https://www.muo.com/tag/mac-really-need-firewall/

https://blog.neilsabol.site/post/quickly-easily-adding-pf-packet-filter-firewall-rules-macos-osx/

http://marckerr.com/a-simple-guild-to-the-mac-pf-firewall/

https://blog.scottlowe.org/2013/05/15/using-pf-on-os-x-mountain-lion/

2.5.2.1 Ensure Firewall Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

A firewall is a piece of software that blocks unwanted incoming connections to a system. Apple has posted general documentation about the application firewall:

http://support.apple.com/en-us/HT201642

Rationale:

A firewall minimizes the threat of unauthorized users gaining access to your system while connected to a network or the Internet.

Impact:

The firewall may block legitimate traffic. Applications that are unsigned will require special handling.

Audit:

Graphical Method:
Perform the following steps to ensure the firewall is enabled:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Firewall`
4. Verify that `Firewall` is `On`

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Firewall` set to `Enabled`

Terminal Method:
Run the following command to verify that the firewall is enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.alf')\
  .objectForKey('globalstate'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.security.fire
wall')\
  .objectForKey('EnableFirewall'))
  if ( ( pref1 == 1 ) || ( pref1 == 2 ) || ( pref2 == "true" ) ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to turn the firewall on:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Firewall`
4. Select `Turn On Firewall`

Terminal Method:
Run the following command to enable the firewall:

```
$ /usr/bin/sudo /usr/bin/defaults write /Library/Preferences/com.apple.alf
globalstate -int <value>
```

For the `<value>`, use either `1`, specific services, or `2`, essential services only.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.security.firewall`
2. The key to include is `EnableFirewall`
3. The key must be set to `<true/>`

References:

1. https://support.apple.com/en-ca/guide/security/seca0e83763f/web

Additional Information:

Your organization might want to verify and limit specific applications that allow incoming connectivity.

To verify those applications:

Graphical Method:

Perform the following steps to ensure the system is configured as prescribed:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Firewall Options`
4. Verify that your organizations necessary rules are set

Terminal Method:

Run the following command to verify the what applications are allowing incoming connections:

```
$ /usr/bin/sudo /usr/libexec/ApplicationFirewall/socketfilterfw --listapps
```

The output will show any applications, and their path, and their associated rule.

example:

```
$ /usr/bin/sudo /usr/libexec/ApplicationFirewall/socketfilterfw --listapps
ALF: total number of apps = 3

1 :  /System/Library/CoreServices/RemoteManagement/ARDAgent.app
         ( Allow incoming connections )

2 :  /Applications/Chess.app
         ( Allow incoming connections )

3 :  /Applications/Contacts.app
         ( Block incoming connections )
```

To remove those applications:

Graphical Method:

Perform the following steps to remove unnecessary firewall rules:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Firewall Options`
4. Select unneeded rule(s)
5. Select the – below to delete them

Terminal Method:

Run the following command to remove specific applications:

```
$ /usr/bin/sudo /usr/libexec/ApplicationFirewall/socketfilterfw --remove
</path/application name>

Application at path ( </path/application name> ) removed from firewall
```

The `</path/application name>` is the one to be removed from the previous listing.

example:

```
$ /usr/bin/sudo /usr/libexec/ApplicationFirewall/socketfilterfw --listapps
ALF: total number of apps = 3

1 :  /System/Library/CoreServices/RemoteManagement/ARDAgent.app
         ( Allow incoming connections )

2 :  /Applications/Chess.app
         ( Allow incoming connections )

3 :  /Applications/Contacts.app
         ( Block incoming connections )

$ sudo /usr/libexec/ApplicationFirewall/socketfilterfw --remove
/Applications/Chess.app

Application at path ( /Applications/Chess.app ) removed from firewall
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.5 Implement and Manage a Firewall on End-User Devices Implement and manage a host-based firewall or port-filtering tool on end-user devices, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.	●	●	●
v8	13.1 Centralize Security Event Alerting Centralize security event alerting across enterprise assets for log correlation and analysis. Best practice implementation requires the use of a SIEM, which includes vendor-defined event correlation alerts. A log analytics platform configured with security-relevant correlation alerts also satisfies this Safeguard.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.	●	●	●
v7	9.5 Implement Application Firewalls Place application firewalls in front of any critical servers to verify and validate the traffic going to the server. Any unauthorized traffic should be blocked and logged.			●

2.5.2.2 Ensure Firewall Stealth Mode Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

While in Stealth mode, the computer will not respond to unsolicited probes, dropping that traffic.

http://support.apple.com/en-us/HT201642

Rationale:

Stealth mode on the firewall minimizes the threat of system discovery tools while connected to a network or the Internet.

Impact:

Traditional network discovery tools like ping will not succeed. Other network tools that measure activity and approved applications will work as expected.

This control aligns with the primary macOS use case of a laptop that is often connected to untrusted networks where host segregation may be non-existent. In that use case, hiding from the other inmates is likely more than desirable. In use cases where use is only on trusted LANs with static IP addresses, stealth mode may not be desirable.

Audit:

Graphical Method:
Perform the following steps to verify the firewall has stealth mode enabled:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Firewall Options...`
4. Verify that `Enable stealth mode` is enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Stealth mode` set to `Enabled`

Terminal Method:
Run the following command to verify that stealth mode is enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.alf')\
  .objectForKey('stealthenabled'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.security.fire
wall')\
  .objectForKey('EnableStealthMode'))
  if ( ( pref1 == 1 ) || ( pref2 == "true" ) ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable stealth mode:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Firewall Options...`
4. Set `Enable stealth mode` to enabled

Terminal Method:
Run the following command to enable stealth mode:

```
$ /usr/bin/sudo /usr/libexec/ApplicationFirewall/socketfilterfw --
setstealthmode on

Stealth mode enabled
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.security.firewall`
2. The key to include is `EnableStealthMode`
3. The key must be set to `<true/>`

Note: This key must be set in the same configuration profile with `EnableFirewall` set to `<true/>`. If it is set in its own configuration profile, it will fail.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.5 Implement and Manage a Firewall on End-User Devices Implement and manage a host-based firewall or port-filtering tool on end-user devices, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.	●	●	●

2.5.3 Ensure Location Services Is Enabled (Automated)

Profile Applicability:

- Level 2

Description:

macOS uses location information gathered through local Wi-Fi networks to enable applications to supply relevant information to users. With the operating system verifying the location, users do not need to change the time or the time zone. The computer will change them based on the user's location. They do not need to specify their location for weather or travel times, and they will receive alerts on travel times to meetings and appointment where location information is supplied.

Location Services simplify some processes with mobile computers, such as asset management and time or log management.

There are some use cases where it is important that the computer not be able to report its exact location. While the general use case is to enable Location Services, it should not be allowed if the physical location of the computer and the user should not be public knowledge.

Rationale:

Location Services are helpful in most use cases and can simplify log and time management where computers change time zones.

Audit:

Graphical Method:
Perform the following steps to ensure that Location Services is enabled:

1. Open `System Preferences
2. Select `Security & Privacy`
3. Select `Privacy`
4. Verify `Location Services` is enabled

Terminal Method:
Run the following command to verify that Location Services are enabled:

```
$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep -c com.apple.locationd

1

$ /usr/bin/sudo -u _locationd /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.locationd').objectForKey(
'LocationServicesEnabled').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable Location Services:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Privacy`
4. Set `Enable Location Services` to enabled

Terminal Method:
Run the following command to enable Location Services:

```
$ /usr/bin/sudo /bin/launchctl load -w
/System/Library/LaunchDaemons/com.apple.locationd.plist
```

If the `com.apple.locationd.plist` outputs `0`, run the following command to also ensure Location Services is running:

```
$ /usr/bin/sudo /usr/bin/defaults write
/var/db/locationd/Library/Preferences/ByHost/com.apple.locationd
LocationServicesEnabled -bool false

$ /usr/bin/sudo /bin/launchctl kickstart -k system/com.apple.locationd
```

Note: In some use cases, organizations may not want Location Services running. To disable Location Services, System Integrity Protection must be disabled.

References:

1. https://support.apple.com/en-us/HT204690

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

2.5.4 Audit Location Services Access (Manual)

Profile Applicability:

- Level 2

Description:

macOS uses location information gathered through local Wi-Fi networks to enable applications to supply relevant information to users. While Location Services may be very useful, it may not be desirable to allow all applications that can use Location Services to use your location for Internet queries in order to provide tailored content based on your current location.

Ensure applications that can use Location Services are authorized and provide that information where the application interacts with external systems. Apple offers feedback within System Preferences and may be enabled to supply information on the menu bar when Location Services are used.

Safari can deny access from websites or prompt for access.

Applications that support Location Services can be individually controlled in the Privacy tab in Security & Privacy under System Preferences.

Access should be evaluated to ensure that privacy controls are as expected.

Rationale:

Privacy controls should be monitored for appropriate settings.

Impact:

Many macOS features rely on Location Services for tailored information. Users expect their time zone and weather to be relevant to where they are without manual intervention. Find my Mac needs to know where your Mac is actually located. Where possible, the tolerance between location privacy and convenience may be best left to the user when the location itself is not sensitive. If facility locations are not public, location information should be tightly controlled.

Audit:

Graphical Method:
Perform the following steps to verify what applications are enabled for Location Services:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Privacy`
4. Select `Location Services`
5. Verify what applications are set for Location Service information

Perform the following to verify what websites are enabled to ask for access to Location Services:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Websites`
5. Select `Location`
6. Verify that `When visiting other websites` is set to `Ask or Deny`

Terminal Method:
Run the following command to evaluate the applications that are enabled to use Location Services:

```
$ /usr/bin/sudo /usr/bin/defaults read /var/db/locationd/clients.plist
```

Ensure that all applications listed have been authorized to access location information.

Remediation:

Graphical Method:
Perform the following to disable unnecessary applications from accessing Location Services:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Privacy`
4. Select `Location Services`
5. Set any applications that are not approved for access to Location Service information to disabled

Perform the following to set websites to ask for permission to access Location Services:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Websites`
5. Select `Location`
6. Set When visiting other websites to Ask or Deny

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.3 Address Unauthorized Software Ensure that unauthorized software is either removed from use on enterprise assets or receives a documented exception. Review monthly, or more frequently.	●	●	●
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	2.6 Address unapproved software Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

2.5.5 Ensure Sending Diagnostic and Usage Data to Apple Is Disabled (Automated)

Profile Applicability:

- Level 2

Description:

Apple provides a mechanism to send diagnostic and analytics data back to Apple to help them improve the platform. Information sent to Apple may contain internal organizational information that should be controlled and not available for processing by Apple. Turn off all Analytics and Improvements sharing.

Share Mac Analytics (Share with App Developers dependent on Mac Analytic sharing)

- Includes diagnostics, usage and location data

Share iCloud Analytics

- Includes iCloud data and usage information

Rationale:

Organizations should have knowledge of what is shared with the vendor and that this setting automatically forwards information to Apple.

Audit:

Graphical Method:
Perform the following steps to verify that diagnostic data is not being send to Apple:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Privacy`
4. Select `Analytics & Improvements`
5. Verify that `Share Mac Analytics` is disabled
6. Verify that `Share with App Developers` is disabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Allow Diagnostic Submission` set to `False`

Terminal Method:
Run the following command to verify that sending diagnostic and usage data to Apple is disabled:

```
$ /usr/bin/sudo /usr/bin/defaults read /Library/Application\
Support/CrashReporter/DiagnosticMessagesHistory.plist AutoSubmit

0

$ /usr/bin/sudo /usr/bin/defaults read /Library/Application\
Support/CrashReporter/DiagnosticMessagesHistory.plist ThirdPartyDataSubmit

0

$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Preferences/com.apple.assistant.support "Siri Data
Sharing Opt-In Status"

2
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Preferences/com.apple.assistant.support "Siri Data
Sharing Opt-In Status"

2

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read
/Users/seconduser/Library/Preferences/com.apple.assistant.support "Siri Data
Sharing Opt-In Status"

1
```

or
Run the following command to verify that a profile is installed that disables sending diagnostic and usage data to Apple:

```
/usr/bin/osascript -l JavaScript << EOS
function run() {
let pref1 =
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.SubmitDiagInfo')\
.objectForKey('AutoSubmit').js
let pref2 =
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('allowDiagnosticSubmission').js
let pref3 =
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('Siri Data Sharing Opt-In Status').js

if ( pref1 == false && pref2 == false && pref3 == 2){
    return("true")
} else {
    return("false")
}
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to disable diagnostic data being sent to Apple:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Privacy`
4. Select `Analytics & Improvements`
5. Set `Share Mac Analytics` to disabled
6. Set `Share with App Developers` to disabled

Terminal Method:
Run the following commands to disable the sending of diagnostic data to Apple:

```
$ /usr/bin/sudo /usr/bin/defaults write /Library/Application\
Support/CrashReporter/DiagnosticMessagesHistory.plist AutoSubmit -bool false

/usr/bin/sudo /usr/bin/defaults write /Library/Application\
Support/CrashReporter/DiagnosticMessagesHistory.plist ThirdPartyDataSubmit -
bool false

$ /usr/bin/sudo /bin/chmod 644 /Library/Application\
Support/CrashReporter/DiagnosticMessagesHistory.plist

$ /usr/bin/sudo /usr/sbin/chgrp admin /Library/Application\
Support/CrashReporter/DiagnosticMessagesHistory.plist

$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Preferences/com.apple.assistant.support "Siri Data
Sharing Opt-In Status" -int 2
```

example:

```
$ /usr/bin/sudo -u seconduser /usr/bin/defaults write
/Users/seconduser/Library/Preferences/com.apple.assistant.support "Siri Data
Sharing Opt-In Status" -int 2
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowDiagnosticSubmission`
3. The key must be set to `<false/>`
4. The key to also include is `Siri Data Sharing Opt-In Status`
5. The key must be set to `<integer>2<integer/>`
6. There must also be a second PayloadType string of `com.apple.SubmitDiagInfo`
7. The key to include is `AutoSubmit`
8. The key must be set to `<false/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://support.apple.com/en-ca/guide/mac-help/mh27990/mac

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.5.6 Ensure Limit Ad Tracking Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Apple provides a framework that allows advertisers to target Apple users and end-users with advertisements. While many people prefer to see advertising that is relevant to them and their interests, the detailed information that is collected, correlated, and available to advertisers in repositories via data mining is often disconcerting. This information is valuable to both advertisers and attackers, and has been used with other metadata to reveal users' identities.

Organizations should manage advertising settings on computers rather than allow users to configure the settings.

Apple Information

Ad tracking should be limited on 10.15 and prior.

Rationale:

Organizations should manage user privacy settings on managed devices to align with organizational policies and user data protection requirements.

Impact:

Uses will see generic advertising rather than targeted advertising. Apple warns that this will reduce the number of relevant ads.

Audit:

Graphical Method:
Perform the following steps to verify that limited ad tracking is set:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Privacy`
4. Select `Apple Advertising`
5. Verify that `Personalized Ads` is not enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `allowApplePersonalizedAdvertising` set to `0`

Terminal Method:
For each user, run the following command to verify that ad tracking is limited:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Preferences/com.apple.AdLib.plist
allowApplePersonalizedAdvertising

0
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Preferences/com.apple.AdLib.plist
allowApplePersonalizedAdvertising

0

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read
/Users/seconduser/Library/Preferences/com.apple.AdLib.plist
allowApplePersonalizedAdvertising

1
```

In this example, firstuser is compliant and seconduser is not.
or
Run the following command to verify that a profile is installed that enables Limit Ad Tracking:

```
$ sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('allowApplePersonalizedAdvertising').js
EOS

false
```

Remediation:

Graphical Method:
Perform the following steps to set limited ad tracking:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Privacy`
4. Select `Apple Advertising`
5. Set `Personalized Ads` to disabled

Terminal Method:
For each needed user, run the following command to enable limited ad tracking:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Preferences/com.apple.Adlib.plist
allowApplePersonalizedAdvertising -bool false
```

example:

```
$ /usr/bin/sudo -u seconduser /usr/bin/defaults write
/Users/seconduser/Library/Preferences/com.apple.Adlib.plist
allowApplePersonalizedAdvertising -bool false
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowApplePersonalizedAdvertising`
3. The key must be set to `<false/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.5.7 Ensure Gatekeeper Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Gatekeeper is Apple's application that utilizes allowlisting to restrict downloaded applications from launching. It functions as a control to limit applications from unverified sources from running without authorization.

Rationale:

Disallowing unsigned software will reduce the risk of unauthorized or malicious applications from running on the system.

Audit:

Graphical Method:
Perform the following steps to ensure that Gatekeeper is enabled:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `General`
4. Verify that 'Allow apps downloaded from:' is set to 'App Store and identified developers'

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Policies` set to `Enable`
4. Verify that an installed profile has `Identified Developers` set to `Allow`

Terminal Method:
Run the following command to verify that Gatekeeper is enabled:

```
$ /usr/bin/sudo /usr/sbin/spctl --status

assessments enabled
```

Remediation:

Graphical Method:
Perform the following steps to enable Gatekeeper:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `General`
4. Set 'Allow apps downloaded from:' to 'App Store and identified developers'

Terminal Method:
Run the following command to enable Gatekeeper to allow applications from App Store and identified developers:

```
$ /usr/bin/sudo /usr/sbin/spctl --master-enable
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.systempolicy.control`
2. The key to include is `AllowIdentifiedDevelopers`
3. The key must be set to `<true/>`
4. The key to also include is `EnableAssessment`
5. The key must be set to `<true/>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	10.1 Deploy and Maintain Anti-Malware Software Deploy and maintain anti-malware software on all enterprise assets.	●	●	●
v8	10.2 Configure Automatic Anti-Malware Signature Updates Configure automatic updates for anti-malware signature files on all enterprise assets.	●	●	●
v8	10.5 Enable Anti-Exploitation Features Enable anti-exploitation features on enterprise assets and software, where possible, such as Microsoft® Data Execution Prevention (DEP), Windows® Defender Exploit Guard (WDEG), or Apple® System Integrity Protection (SIP) and Gatekeeper™.		●	●
v7	8.2 Ensure Anti-Malware Software and Signatures are Updated Ensure that the organization's anti-malware software updates its scanning engine and signature database on a regular basis.	●	●	●
v7	8.4 Configure Anti-Malware Scanning of Removable Devices Configure devices so that they automatically conduct an anti-malware scan of removable media when inserted or connected.	●	●	●

2.5.8 Ensure a Custom Message for the Login Screen Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

An access warning informs the user that the system is reserved for authorized use only, and that the use of the system may be monitored.

Rationale:

An access warning may reduce a casual attacker's tendency to target the system. Access warnings may also aid in the prosecution of an attacker by evincing the attacker's knowledge of the system's private status, acceptable use policy, and authorization requirements.

Impact:

If users are not informed of their responsibilities, unapproved activities may occur. Users that are not approved for access may take the lack of a warning banner as implied consent to access.

Audit:

*Graphical Method:**
Perform the following steps to ensure that the a login banner is configured:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Verify `Show a message when the screen is locked` is enabled
4. Select `Set Lock Message`
5. Verify that the message displayed is configured to your organization's required text

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Banner Text` is configured to your organization's required text

Terminal Method:
Run the following command to verify that a custom message on the login screen is configured:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.loginwindow')\
.objectForKey('LoginwindowText').js
EOS
```

The output should be a message that is configured to your organization's required text.
example:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.loginwindow')\
.objectForKey('LoginwindowText').js
EOS

Center for Internet Security Test Message
```

Remediation:

Graphical Method:
Perform the following steps to enable a login banner set to your organization's required text:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `Show a message when the screen is locked`
4. Select `Set Lock Message`
5. Insert text in the `Set a message to appear on the lock screen` that matches your organization's required text

Terminal Method:
Run the following command to enable a custom login screen message:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.loginwindow LoginwindowText "<custom message>"
```

example:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.loginwindow LoginwindowText "Center for
Internet Security Test Message"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.loginwindow`
2. The key to include is `LoginwindowText`
3. The key must be set to `<string><Your organization's required text></string>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

2.5.9 Ensure an Administrator Password Is Required to Access System-Wide Preferences (Automated)

Profile Applicability:

- Level 1

Description:

System Preferences controls system and user settings on a macOS Computer. System Preferences allows the user to tailor their experience on the computer as well as allowing the System Administrator to configure global security settings. Some of the settings should only be altered by the person responsible for the computer.

Rationale:

By requiring a password to unlock system-wide System Preferences, the risk is mitigated of a user changing configurations that affect the entire system and requires an admin user to re-authenticate to make changes

Impact:

Users will need to enter their password to unlock some additional preference panes that are unlocked by default like Network, Startup and Printers & Scanners.

Audit:

Graphical Method:
Perform the following steps to verify that an administrator password is required to access system-wide preferences:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `General`
4. Select `Advanced...`
5. Verify that `Require an administrator password to access system-wide preferences` is enabled

Terminal Method:
Run the following command to verify that accessing system-wide preferences requires an administrator password:

```
$ /usr/bin/sudo /usr/bin/security authorizationdb read system.preferences 2>
/dev/null | grep -A1 shared | grep false

<false/>
```

Remediation:

Graphical Method:
Perform the following steps to verify that an administrator password is required to access system-wide preferences:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `General`
4. Select `Advanced...`
5. Set `Require an administrator password to access system-wide preferences` to enabled

Terminal Method:
The authorizationdb settings cannot be written to directly, so the plist must be exported out to temporary file. Changes can be made to the temporary plist, then imported back into the authorizationdb settings.
Run the following commands to enable that an administrator password is required to access system-wide preferences:

```
$ /usr/bin/sudo /usr/bin/security authorizationdb read system.preferences >
/tmp/system.preferences.plist

YES (0)

$ /usr/bin/sudo /usr/bin/defaults write /tmp/system.preferences.plist shared
-bool false

$ /usr/bin/sudo /usr/bin/security authorizationdb write system.preferences <
/tmp/system.preferences.plist

YES (0)
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

2.5.10 Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Sleep and screen saver modes are low power modes that reduce electrical consumption while the system is not in use.

Rationale:

Prompting for a password when waking from sleep or screen saver mode mitigates the threat of an unauthorized person gaining access to a system in the user's absence.

Impact:

Without a screenlock in place anyone with physical access to the computer would be logged in and able to use the active user's session.

Audit:

Graphical Method:
Perform the following steps to verify that a password is required to wake from sleep or screen saver:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `General`
4. Verify that `Require password after or screensaver begins` is set with `immediately` or `5 seconds`

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Ask For Password` set to `True`
4. Verify that the same installed profile has `Ask For Password Delay` set to `<0,5>`

Terminal Method:
Run the following command to verify that a profile is installed that requires a password to wake the computer from sleep or from the screen saver:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.screensaver')
\
  .objectForKey('askForPassword'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.screensaver')
\
  .objectForKey('askForPasswordDelay'))
  if ( pref1 == 1 && pref2 <= 5 ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable a password for unlock after a screen saver begins or after sleep:

1. Open `System Preferences`
2. Select `Security & Privacy`
3. Select `General`
4. Set `Require password after or screensaver begins` with a time of `immediately` or `5 seconds`

Terminal Method:
Run the following command to require a password to unlock the computer after the screen saver engages or the computer sleeps:

```
$ /usr/bin/sudo /usr/sbin/sysadminctl -screenLock immediate -password
<administrator password>
```

or

```
$ /usr/bin/sudo /usr/sbin/sysadminctl -screenLock 5 seconds -password
<administrator password>
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.screensaver`
2. The key to include is `askForPassword`
3. The key must be set to `<true/>`
4. The key to also include is `askForPasswordDelay`
5. The key must be set to `<integer><0,5></integer>`

References:

1. https://blog.kolide.com/screensaver-security-on-macos-10-13-is-broken-a385726e2ae2
2. https://github.com/rtrouton/profiles/blob/master/SetDefaultScreensaver/SetDefaultScreensaver.mobileconfig

Additional Information:

This only protects the system when the screen saver is running.

Note: The command line check in previous versions of the Benchmark does not work as expected here. The use of a profile is recommended for both implementation and auditing on a 10.13 system or later.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.7 Manage Default Accounts on Enterprise Assets and Software Manage default accounts on enterprise assets and software, such as root, administrator, and other pre-configured vendor accounts. Example implementations can include: disabling default accounts or making them unusable.	●	●	●
v7	4.2 Change Default Passwords Before deploying any new asset, change all default passwords to have values consistent with administrative level accounts.	●	●	●

2.6 Apple ID

Apple is a hardware manufacturer that develops operating systems for the hardware it creates. Apple is also a cloud service provider and those services include applications, music, books, television, cloud storage, etc. Apple simplifies the process to ensure that all user devices are entitled to content where the user has purchased access, or is part of an Apple basic level of entitlement (BLE) for purchasing an Apple device. The use of an Apple ID allows for a consistent access and experience across all Apple devices. An Apple ID functions as Single Sign-On access to all Apple provided services. It is critical that each user's account is protected appropriately so that unauthorized access risk is heavily mitigated.

https://support.apple.com/en-us/HT203993

https://en.wikipedia.org/wiki/Apple_ID

https://www.lifewire.com/what-is-an-apple-id-1994330

https://support.apple.com/en-us/HT201303

2.6.1 iCloud

iCloud is Apple's service for synchronizing, storing, and backing up data from Apple applications in both macOS and iOS.

macOS controls for iCloud are part of the Apple ID settings in macOS. The configuration options in macOS resemble the options in iOS.

Apple's iCloud is a consumer-oriented service that allows a user to store data as well as find, control, and back up devices that are associated with their Apple ID (Apple account). The use of iCloud on Enterprise devices should align with the acceptable use policy for devices that are managed, as well as confidentiality requirements for data handled by the user. If iCloud is allowed, the data that is copied to Apple servers will likely be duplicated on both personal as well as Enterprise devices.

For many users, the Enterprise email system may replace many of the available features in iCloud. Calendars, notes, and contacts can sync to the official Enterprise repository and be available through multiple devices if using either an Exchange or Google environment email.

Depending on workplace requirements, it may not be appropriate to intermingle Enterprise and personal bookmarks, photos, and documents. Since the service allows every device associated with the user's ID to synchronize and have access to the cloud storage, the concern is not just about having sensitive data on Apple's servers, but also having that same data on the phone of the teenage son or daughter of an employee. The use of family sharing options can reduce the risk.

Apple's iCloud is just one of many cloud-based solutions being used for data synchronization across multiple platforms, and it should be controlled consistently with other cloud services in your environment. Work with your employees and configure the access to best enable data protection for your mission.

2.6.1.1 Audit iCloud Keychain (Manual)

Profile Applicability:

- Level 2

Description:

The iCloud keychain is Apple's password manager that works with macOS and iOS. The capability allows users to store passwords in either iOS or macOS for use in Safari on both platforms and other iOS-integrated applications. The most pervasive use is driven by iOS use rather than macOS. The passwords stored in a macOS keychain on an Enterprise-managed computer could be stored in Apple's cloud and then be available on a personal computer using the same account. The stored passwords could be for organizational as well as for personal accounts.

If passwords are no longer being used as organizational tokens, they are not in scope for iCloud keychain storage.

Rationale:

Ensure that the iCloud keychain is used consistently with organizational requirements.

Audit:

Graphical Method:
Perform the following steps to verify the iCloud keychain sync service:

1. Open `System Preferences`
2. Select `Apple ID`
3. Select `iCloud`
4. Verify that `Keychain` is set to your organization's requirements

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Disallow iCloud Keychain Sync` is set to your organization's requirements

Terminal Method:
For each user, run this command to verify the iCloud keychain sync services:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Preferences/MobileMeAccounts | grep -B 1
KEYCHAIN_SYNC

Enabled = <0,1>;
Name = "KEYCHAIN_SYNC";
```

The output will be either a `0`, disabled, or `1`, enabled. Verify if the setting meets your organizations requirements
example:

```
$ /usr/bin/sudo -u seconduser /usr/bin/defaults read
/Users/seconduser/Library/Preferences/MobileMeAccounts | grep -B 1
KEYCHAIN_SYNC
                    Enabled = 0;
                    Name = "KEYCHAIN_SYNC";
```

or
Run the following command to verify that a profile is installed that sets iCloud Keychain sync to your organizations settings:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('allowCloudKeychainSync').js
EOS
```

If the output is `false`, iCloud Keychain Sync is disbaled. If the output is `true` iCloud Keychain sync is enabled.

Remediation:

Graphical Method:
Perform the following steps to set iCloud keychain sync based on your organization's requirements:

1. Open `System Preferences`
2. Select `Apple ID`
3. Select `iCloud`
4. Set `Keychain` to meet your organization's requirements

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowCloudKeychainSync`
3. The key should be set `<true/>`, to allow iCloud keychain syncing, or `<false/>`, to disable it, based on your organization's requirements

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	15.3 Classify Service Providers Classify service providers. Classification consideration may include one or more characteristics, such as data sensitivity, data volume, availability requirements, applicable regulations, inherent risk, and mitigated risk. Update and review classifications annually, or when significant enterprise changes occur that could impact this Safeguard.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.6.1.2 Audit iCloud Drive (Manual)

Profile Applicability:

- Level 2

Description:

iCloud Drive is Apple's storage solution for applications on both macOS and iOS to use the same files that are resident in Apple's cloud storage. The iCloud Drive folder is available much like Dropbox, Microsoft OneDrive, or Google Drive.

One of the concerns in public cloud storage is that proprietary data may be inappropriately stored in an end user's personal repository. Organizations that need specific controls on information should ensure that this service is turned off or the user knows what information must be stored on services that are approved for storage of controlled information.

Rationale:

Organizations should review third party storage solutions pertaining to existing data confidentiality and integrity requirements.

Impact:

Users will not be able to use continuity on macOS to resume the use of newly composed but unsaved files.

Audit:

Graphical Method:
Perform the following steps to verify if iCloud Drive is enabled:

1. Open `System Preferences`
2. Select `Apple ID`
3. Select `iCloud`
4. Verify that `iCloud Drive` is set within your organization's requirements

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Disallow iCloud Drive` is set to your organization's requirements

Terminal Method:
Run the following command to verify that iCloud Drive is set to your organizations specifications:

```
$ /usr/bin/sudosudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Preferences/MobileMeAccounts | /usr/bin/grep -B 1
MOBILE_DOCUMENTS
```

The output will include `Enabled =` and iCloud Drive is either enabled, `1`, or disabled, `0`. Verify that the service is set to your organization's requirements.
example:

```
$ /usr/bin/sudosudo -u seconduser /usr/bin/defaults read
/Users/seconduser/Library/Preferences/MobileMeAccounts | /usr/bin/grep -B 1
MOBILE_DOCUMENTS

Enabled = 0;
Name = "MOBILE_DOCUMENTS";
```

or
Run the following command to verify that a profile is installed that sets iCloud Drive sync to your organizations settings:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('allowCloudDocumentSync').js
EOS
```

If the output is `false`, iCloud Drive Sync is disbaled. If the output is `true` iCloud Drive sync is enabled.

Remediation:

Graphical Method:
Perform the following steps to set iCloud Drive to your organization's requirements:

1. Open `System Preferences`
2. Select `Apple ID`
3. Select `iCloud`
4. Set `iCloud Drive` to for your organization's requirements

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowCloudDocumentSync`
3. The key should be set `<true/>`, to allow iCloud Drive, or `<false/>`, to disable it, based on your organization's requirements

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://developer.apple.com/documentation/devicemanagement/restrictions

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	15.3 Classify Service Providers Classify service providers. Classification consideration may include one or more characteristics, such as data sensitivity, data volume, availability requirements, applicable regulations, inherent risk, and mitigated risk. Update and review classifications annually, or when significant enterprise changes occur that could impact this Safeguard.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

2.6.1.3 Ensure iCloud Drive Document and Desktop Sync Is Disabled (Automated)

Profile Applicability:

- Level 2

Description:

With macOS 10.12, Apple introduced the capability to have a user's Desktop and Documents folders automatically synchronize to the user's iCloud Drive, provided they have enough room purchased through Apple on their iCloud Drive. This capability mirrors what Microsoft is doing with the use of OneDrive and Office 365. There are concerns with using this capability.

The storage space that Apple provides for free is used by users with iCloud mail, all of a user's Photo Library created with the ever larger Multi-Pixel iPhone cameras, and all iOS Backups. Adding a synchronization capability for users who have files going back a decade or more, storage may be tight using the free 5GB provided without purchasing much larger storage capacity from Apple. Users with multiple computers running 10.12 and above with unique content on each will have issues as well.

Enterprise users may not be allowed to store Enterprise information in a third-party public cloud. In previous implementations, such as iCloud Drive or DropBox, the user selected what files were synchronized even if there were no other controls. The new feature synchronizes all files in a folder widely used to put working files.

The automatic synchronization of all files in a user's Desktop and Documents folders should be disabled.

https://derflounder.wordpress.com/2016/09/23/icloud-desktop-and-documents-in-macos-sierra-the-good-the-bad-and-the-ugly/

Rationale:

Automated Document synchronization should be planned and controlled to approved storage.

Impact:

Users will not be able to use iCloud for the automatic sync of the Desktop and Documents folders.

Audit:

Graphical Method:
Perform the following steps to verify if Desktop and Documents in iCloud Drive is enabled:

1. Open `System Preferences
2. Select `Apple ID
3. Select `iCloud
4. Verify that `iCloud Drive` is not set
5. If `iCloud Drive` is set, select `Options...`
6. Verify that `Desktop & Documents Folders` is not enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Disallow iCloud Desktop & Documents Sync` is set to `True`

Terminal Method:
For each user, run the following command to verify that the Documents and Desktop folders are not syncing to iCloud:

```
$ /usr/bin/sudo -u <username> /bin/ls -l /Users/<username>/Library/Mobile\
Documents/com~apple~CloudDocs/Documents/ | /usr/bin/grep total

$ /usr/bin/sudo -u <username> /bin/ls -l /Users/<username>/Library/Mobile\
Documents/com~apple~CloudDocs/Desktop/ | /usr/bin/grep total
```

example:

```
$ /usr/bin/sudo -u seconduser /bin/ls -l /Users/seconduser/Library/Mobile\
Documents/com~apple~CloudDocs/Documents/ | /usr/bin/grep total

$ /usr/bin/sudo -u seconduser /bin/ls -l /Users/seconduser/Library/Mobile\
Documents/com~apple~CloudDocs/Desktop/ | /usr/bin/grep total

total 8
```

In the above example, there is an output so the machine is not compliant.
or
Run the following command to verify that a profile is installed that disables iCloud Document and Desktop Sync:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('allowCloudDesktopAndDocuments').js
EOS

false
```

Remediation:

Graphical Method:
Perform the following steps to disable iCloud Desktop and Document syncing:

1. Open `System Preferences
2. Select `Apple ID
3. Select `iCloud
4. Select `Options...` next to `iCloud Drive`
5. Set `Desktop & Documents Folders` to disabled

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowCloudDesktopAndDocuments`
3. The key must be set to `<false/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://developer.apple.com/documentation/devicemanagement/restrictions

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	15.3 Classify Service Providers Classify service providers. Classification consideration may include one or more characteristics, such as data sensitivity, data volume, availability requirements, applicable regulations, inherent risk, and mitigated risk. Update and review classifications annually, or when significant enterprise changes occur that could impact this Safeguard.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

2.6.2 Audit App Store Password Settings (Manual)

Profile Applicability:

- Level 2

Description:

With OS X 10.11, Apple added settings for password storage for the App Store in macOS. These settings parallel the settings in iOS. As with iOS, the choices are a requirement to provide a password after every purchase or to have a 15-minute grace period, and whether or not to require a password for free purchases. The response to this setting is stored in a cookie and processed by iCloud.

There is plenty of risk information on the wisdom of this setting for parents with children buying games on iPhones and iPads. The most relevant information here is the likelihood that users who are not authorized to download software may have physical access to an unlocked computer where someone who is authorized recently made a purchase. If that is a concern, a password should be required at all times for App Store access in the Password Settings controls.

Rationale:

Audit:

Graphical Method:
Perform the following steps to verify that App Store Passwords are set to your organization's requirements:

1. Open `System Preferences
2. Select `Apple ID
3. Select `Media & Purchases
4. Verify that `Free Downloads` is set to your organization's requirements
5. Verify that `Purchases and In-App Purchases` is set to your organization's requirements

Remediation:

Graphical Method:
Perform the following steps to set App Store Passwords to your organization's requirements:

1. Open `System Preferences`
2. Select `Apple ID`
3. Select `Media & Purchases`
4. Select the setting for `Free Downloads` that are withing your organization's requirements
5. Select the setting for `Purchases and In-App Purchases` that are within your organization's requirements

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

2.7 Time Machine

One of the most important IT Operational concerns is to ensure that information is protected against loss or tampering. The purpose of the IT devices is to process the data, after all. At one time the cost of IT equipment and the volume of the data might make protection of the equipment itself more important. At this point, the vast size of data archives and the lower cost of end-user equipment makes data protection central to operational planning. Backup strategies are generally focused on ensuring that there are multiple copies of relevant versions of user files. The plan is that no single hardware or software loss or failure will result in major data loss.

Apple does not provide a native remote logging capability that encrypts data in transit (DIT). If no third party tool or agent is installed on organizational-manned Macs, it is even more important to ensure that backup processes are implemented with log backups as part of the architecture.

In recent years the criticality of information backup, protection of data, and data backups has become even more important with the rise of cybercriminals that not only commit denial-of-service attacks using ransomware to encrypt your working data to make it inaccessible, but also encrypt the backups if reachable. Newer threats include blackmail to compromise data confidentiality. A comprehensive plan to protect data from compromise is even more vital with current threats. The Time Machine controls are only recommended best practices to assist in ease of frequent backups and the encryption of backup volumes.

Apple introduced Time Machine in 2007 as a simple-to-use, built-in mechanism for users to ensure that their machine was backed up, and if there was a mistake or loss, that information could be easily recovered. There are other solutions to ensure information is protected, including several Enterprise solutions and simple drive or directory cloning.

The controls in this section are specifically about Time Machine. The general ideas are applicable to any data backup solution. These controls are only pertinent to organizations already using Time Machine as part of their backup solutions to ensure the included Apple backup solution is being used effectively. We are not endorsing that Time Machine should be used exclusively or as part of the Enterprise backup solution. The controls first check that Time Machine is actually enabled.

To enable Time Machine, follow the instructions here: https://support.apple.com/en-us/HT201250

For more details on Time Machine:

- https://eclecticlight.co/tag/time-machine/
- https://www.pcmag.com/how-to/how-to-back-up-your-mac-with-time-machine

2.7.1 Ensure Backup Automatically is Enabled If Time Machine Is Enabled (Automated)

Profile Applicability:

- Level 2

Description:

Backup solutions are only effective if the backups run on a regular basis. The time to check for backups is before the hard drive fails or the computer goes missing. In order to simplify the user experience so that backups are more likely to occur, Time Machine should be on and set to Back Up Automatically whenever the target volume is available.

Operational staff should ensure that backups complete on a regular basis and the backups are tested to ensure that file restoration from backup is possible when needed.

Backup dates are available even when the target volume is not available in the Time Machine plist.

SnapshotDates = (

"2012-08-20 12:10:22 +0000",

"2013-02-03 23:43:22 +0000",

"2014-02-19 21:37:21 +0000",

"2015-02-22 13:07:25 +0000",

"2016-08-20 14:07:14 +0000"

When the backup volume is connected to the computer more extensive information is available through tmutil. See man tmutil

Rationale:

Backups should automatically run whenever the backup drive is available.

Impact:

The backup will run periodically in the background and could have user impact while running.

Audit:

Graphical Method:
Perform the following steps to ensure that automatic backups are set if Time Machine is enabled:

1. Open `System Preferences`
2. Select `Time Machine`
3. Verify that `Back Up Automatically` is enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `AutoBackup=1`

Terminal Method:
Run the following command to verify that Time Machine is set to automatically backup if Time Machine is enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.TimeMachine')
\
  .objectForKey('AutoBackup'))
  if ( pref1  == null ) {
    return("Preference Not Set")
  }  else if ( pref1 == 1 ) {
    return("true")
  }  else {
    return("false")
  }
}
EOS
```

The output should either be `Preference Not Set` or `true`. If it is `false`, then the computer is not in compliance
Run the following command to check the snapshot dates to verify that the dates meet your organization's approved backup frequency:

```
$ /usr/bin/sudo /usr/bin/defaults read
/Library/Preferences/com.apple.TimeMachine.plist Destinations
```

The output will contain all the Time Machine backups in the format `"YYYY-MM-DD HH:MM:SS +0000"`
example:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.TimeMachine')
\
  .objectForKey('AutoBackup'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.TimeMachine')
\
  .objectForKey('LastDestinationID'))
  if ( pref2  == null ) {
    return("Preference Not Set")
  }  else if ( pref1 == 1 ) {
    return("true")
  }  else {
    return("false")
  }
}
EOS

true

$ /usr/bin/sudo /usr/bin/defaults read read
/Library/Preferences/com.apple.TimeMachine.plist Destinations

(
        {
        BackupAlias = {length = 348, bytes = 0x00000000 015c0002 00011442
61636b75 ... 20564d00 ffff0000 };
        BytesAvailable = 4855873536;
        BytesUsed = 5125054464;
        ConsistencyScanDate = "2022-09-22 18:21:01 +0000";
        DestinationID = "A64EA502-30DD-480C-9F7B-4F3EEDD0D186";
        DestinationUUIDs =         (
            "0D946E5D-68ED-4F63-BCBD-CE7FC94F47C0"
        );
        FilesystemTypeName = apfs;
        HealthCheckDecision = 0;
        InheritanceDecision = 0;
        LastKnownEncryptionState = Encrypted;
        RESULT = 0;
        ReferenceLocalSnapshotDate = "2022-09-22 18:21:53 +0000";
        SnapshotDates =         (
            "2022-09-22 18:21:01 +0000",
            "2022-09-22 18:21:32 +0000",
            "2022-09-22 18:21:57 +0000"
        );
    }
)
```

Remediation:

Graphical Method:
Perform the following steps to enable to backup automatically if Time Machine is enabled:

1. Open `System Preferences`
2. Select `Time Machine`
3. Verify that `Time Machine` is enabled
4. Set `Back Up Automatically` to enabled

Terminal Method:
Run the following command to enable automatic backups if Time Machine is enabled:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.TimeMachine.plist AutoBackup -bool true
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.TimeMachine`
2. The key to include is `Forced`
3. The key must be set to:

```
<array>
        <dict>
                <key>mcx_preference_settings</key>
                        <dict>
                                <key>AutoBackup</key>
                                <true/>
                        </dict>
        </dict>
</array>
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	11.2 Perform Automated Backups Perform automated backups of in-scope enterprise assets. Run backups weekly, or more frequently, based on the sensitivity of the data.	●	●	●
v7	10.1 Ensure Regular Automated Back Ups Ensure that all system data is automatically backed up on regular basis.	●	●	●

2.7.2 Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

One of the most important security tools for data protection on macOS is FileVault. With encryption in place it makes it difficult for an outside party to access your data if they get physical possession of the computer. One very large weakness in data protection with FileVault is the level of protection on backup volumes. If the internal drive is encrypted but the external backup volume that goes home in the same laptop bag is not it is self-defeating. Apple tries to make this mistake easily avoided by providing a checkbox to enable encryption when setting-up a Time Machine backup. Using this option does require some password management, particularly if a large drive is used with multiple computers. A unique complex password to unlock the drive can be stored in keychains on multiple systems for ease of use.

While some portable drives may contain non-sensitive data and encryption may make interoperability with other systems difficult backup volumes should be protected just like boot volumes.

Rationale:

Backup volumes need to be encrypted.

Audit:

Graphical Method:
Perform the following steps to ensure the drive used for Time Machine is encrypted if Time Machine is enabled:

1. Open `System Preferences`
2. Select `Time Machine`
3. Select `Backup Disk...`
4. Select the Time Machine backup drive
5. Verify that `Encrypt backups` is set

Terminal Method:
Run the following command to verify if the Time Machine disk encryption is enabled:

```
$ /usr/bin/sudo /usr/bin/defaults read
/Library/Preferences/com.apple.TimeMachine.plist | grep -c NotEncrypted

0
```

Remediation:

Graphical Method:
Perform the following steps to enable encryption on the Time Machine drive:

1. Open `System Preferences
2. Select `Time Machine
3. Select `Backup Disk...`
4. Select the existing Time Machine backup drive from the Available Drive list
5. Set `Encrypt backups` to enabled
6. Select `Use Disk`

Note: You can set encryption through Disk Utility or `/usr/bin/sudo /usr/sbin/diskutil` in terminal, but it is advised to use the GUI.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.6 Encrypt Data on End-User Devices Encrypt data on end-user devices containing sensitive data. Example implementations can include: Windows BitLocker®, Apple FileVault®, Linux® dm-crypt.	●	●	●
v8	3.11 Encrypt Sensitive Data at Rest Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.		●	●
v8	11.3 Protect Recovery Data Protect recovery data with equivalent controls to the original data. Reference encryption or data separation, based on requirements.	●	●	●
v7	10.4 Ensure Protection of Backups Ensure that backups are properly protected via physical security or encryption when they are stored, as well as when they are moved across the network. This includes remote backups and cloud services.	●	●	●
v7	13.6 Encrypt the Hard Drive of All Mobile Devices. Utilize approved whole disk encryption software to encrypt the hard drive of all mobile devices.	●	●	●
v7	14.8 Encrypt Sensitive Information at Rest Encrypt all sensitive information at rest using a tool that requires a secondary authentication mechanism not integrated into the operating system, in order to access the information.			●

2.8 Energy Saver

2.8.1 Ensure Wake for Network Access Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

This feature allows the computer to take action when the user is not present and the computer is in energy saving mode. These tools require FileVault to remain unlocked and fully rejoin known networks. This macOS feature is meant to allow the computer to resume activity as needed regardless of physical security controls.

This feature allows other users to be able to access your computer's shared resources, such as shared printers or Apple Music playlists, even when your computer is in sleep mode. In a closed network when only authorized devices could wake a computer, it could be valuable to wake computers in order to do management push activity. Where mobile workstations and agents exist, the device will more likely check in to receive updates when already awake. Mobile devices should not be listening for signals on any unmanaged network or where untrusted devices exist that could send wake signals.

Rationale:

Disabling this feature mitigates the risk of an attacker remotely waking the system and gaining access.

Impact:

Management programs like Apple Remote Desktop Administrator use wake-on-LAN to connect with computers. If turned off, such management programs will not be able to wake a computer over the LAN. If the wake-on-LAN feature is needed, do not turn off this feature.

The control to prevent computer sleep has been retired for this version of the Benchmark. Forcing the computer to stay on and use energy in case a management push is needed is contrary to most current management processes. Only keep computers unslept if after hours pushes are required on closed LANs.

Audit:

Graphical Method:
Perform the following steps to verify that Wake for network access is not enabled:

1. Open `System Preferences`
2. Select `Battery`
3. Select `Power Adapter`
4. Verify that `Wake for network access` is not enabled

Terminal Method:
Run the following command verify if Wake for network access is disabled:

```
$ /usr/bin/sudo /usr/bin/pmset -g custom | /usr/bin/grep -e womp

womp                 0
```

or
Run the following command to verify that a profile is installed that enables App Store updates to be automatically installed:

```
$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep "Wake On LAN"

                            "Wake On LAN" = 0;
                            "Wake On LAN" = 0;
                            "Wake On LAN" = 0;

$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep "Wake On Modem
Ring"

                            "Wake On Modem Ring" = 0;
                            "Wake On Modem Ring" = 0;
                            "Wake On Modem Ring" = 0;
```

Remediation:

Graphical Method:
Perform the following steps to disable Wake for network access:

1. Open `System Preferences`
2. Select `Battery`
3. Select `Power Adapter`
4. Set `Wake for network access` to disabled

Terminal Method:
Run the following command to disable Wake for network access:

```
$ /usr/bin/sudo /usr/bin/pmset -a womp 0
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.MCX`
2. The key to include is `com.apple.EnergySaver.desktop.ACPower`
3. The key must be set to:

```
<dict>
        <key>Wake On LAN</key>
        <integer>0</integer>
        <key>Wake On Modem Ring</key>
        <integer>0</integer>
</dict>
```

4. The key to also include is `com.apple.EnergySaver.portable.ACPower`
5. The key must be set to:

```
<dict>
        <key>Wake On LAN</key>
        <integer>0</integer>
        <key>Wake On Modem Ring</key>
        <integer>0</integer>
</dict>
```

6. The key to also include is `com.apple.EnergySaver.portable.BatteryPower`
7. The key must be set to:

```
<dict>
        <key>Wake On LAN</key>
        <integer>0</integer>
        <key>Wake On Modem Ring</key>
        <integer>0</integer>
</dict>
```

Note: Both `Wake on LAN` and `Wake on Modem Ring` need to be set. Only setting `Wake On LAN` will allow the profile to install but not set any settings. This profile will only apply the setting at installation and is not sticky.

Additional Information:

`man pmset`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.8.2 Ensure Power Nap Is Disabled for Intel Macs (Automated)

Profile Applicability:

- Level 1

Description:

Power Nap allows the system to stay in low power mode, especially while on battery power, and periodically connect to previously known networks with stored credentials for user applications to phone home and get updates. This capability requires FileVault to remain unlocked and the use of previously joined networks to be risk accepted based on the SSID without user input.

This control has been updated to check the status on both battery and AC Power. The presence of an electrical outlet does not completely correlate with logical and physical security of the device or available networks.

Rationale:

Disabling this feature mitigates the risk of an attacker remotely waking the system and gaining access.

The use of Power Nap adds to the risk of compromised physical and logical security. The user should be able to decrypt FileVault and have the applications download what is required when the computer is actively used.

The control to prevent computer sleep has been retired for this version of the Benchmark. Forcing the computer to stay on and use energy in case a management push is needed is contrary to most current management processes. Only keep computers unslept if after hours pushes are required on closed LANs.

Impact:

Power Nap exists for unattended user application updates like email and social media clients. With Power Nap disabled, the computer will not wake and reconnect to known wireless SSIDs intermittently when slept.

Audit:

Graphical Method:
Perform the following steps to verify that Power Nap is not enabled:

1. Open `System Preferences`
2. Select `Battery`
3. Select `Battery`
4. Verify that `Power Nap` is not enabled
5. Select `Power Adapter`
6. Verify that `Power Nap` is not enabled
7. Select `UPS`
8. Verify that `Power Nap` is not enabled

Terminal Method:
Run the following command to verify if Power Nap is disabled:

```
$ /usr/bin/sudo /usr/bin/pmset -g custom | /usr/bin/grep -c 'powernap
1'

0
```

Remediation:

Graphical Method:
Perform the following steps to disable Power Nap:

1. Open `System Preferences`
2. Select `Battery`
3. Select `Battery`
4. Set `Enable Power Nap` to enabled
5. Select `Power Adapter`
6. Set `Enable Power Nap` to enabled
7. Select `UPS`
8. Set `Enable Power Nap` to enabled

Terminal Method:
Run the following command to disable Power Nap:

```
$ /usr/bin/sudo /usr/bin/pmset -a powernap 0
```

Additional Information:

`man pmset`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.8.3 Ensure the OS is not Activate When Resuming from Sleep (Automated)

Profile Applicability:

- Level 2

Description:

In order to use a computer with Full Disk Encryption (FDE), macOS must keep encryption keys in memory to allow the use of the disk that has been FileVault protected. The storage volume has been unlocked and acts as if it were not encrypted. When the system is not in use, the volume is protected through encryption. When the system is sleeping and available to quickly resume, the encryption keys remain in memory.

If an unauthorized party has possession of the computer and the computer is only slept, there are known attack vectors that can be attempted against the RAM that has the encryption keys or the running operating system protected by a login screen. Network attacks if network interfaces are on, as well as USB or other open device ports, are possible. Most of these attacks require knowledge of unpatched vulnerabilities or a high level of sophistication if all the other controls function as intended.

There is little impact on hibernating the system rather than sleeping after an appropriate time period to remediate the risk of OS level attacks. Hibernation writes the keys to disk and requires FileVault to be unlocked prior to the OS being available. In the case of unauthorized personnel with access to the computer, encryption would have to be broken prior to attacking the operating system in order to recover data from the system.

https://www.helpnetsecurity.com/2018/08/20/laptop-sleep-security/

Mac systems should be set to hibernate after sleeping for a risk-acceptable time period. The default value for "standbydelay" is three hours (10800 seconds). This value is likely appropriate for most desktops. If Mac desktops are deployed in unmonitored, less physically secure areas with confidential data, this value might be adjusted. The desktop would have to retain power, however, so that the running OS or physical RAM could be attacked.

MacBooks should be set so that the standbydelay is 15 minutes (900 seconds) or less. This setting should allow laptop users in most cases to stay within physically secured areas while going to a conference room, auditorium, or other internal location without having to unlock the encryption. When the user goes home at night, the laptop will auto-hibernate after 15 minutes and require the FileVault password to unlock prior to logging back into system when it resumes.

MacBooks should also be set to a hibernate mode that removes power from the RAM. This will stop the possibility of cold boot attacks on the system.

Rationale:

To mitigate the risk of data loss, the system should power down and lock the encrypted drive after a specified time. Laptops should hibernate 15 minutes or less after sleeping.

Impact:

The laptop will take additional time to resume normal operation then if only sleeping rather than hibernating.

Audit:

Terminal Method:
Run the following command to verify the hibernation settings and that FileVault keys are destroyed on standby:

```
$ /usr/bin/sudo /usr/sbin/system_profiler SPHardwareDataType | /usr/bin/grep
-e MacBook
```

If there is an output, run the following:

```
$ /usr/bin/sudo /usr/bin/pmset -b -g | grep -e standby
```

The output should include a `standbydelaylow` value ≤ 900, a `standbydelayhigh` value ≤ 900, and a `highstandbythreshold` value ≥ 90.

```
$ /usr/bin/sudo /usr/bin/pmset -b -g | /usr/bin/grep DestroyFVKeyOnStandby

 DestroyFVKeyOnStandby         1

$ /usr/bin/sudo /usr/bin/pmset -b -g | /usr/bin/grep hibernatemode

 hibernatemode        25
```

example:

```
$ /usr/bin/sudo /usr/sbin/system_profiler SPHardwareDataType | /usr/bin/grep
-e MacBook

      Model Name: MacBook Pro
      Model Identifier: MacBookPro13,1

$ /usr/bin/sudo /usr/bin/pmset -b -g | /usr/bin/grep -e standby

 standbydelaylow      600
 standby              1
 standbydelayhigh     600
 highstandbythreshold 50

$ /usr/bin/sudo /usr/bin/pmset -b -g | /usr/bin/grep DestroyFVKeyOnStandby

 DestroyFVKeyOnStandby         1

$ /usr/bin/sudo /usr/bin/pmset -b -g | /usr/bin/grep hibernatemode

 hibernatemode        25
```

Remediation:

Terminal Method:
Run the following command to set the hibernate delays and to ensure the FileVault keys are set to be destroyed on standby:

```
$ /usr/bin/sudo /usr/bin/pmset -a standbydelaylow <value≤900>
$ /usr/bin/sudo /usr/bin/pmset -a standbydelayhigh <value≤900>
$ /usr/bin/sudo /usr/bin/pmset -a highstandbythreshold <value≥90>
$ /usr/bin/sudo /usr/bin/pmset -a destroyfvkeyonstandby 1
$ /usr/bin/sudo /usr/bin/pmset -a hibernatemode 25
```

example:

```
$ /usr/bin/sudo /usr/bin/pmset -a standbydelaylow 500
$ /usr/bin/sudo /usr/bin/pmset -a standbydelayhigh 500
$ /usr/bin/sudo /usr/bin/pmset -a highstandbythreshold 100
$ /usr/bin/sudo /usr/bin/pmset -a destroyfvkeyonstandby 1
$ /usr/bin/sudo /usr/bin/pmset -a hibernatemode 25
```

References:

1. https://www.lifewire.com/change-mac-sleep-settings-2260804
2. https://www.zdziarski.com/blog/?p=6705
3. https://www.howtogeek.com/260478/how-to-choose-when-your-mac-hibernates-or-enters-standby/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	16.11 Lock Workstation Sessions After Inactivity Automatically lock workstation sessions after a standard period of inactivity.	●	●	●

2.9 Ensure Legacy EFI Is Valid and Updating (Automated)

Profile Applicability:

- Level 1

Description:

In order to mitigate firmware attacks, Apple has created an automated Firmware check to ensure that the EFI version running is a known good version from Apple. There is also an automated process to check it every seven days.

This check is only valid on T1 chips and prior. Neither T2 chips nor Apple silicon require this control check

Rationale:

If the Firmware of a computer has been compromised, the Operating System that the Firmware loads cannot be trusted, either.

Audit:

Terminal Method:
Perform the following to verify that the computer has up-to-date firmware if running a pre-T2 Mac:

```
$ /usr/bin/sudo /usr/sbin/system_profiler SPiBridgeDataType | grep "T2"
```

If the output is `Model Name: Apple T2 Security Chip`, the computer us compliant.
If there is no output, run the following to determine if the EFI firmware is up-to-date:

```
$ /usr/bin/sudo /usr/libexec/firmwarecheckers/eficheck/eficheck --integrity-check
```

The output should include `Primary allowlist version match found. No changes detected in primary hashes.` as well as the model and version in this format <Model Number>.xxx.xxxx.xxx.xxxxxxxxx.
Then run the following command to verify that the EFI check system daemon is running:

```
$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep com.apple.driver.eficheck

Result: -       0       com.apple.driver.eficheck
```

Note: The command to verify if the EFI check daemon is running can also be ran on Macs with a T2 Security Chip.
example:

```
$ /usr/bin/sudo /usr/libexec/firmwarecheckers/eficheck/eficheck --integrity-check

EFI Version: MB101.88Z.F000.B00.2203161719  (Boot ROM Version: 451.120.7.0.0)
Primary allowlist version match found. No changes detected in primary hashes.

$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep com.apple.driver.eficheck

Result: -       0       com.apple.driver.eficheck
```

Remediation:

If EFI does not pass the integrity check, you may send a report to Apple. Backing up files and clean installing a known good Operating System and Firmware is recommended.

Additional Information:

EFI is the software link between the motherboard hardware and the software operating system. EFI determines which partition or disk to load macOS from, and it also determines whether the user can enter single user mode. The main reasons to set a firmware password have been protections against an alternative boot disk, protection against a passwordless root shell through single user mode, and protection against firewire DMA attacks. While it was easier in the past to reset the firmware password by removing RAM, it did make tampering slightly harder because having to remove RAM remediated memory scraping attacks through DMA. It has always been difficult to manage the firmware password on macOS computers, though some tools did make it much easier.

The EFI password management capability has been replaced in new Apple silicon Macs. The security features are replaced in the Silicon Mac recoveryOS. Long-term it appears that macOS EFI management is a deprecated technology in mixed Intel/Apple Silicon environments.

Apple patched OS X in 10.7 to mitigate the DMA attacks, and the use of FileVault 2 Full-Disk Encryption mitigates the risk of damage to the boot volume if an unauthorized user uses a different boot volume or uses single user mode. Apple's reliance on the recovery partition and the additional features it provides make controls that do not allow the user to boot into the recovery partition less attractive.

Starting in late 2010 with the MacBook Air, Apple has slowly updated the requirements to recover from a lost firmware password. Apple only supports taking the computer to an Apple authorized service provider. This change makes managing the firmware password effectively more critical if it is used.

Setting the firmware password may be a good practice in some environments. We cannot recommend it as a standard security practice at this time.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.2 Ensure Authorized Software is Currently Supported Ensure that only currently supported software is designated as authorized in the software inventory for enterprise assets. If software is unsupported, yet necessary for the fulfillment of the enterprise's mission, document an exception detailing mitigating controls and residual risk acceptance. For any unsupported software without an exception documentation, designate as unauthorized. Review the software list to verify software support at least monthly, or more frequently.	●	●	●
v7	2.2 Ensure Software is Supported by Vendor Ensure that only software applications or operating systems currently supported by the software's vendor are added to the organization's authorized software inventory. Unsupported software should be tagged as unsupported in the inventory system.	●	●	●

2.10 Audit Siri Settings (Manual)

Profile Applicability:

- Level 1

Description:

With macOS 10.12 Sierra, Apple has introduced Siri from iOS to macOS. While there are data spillage concerns with the use of data-gathering personal assistant software, the risk here does not seem greater in sending queries to Apple through Siri than in sending search terms in a browser to Google or Microsoft. While it is possible that Siri will be used for local actions rather than Internet searches, Siri could, in theory, tell Apple about confidential Programs and Projects that should not be revealed. This appears be a usage edge case.

In cases where sensitive or protected data is processed and Siri could expose that information through assisting a user in navigating their machine, it should be disabled. Siri does need to phone home to Apple, so it should not be available from air-gapped networks as part of its requirements.

Most of the use case data published has shown that Siri is a tremendous time saver on iOS where multiple screens and menus need to be navigated through. Information like sports scores, weather, movie times, and simple to-do items on existing calendars can be easily found with Siri. None of the standard use cases should be more risky than already approved activity.

For information on Apple's privacy policy for Siri, click here.

Rationale:

Where "normal" user activity is already limited, Siri use should be controlled as well.

Audit:

Graphical Method:
Perform the following steps to verify Siri settings:

1. Open `System Preferences`
2. Select `Siri`
3. Verify the settings are within your organization's parameters
4. Select `Show All`
5. Select `Accessibility`
6. Select `Siri`
7. Verify `Enable Type to Siri` is set to your organization's parameters

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Allow Assistant` is set to your organization's parameters

Terminal Method:
Run the following commands to verify the Siri settings:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
com.apple.assistant.support.plist 'Assistant Enabled'
```

The output will be either `0`, Siri is disabled, or `1`, Siri is enabled.

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read com.apple.Siri.plist
```

The output will be either `0`, disabled, or `1` for the following Siri options:

1. LockscreenEnabled - Is Siri enabled when the system is locked?
2. StatusMenuVisible - Is Siri visible in the menu bar?
3. TypeToSiriEnabled - Is Siri enabled to accept typed requests versus spoken ones
4. VoiceTriggerUserEnabled - Is "Hey Siri" enabled?

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
com.apple.assistant.support.plist 'Assistant Enabled'

0

$ /usr/bin/sudo -u firstuser /usr/bin/defaults read com.apple.Siri.plist

{
    LockscreenEnabled = 0;
    StatusMenuVisible = 0;
    TypeToSiriEnabled = 0;
    VoiceTriggerUserEnabled = 0;
}

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read
com.apple.assistant.support.plist 'Assistant Enabled'

1

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read com.apple.Siri.plist

{
    LockscreenEnabled = 0;
    StatusMenuVisible = 1;
    TypeToSiriEnabled = 0;
    VoiceTriggerUserEnabled = 1;
}

$ /usr/bin/sudo -u thirduser /usr/bin/defaults read
com.apple.assistant.support.plist 'Assistant Enabled'

1

$ /usr/bin/sudo -u thirduser /usr/bin/defaults read com.apple.Siri.plist

{
    LockscreenEnabled = 1;
    StatusMenuVisible = 0;
    TypeToSiriEnabled = 1;
    VoiceTriggerUserEnabled = 1;
}
```

or

Run the following command to verify that a profile is installed that sets Siri to your organization's setting:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.applicationaccess')\
.objectForKey('allowAssistant').js
EOS
```

The output will be `true` if Siri is enabled with a profile or `false` if is disabled with a profile.

Note: Siri can only be enabled or disabled through configuration profiles. Any additional settings need to be set through either `System Settings` or `Terminal`.

Remediation:

Graphical Method:
Perform the following steps to set Siri to your organization's parameters:

1. Open `System Preferences`
2. Select `Siri`
3. Select the settings that are within your organization's requirements
4. Select `Show All`
5. Select `Accessibility`
6. Select `Siri`
   7 Select `Enable Type to Siri` to your organization's requirements

Terminal Method:
Run the following commands to enable or disable Siri settings:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
com.apple.assistant.support.plist 'Assistant Enabled' -bool <true/false>

$ /usr/bin/sudo -u <username> /usr/bin/defaults write com.apple.Siri.plist
LockscreenEnabled -bool <true/false>

$ /usr/bin/sudo -u <username> /usr/bin/defaults write com.apple.Siri.plist
StatusMenuVisible -bool <true/false>

$ /usr/bin/sudo -u <username> /usr/bin/defaults write com.apple.Siri.plist
TypeToSiriEnabled -bool <true/false>

$ /usr/bin/sudo -u <username> /usr/bin/defaults write com.apple.Siri.plist
VoiceTriggerUserEnabled -bool <true/false>
```

After running the default writes, the WindowServer needs to be restarted and the caches cleared. Run the following commands to perform that action:

```
$ /usr/bin/sudo /usr/bin/killall -HUP cfprefsd

$ /usr/bin/sudo /usr/bin/killall SystemUIServer
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
com.apple.assistant.support.plist 'Assistant Enabled' -bool true

$ /usr/bin/sudo -u firstuser /usr/bin/defaults write com.apple.Siri.plist
StatusMenuVisible -bool true

$ /usr/bin/sudo -u firstuser /usr/bin/defaults write com.apple.Siri.plist
LockscreenEnabled -bool false

$ /usr/bin/sudo /usr/bin/killall -HUP cfprefsd

$ /usr/bin/sudo /usr/bin/killall SystemUIServer

$ /usr/bin/sudo -u seconduser /usr/bin/defaults write
com.apple.assistant.support.plist 'Assistant Enabled' -bool false

$ /usr/bin/sudo /usr/bin/killall -HUP cfprefsd

$ /usr/bin/sudo /usr/bin/killall SystemUIServer

$ /usr/bin/sudo -u thirduser /usr/bin/defaults write com.apple.Siri.plist
VoiceTriggerUserEnabled -bool false

$ /usr/bin/sudo -u thirduser /usr/bin/defaults write com.apple.Siri.plist
TypeToSiriEnabled -bool false

$ /usr/bin/sudo /usr/bin/killall -HUP cfprefsd

$ /usr/bin/sudo /usr/bin/killall SystemUIServer
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.applicationaccess`
2. The key to include is `allowAssistant`
3. Set the key to `<true/>` or `<false/>` based on your organization's requirements

Note: Siri can only be enabled or disabled through configuration profiles. Any additional settings need to be set through either `System Settings` or `Terminal`.

References:

1. https://support.apple.com/en-us/HT210657

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.11 Audit Sidecar Settings (Manual)

Profile Applicability:

- Level 1

Description:

Apple introduced a technology called Sidecar with the release of mac OS 10.15 "Catalina" that allows the use of an Apple iPad as an additional screen. There are no known security issues with the use of Sidecar at the time of the publication of this Benchmark. There are security concerns with some of the underlying technology that allows this feature to work. The Apple support article below has the additional requirements that are reproduced below. So while Sidecar may not have an explicit security concern some organizations may have requirements that block the use of the features required to allow Sidecar to work.

https://support.apple.com/en-afri/HT210380

Additional requirements

- Both devices must be signed in to iCloud with the same Apple ID using two-factor authentication.
- To use Sidecar wirelessly, both devices must be within 10 meters (30 feet) of each other and have Bluetooth, Wi-Fi, and Handoff turned on. Also make sure that the iPad is not sharing its cellular connection and the Mac is not sharing its Internet connection.
- To use Sidecar over USB, make sure that your iPad is set to trust your Mac.

Organizations that do not allow the use of iCloud and more specifically Handoff will not be able to use Sidecar.

Some organizations may not allow the use of mixed ownership for P2P wireless or USB connections so that unless the organization controls both the Mac and the iPad connections may not be approved and the use of a single Apple ID for distinctly managed devices may be prohibited.

Rationale:

Organizations need to have an understanding of integration of organizational and personal inventory in the work environment.

Audit:

Graphical Method:
Perform the following steps to verify Sidecar's setting:

1. Open `System Preferences`
2. Select `Sidecar`
3. Verify the settings are within your organization's parameters

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `AllowAllDevices` to your organization's parameters
4. Verify that an installed profile has `hasShownPref` to your organization's parameters

Terminal Method:
Run the following commands to verify if Sidecar is enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.sidecar.display')\
.objectForKey('AllowAllDevices').js
EOS
```

The output will be either `false`, Sidecar is disabled, or `true`, Sidecar is enabled.

Remediation:

Graphical Method:
Perform the following steps to set Sidecar to your organization's parameters:

1. Open `System Preferences`
2. Select `Sidecar`
3. Select the settings that are within your organization's parameters

Terminal Method:
Run the following to enable or disable Sidecar settings:

```
$ /usr/bin/sudo /usr/bin/defaults write com.apple.sidecar.display
AllowAllDevices -bool <true/false>

$ /usr/bin/sudo /usr/bin/defaults write com.apple.sidecar.display
hasShownPref -bool <true/false>
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.sidecar.display`
2. The key to include is `Forced`
3. The key must be set to:

```
<dict>
        <key>Forced</key>
                <array>
                        <dict>
                                <key>mcx_preference_settings</key>
                                        <dict>
                                                <key>AllowAllDevices</key>
                                                <<true/false>/>
                                                <key>hasShownPref</key>
                                                <<true/false>/>
                                        </dict>
                        </dict>
                </array>
</dict>
```

Note: Using the Terminal and Profile Methods will not display in System Preferences, but will disable the underlying service.

References:

1. https://www.apple.com/macos/catalina/docs/Sidecar_Tech_Brief_Oct_2019.pdf
2. https://www.pocket-lint.com/laptops/news/apple/148262-apple-sidecar-macos-ipados-features-explained

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

2.12 Audit Touch ID and Wallet & Apple Pay Settings (Manual)

Profile Applicability:

- Level 1

Description:

Apple has integrated Touch ID with macOS and allows fingerprint use for many common operations. All use of Touch ID requires the presence of a password and the use of that password after every reboot, or when more than 48 hours has elapsed since the device was last unlocked.

Touch ID is a prerequisite for using Apple Pay and Wallet on macOS. Apple Pay allows an Apple account holder to enroll their credit cards in Apple Pay and pay enrolled vendors without using the physical card or number. Apple's service eliminates the requirement to send the credit card number itself to the vendor. Apple Pay on a Mac allows the use of credit cards the user has already enrolled and reduces user risk for credit card purchases.

Rationale:

Touch ID allows for an account-enrolled fingerprint to access a key that uses a previously provided password.

Some environments may have rules around purchases from organizationally managed computers and may want to discourage shopping from them. It is difficult to block access to websites that allow purchases, and Apple Pay has more controls for user protection than the manual entry of credit card information.

Impact:

Touch ID is more convenient for use with aggressive screen lock controls.

Audit:

Graphical Method:
Perform the following steps to verify Touch ID settings:

1. Open `System Preferences`
2. Select `Touch ID`
3. Verify the `Touch ID` settings match your organization's settings
4. Open `System Preferences`
5. Select `Wallet & Apple Pay`
6. Verify the `Wallet & Apple Pay` settings match your organization's settings

Remediation:

Graphical Method:
Perform the following steps to set Touch ID to your organization's settings:

1. Open `System Preferences`
2. Select `Touch ID`
3. Select the `Touch ID` settings match your organization's settings
4. Open `System Preferences`
5. Select `Wallet & Apple Pay`
6. Select the `Wallet & Apple Pay` settings match your organization's settings

References:

1. https://support.apple.com/guide/mac-help/use-wallet-apple-pay-on-mac-mchl4773988b/mac
2. https://www.apple.com/apple-pay/
3. https://support.apple.com/guide/mac-help/touch-id-mchl16fbf90a/mac

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

3 Logging and Auditing

This section provides guidance on configuring the logging and auditing facilities available in macOS. Starting with macOS 10.12, Apple introduced unified logging. This capability replaces the previous logging methodology with centralized, system-wide common controls. A full explanation of macOS logging behavior is beyond the scope of this Benchmark. These changes impact previous logging controls from macOS Benchmarks. At this point, many of the syslog controls have been or are being removed since the old logging methods have been deprecated. Controls that still appear useful will be retained. Some legacy controls have been removed for this release.

More info https://developer.apple.com/documentation/os/logging
https://eclecticlight.co/2018/03/19/macos-unified-log-1-why-what-and-how/

3.1 Ensure Security Auditing Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

macOS's audit facility, `auditd`, receives notifications from the kernel when certain system calls, such as `open`, `fork`, and `exit`, are made. These notifications are captured and written to an audit log.

Rationale:

Logs generated by `auditd` may be useful when investigating a security incident as they may help reveal the vulnerable application and the actions taken by a malicious actor.

Audit:

Terminal Method:
Perform the following to verify that security auditing is enabled:
Run the following command to verify auditd:

```
$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep -i auditd

-       0       com.apple.auditd
```

Remediation:

Terminal Method:
Perform the following to enable security auditing:
Run the following command to load auditd:

```
$ /usr/bin/sudo /bin/launchctl load -w
/System/Library/LaunchDaemons/com.apple.auditd.plist
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 Collect Audit Logs Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.	●	●	●
v8	8.5 Collect Detailed Audit Logs Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.		●	●
v7	4.9 Log and Alert on Unsuccessful Administrative Account Login Configure systems to issue a log entry and alert on unsuccessful logins to an administrative account.		●	●
v7	6.2 Activate audit logging Ensure that local logging has been enabled on all systems and networking devices.	●	●	●

3.2 Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements (Automated)

Profile Applicability:

- Level 2

Description:

Auditing is the capture and maintenance of information about security-related events. Auditable events often depend on differing organizational requirements.

Rationale:

Maintaining an audit trail of system activity logs can help identify configuration errors, troubleshoot service disruptions, and analyze compromises or attacks that have occurred, have begun, or are about to begin. Audit logs are necessary to provide a trail of evidence in case the system or network is compromised.

Depending on the governing authority, organizations can have vastly different auditing requirements. In this control we have selected a minimal set of audit flags that should be a part of any organizational requirements. The flags selected below may not adequately meet organizational requirements for users of this benchmark. The auditing checks for the flags proposed here will not impact additional flags that are selected.

Audit:

Terminal Method:
Run the following command to verify the Security Auditing Flags that are enabled:

```
$ /usr/bin/sudo /usr/bin/grep -e "^flags:" /etc/security/audit_control
```

The output should include the following flags:

- `-fm` - audit failed file attribute modification events
- `ad` - audit successful/failed administrative events
- `-ex` - audit failed program execution
- `aa` - audit all authorization and authentication events
- `-fr` - audit all failed read actions where enforcement stops a read of a file
- `lo` - audit successful/failed login/logout events
- `-fw` - audit all failed write actions where enforcement stopped a file write

The `-all` flag will capture all failed events across all audit classes and can be used to supersede the individual flags for failed events.
Note: Excluding potentially noisy audit events may be ideal, depending on your use-case.
Note: Historical audit flags are listed below as preliminary guidance.

Remediation:

Terminal Method:
Perform the following to set the required Security Auditing Flags:
Edit the `/etc/security/audit_control` file and add `-fm`, `ad`, `-ex`, `aa`, `-fr`, `lo`, and `-fw` to `flags`. You can also substitute `-all` for `-fm`, `-ex`, `-fr`, and `-fw`.

References:

1. https://derflounder.wordpress.com/2012/01/30/openbsm-auditing-on-mac-os-x/
2. https://csrc.nist.gov/CSRC/media/Publications/sp/800-179/rev-1/draft/documents/sp800-179r1-draft.pdf
3. https://www.scip.ch/en/?labs.20150108
4. https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-171r2.pdf
5. https://www.whitehouse.gov/wp-content/uploads/2021/08/M-21-31-Improving-the-Federal-Governments-Investigative-and-Remediation-Capabilities-Related-to-Cybersecurity-Incidents.pdf

Additional Information:

Flag settings are currently based on the guidance provided by the NIST through the macOS Security guidance they are providing in their GitHub repository. You can find that guidance here.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.14 Log Sensitive Data Access Log sensitive data access, including modification and disposal.			●
v8	8.2 Collect Audit Logs Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.	●	●	●
v8	8.5 Collect Detailed Audit Logs Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.		●	●
v7	6.2 Activate audit logging Ensure that local logging has been enabled on all systems and networking devices.	●	●	●
v7	14.9 Enforce Detail Logging for Access or Changes to Sensitive Data Enforce detailed audit logging for access to sensitive data or changes to sensitive data (utilizing tools such as File Integrity Monitoring or Security Information and Event Monitoring).			●

3.3 Ensure install.log Is Retained for 365 or More Days and No Maximum Size (Automated)

Profile Applicability:

- Level 1

Description:

macOS writes information pertaining to system-related events to the file `/var/log/install.log` and has a configurable retention policy for this file. The default logging setting limits the file size of the logs and the maximum size for all logs. The default allows for an errant application to fill the log files and does not enforce sufficient log retention. The Benchmark recommends a value based on standard use cases. The value should align with local requirements within the organization.

The default value has an "all_max" file limitation, no reference to a minimum retention, and a less precise rotation argument.

The all_max flag control will remove old log entries based only on the size of the log files. Log size can vary widely depending on how verbose installing applications are in their log entries. The decision here is to ensure that logs go back a year, and depending on the applications a size restriction could compromise the ability to store a full year.

While this Benchmark is not scoring for a rotation flag, the default rotation is sequential rather than using a timestamp. Auditors may prefer timestamps in order to simply review specific dates where event information is desired.

Please review the File Rotation section in the man page for more information.

```
man asl.conf
```

- The maximum file size limitation string should be removed "all_max="
- An organization appropriate retention should be added "ttl="
- The rotation should be set with timestamps "rotate=utc" or "rotate=local"

Rationale:

Archiving and retaining `install.log` for at least a year is beneficial in the event of an incident as it will allow the user to view the various changes to the system along with the date and time they occurred.

Impact:

Without log files system maintenance and security forensics cannot be properly performed.

Audit:

Terminal Method:
Run the following command to verify that log files are retained for at least 365 days with no maximum size:

```
$ /usr/bin/sudo /usr/bin/grep -i ttl /etc/asl/com.apple.install
```

The output must include `ttl≥365`

```
$ /usr/bin/sudo -i all_max= /etc/asl/com.apple.install
```

No results should be returned.

Remediation:

Terminal Method:
Perform the following to ensure that install logs are retained for at least 365 days:
Edit the `/etc/asl/com.apple.install` file and add or modify the `ttl` value to `365` or greater on the `file` line. Also, remove the `all_max=` setting and value from the `file` line.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.1 Establish and Maintain an Audit Log Management Process Establish and maintain an audit log management process that defines the enterprise's logging requirements. At a minimum, address the collection, review, and retention of audit logs for enterprise assets. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	8.3 Ensure Adequate Audit Log Storage Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.	●	●	●
v7	6.4 Ensure adequate storage for logs Ensure that all systems that store logs have adequate storage space for the logs generated.		●	●
v7	6.7 Regularly Review Logs On a regular basis, review logs to identify anomalies or abnormal events.		●	●

3.4 Ensure Security Auditing Retention Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

The macOS audit capability contains important information to investigate security or operational issues. This resource is only completely useful if it is retained long enough to allow technical staff to find the root cause of anomalies in the records.

Retention can be set to respect both size and longevity. To retain as much as possible under a certain size, the recommendation is to use the following:

expire-after:60d OR 5G

This recomendation is based on minimum storage for review and investigation. When a third party tool is in use to allow remote logging or the store and forwarding of logs, this local storage requirement is not required.

Rationale:

The audit records need to be retained long enough to be reviewed as necessary.

Impact:

The recommendation is that at least 60 days or 5 gigabytes of audit records are retained. Systems that have very little remaining disk space may have issues retaining sufficient data.

Audit:

Terminal Method:
Run the following command to verify audit retention:

```
$ /usr/bin/sudo /usr/bin/grep -e "^expire-after" /etc/security/audit_control
```

The output value for `expire-after:` should be ≥ `60d OR 5G`
Note: If your organization is offloading your security logs, we recommend following the same guidance (at minimum) for your off-site log storage. Your local storage limit (or time frame) may fail if they are set to lower in this case, but are following the guidance.

Remediation:

Terminal Method:
Perform the following to set the audit retention length:
Edit the `/etc/security/audit_control` file so that `expire-after:` is at least `60d OR 5G`

Default Value:

More info in the man page. To reference the man page use the command `$ man audit_control`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.1 Establish and Maintain an Audit Log Management Process Establish and maintain an audit log management process that defines the enterprise's logging requirements. At a minimum, address the collection, review, and retention of audit logs for enterprise assets. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	8.3 Ensure Adequate Audit Log Storage Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.	●	●	●
v7	6.4 Ensure adequate storage for logs Ensure that all systems that store logs have adequate storage space for the logs generated.		●	●
v7	6.7 Regularly Review Logs On a regular basis, review logs to identify anomalies or abnormal events.		●	●

3.5 Ensure Access to Audit Records Is Controlled (Automated)

Profile Applicability:

- Level 1

Description:

The audit system on macOS writes important operational and security information that can be both useful for an attacker and a place for an attacker to attempt to obfuscate unwanted changes that were recorded. As part of defense-in-depth, the /etc/security/audit_control configuration and the files in /var/audit should be owned only by root with group wheel with read-only rights and no other access allowed. macOS ACLs should not be used for these files.

Rationale:

Audit records should never be changed except by the system daemon posting events. Records may be viewed or extracts manipulated, but the authoritative files should be protected from unauthorized changes.

Impact:

This control is only checking the default configuration to ensure that unwanted access to audit records is not available.

Audit:

Terminal Method:
Run the following commands to check file access rights:

```
$ /usr/bin/sudo /bin/ls -n $(/usr/bin/sudo /usr/bin/grep '^dir'
/etc/security/audit_control | /usr/bin/awk -F: '{print $2}') | /usr/bin/awk
'{s+=$3} END {print s}'

0

$ /usr/bin/sudo /bin/ls -n $(/usr/bin/sudo /usr/bin/grep '^dir'
/etc/security/audit_control | /usr/bin/awk -F: '{print $2}') | /usr/bin/awk
'{s+=$4} END {print s}'

0

$ /usr/bin/sudo /bin/ls -l $(/usr/bin/sudo /usr/bin/grep '^dir'
/etc/security/audit_control | /usr/bin/awk -F: '{print $2}') | /usr/bin/awk
'!/-r--r-----|current|total/{print $1}' | /usr/bin/wc -l | /usr/bin/tr -d ' '

0

$ /usr/bin/sudo /bin/ls -n $(/usr/bin/sudo /usr/bin/grep '^dir' /var/audit/ |
/usr/bin/awk -F: '{print $2}') | /usr/bin/awk '{s+=$3} END {print s}'

0

$ /usr/bin/sudo /bin/ls -n $(/usr/bin/sudo /usr/bin/grep '^dir' /var/audit/ |
/usr/bin/awk -F: '{print $2}') | /usr/bin/awk '{s+=$4} END {print s}'

0

$ /usr/bin/sudo /bin/ls -l $(/usr/bin/sudo /usr/bin/grep '^dir' /var/audit/ |
/usr/bin/awk -F: '{print $2}') | /usr/bin/awk '!/-r--r-----
|current|total/{print $1}' | /usr/bin/wc -l | /usr/bin/tr -d ' '

0
```

Remediation:

Terminal Method:
Run the following to commands to set the audit records to the root user and wheel group:

```
$ /usr/bin/sudo /usr/sbin/chown -R root:wheel /etc/security/audit_control

$ /usr/bin/sudo /bin/chmod -R o-rw /etc/security/audit_control

$ /usr/bin/sudo /usr/sbin/chown -R root:wheel /var/audit/

$ /usr/bin/sudo /bin/chmod -R o-rw /var/audit/
```

Note: It is recommended to do a thorough verification process on why the audit logs have been changed before following the remediation steps. If the system has different access controls on the audit logs, and the changes cannot be traced, a new install may be prudent. Check for signs of file tampering as well as unapproved OS changes.

Additional Information:

From ls man page

```
-e      Print the Access Control List (ACL) associated with the file, if
            present, in long (-l) output.
```

More info:

https://www.techrepublic.com/blog/apple-in-the-enterprise/introduction-to-os-x-access-control-lists-acls/

http://ahaack.net/technology/OS-X-Access-Control-Lists-ACL.html

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

3.6 Ensure Firewall Logging Is Enabled and Configured (Automated)

Profile Applicability:

- Level 1

Description:

The socketfilter Firewall is what is used when the Firewall is turned on in the Security & Privacy Preference Pane. In order to appropriately monitor what access is allowed and denied, logging must be enabled. The logging level must be set to "detailed" to be useful in monitoring connection attempts that the firewall detects. Throttled login is not sufficient for examine Firewall connection attempts.

In depth log monitoring on macOS may require changes to the "Enable-Private-Data" key in SystemLogging.System to ensure more complete logging.

Reviewing macOS Unified Logs

Rationale:

In order to troubleshoot the successes and failures of a Firewall, detailed logging should be enabled.

Impact:

Detailed logging may result in excessive storage.

Audit:

Graphical Method:
Perform the following steps to ensure that Firewall updates install automatically:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Firewall` set to `Enabled`
4. Verify that the same installed profile has `Logging` set to `Enabled`
5. Verify that the same installed profile has `Logging option` set to `Detailed`

Terminal Method:
Run the following command to verify that the Firewall log is enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.security.firewall')\
  .objectForKey('EnableLogging').js
  let pref2 =
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.security.firewall')\
  .objectForKey('LoggingOption').js
  let pref3 = $.NSUserDefaults.alloc.initWithSuiteName('com.apple.alf')\
  .objectForKey('loggingenabled').js
  let pref4 = $.NSUserDefaults.alloc.initWithSuiteName('com.apple.alf')\
  .objectForKey('loggingoption').js
  if ( ( pref1 == true && pref2 == "detail" ) || ( pref3 == 1 && pref4 == 2 )
) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Terminal Method:
Run the following command to enable logging of the firewall:

```
$ /usr/bin/sudo /usr/libexec/ApplicationFirewall/socketfilterfw --
setloggingmode on

Turning on log mode

$ /usr/bin/sudo /usr/libexec/ApplicationFirewall/socketfilterfw --
setloggingopt detail

Setting detail log option
```

Note: If the Firewall settings are set through a configuration profile, then modifications cannot be done through the command line. If attempted, you will receive the message `Firewall settings cannot be modified from command line on managed Mac computers.`

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.security.firewall`
2. The key to include is `EnableFirewall`
3. The key must be set to `<true/>`
4. The key to also include is `EnableLogging`
5. The key must be set to `<true/>`
6. The key to also include is `LoggingOption`

7. The key must be set to `<string>detail</string>`

Note: Firewall Logging must be enabled with this profile. It can either be set with the Firewall and Stealth Mode (2.5.2.2 and 2.5.2.3) or as a separate profile. Setting logging with its own profile will not cause a conflict.

References:

1. https://developer.apple.com/documentation/devicemanagement/firewall?language=objc
2. https://www.mandiant.com/resources/blog/reviewing-macos-unified-logs
3. http://krypted.com/tag/socketfilterfw/
4. https://github.com/mandiant/macos-UnifiedLogs

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.5 Implement and Manage a Firewall on End-User Devices Implement and manage a host-based firewall or port-filtering tool on end-user devices, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.	●	●	●
v8	8.2 Collect Audit Logs Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.	●	●	●
v8	8.5 Collect Detailed Audit Logs Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.		●	●
v7	6.2 Activate audit logging Ensure that local logging has been enabled on all systems and networking devices.	●	●	●
v7	6.3 Enable Detailed Logging Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.		●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

3.7 Audit Software Inventory (Manual)

Profile Applicability:

- Level 2

Description:

With the introduction of Mac OS X 10.6.6, Apple added a new application, App Store, which resides in the Applications directory. This application allows a user with admin privileges and an Apple ID to browse Apple's online App Store, purchase (including no-cost purchases), and install new applications, bypassing Enterprise software inventory controls. Any admin user can install software in the /Applications directory whether from internet downloads, thumb drives, optical media, cloud storage, or even binaries through email. Even standard users can run executables downloaded to their home folder by default. The source of the software is not nearly as important as a consistent audit of all installed software for patch compliance and appropriateness.

A single user desktop where the user, administrator, and the person approving software are all the same person probably does not need to audit software inventory to this extent. It is helpful in the case of stability problems or malware, however.

Scan systems on a monthly basis and determine the number of unauthorized pieces of software that are installed. Verify that if an unauthorized piece of software is found one month, it is removed from the system the next.

Export System Information through the built-in System Information Application or other third-party tools on an organizationally defined timetable.

Rationale:

Part of comprehensive IT security involves device management and ensuring that all software is authorized and patched. Checking for macOS updates and app updates are relatively simple for the end user, and can even be updated with minimal privileges from trusted sources, if enabled. Remote monitoring of the patch status for software maintained through Apple is very well supported by management applications. Neither Apple capabilities nor third-party patch management solutions will cover all mission-necessary software for most organizations. Full visibility into software present on the system enables vulnerability and risk management.

P.S. Don't forget about browser plugins/extensions for all installed software.

Audit:

Graphical Mode:
Perform the following steps to access System Information:

1. Select the `Apple` icon in the `menu bar`
2. Select `About this Mac`
3. Select `System Report`
4. Select `File`
5. Select `Save`
6. Choose the name of the file and location to save the file to

Terminal Method:
Run the following command to view all System Profiler details

```
$ /usr/bin/sudo /usr/sbin/system_profiler SPApplicationsDataType
```

Remediation:

Delete any unnecessary applications from the system.

References:

1. https://support.apple.com/en-us/HT203001
2. https://www.cisecurity.org/controls/inventory-and-control-of-software-assets/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.1 Establish and Maintain a Software Inventory Establish and maintain a detailed inventory of all licensed software installed on enterprise assets. The software inventory must document the title, publisher, initial install/use date, and business purpose for each entry; where appropriate, include the Uniform Resource Locator (URL), app store(s), version(s), deployment mechanism, and decommission date. Review and update the software inventory bi-annually, or more frequently.	●	●	●
v7	2.1 Maintain Inventory of Authorized Software Maintain an up-to-date list of all authorized software that is required in the enterprise for any business purpose on any business system.	●	●	●

4 Network Configurations

This section contains guidance on configuring the networking-related aspects of macOS.

4.1 Ensure Bonjour Advertising Services Is Disabled (Automated)

Profile Applicability:

- Level 2

Description:

Bonjour is an auto-discovery mechanism for TCP/IP devices which enumerate devices and services within a local subnet. DNS on macOS is integrated with Bonjour and should not be turned off, but the Bonjour advertising service can be disabled.

Rationale:

Bonjour can simplify device discovery from an internal rogue or compromised host. An attacker could use Bonjour's multicast DNS feature to discover a vulnerable or poorly-configured service or additional information to aid a targeted attack. Implementing this control disables the continuous broadcasting of "I'm here!" messages. Typical end-user endpoints should not have to advertise services to other computers. This setting does not stop the computer from sending out service discovery messages when looking for services on an internal subnet, if the computer is looking for a printer or server and using service discovery. To block all Bonjour traffic except to approved devices, the pf or other firewall would be needed.

Impact:

Some applications, like Final Cut Studio and AirPort Base Station management, may not operate properly if the `mDNSResponder` is turned off.

Audit:

Graphical Method:
Perform the following steps to ensure that Bonjour Advertising is disabled:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `NoMulticastAdvertisements` set to `1`

Terminal Method:
Run the following command to verify that Bonjour Advertising is not enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.mDNSResponder')\
.objectForKey('NoMulticastAdvertisements').js
EOS

true
```

Remediation:

Terminal Method:
Run the following command to disable Bonjour Advertising services:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.mDNSResponder.plist NoMulticastAdvertisements
-bool true
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mDNSResponder`
2. The key to include is `NoMulticastAdvertisements`

Additional Information:

Anything Bonjour discovers is already available on the network and probably discoverable with network scanning tools. The security benefit of disabling Bonjour for that reason is minimal.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

4.2 Ensure Show Wi-Fi status in Menu Bar Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

The Wi-Fi status in the menu bar indicates if the system's wireless internet capabilities are enabled. If so, the system will scan for available wireless networks in order to connect. At the time of this revision, all computers Apple builds have wireless network capability, which has not always been the case. This control only pertains to systems that have a wireless NIC available. Operating systems running in a virtual environment may not score as expected, either.

Rationale:

Enabling "Show Wi-Fi status in menu bar" is a security awareness method that helps mitigate public area wireless exploits by making the user aware of their wireless connectivity status.

Impact:

The user of the system should have a quick check on their wireless network status available.

Audit:

Graphical Method:
Perform the following steps to verify that the Wi-Fi status shows in the menu bar:

1. Open `System Preferences`
2. Select `Network`
3. Select `Wi-Fi`
4. Verify that `Show Wi-Fi status in menu bar` is enabled

Terminal Method:
For each user, run the following command to verify that Wi-Fi status is enabled in the menu bar:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Preferences/com.apple.systemuiserver.plist
menuExtras | /usr/bin/grep AirPort.menu

"/System/Library/CoreServices/Menu Extras/AirPort.menu"
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Preferences/com.apple.systemuiserver.plist
menuExtras | /usr/bin/grep AirPort.menu

"/System/Library/CoreServices/Menu Extras/AirPort.menu"
```

Remediation:

Graphical Method:
Perform the following steps to enable Wi-Fi status in the menu bar:

1. Open `System Preferences`
2. Select `Network`
3. Select `Wi-Fi`
4. Set `Show Wi-Fi status in menu bar` to enabled

Terminal Method:
For each user, run the following command to enable Wi-Fi status in the menu bar:

```
$ /usr/bin/sudo -u <username> defaults -currentHost write
com.apple.controlcenter.plist WiFi -int 18

$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Preferences/com.apple.systemuiserver menuExtras -
array-add "/System/Library/CoreServices/Menu Extras/AirPort.menu"
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Preferences/com.apple.systemuiserver.plist
menuExtras -array-add "/System/Library/CoreServices/Menu Extras/AirPort.menu"
```

Note: If the remediation is run multiple times, multiple instances of the Wi-Fi status will appear after rebooting the system. Command-click and drag the unwanted icons off the menu bar

Additional Information:

AirPort is Apple's marketing name for its 802.11x based wireless network interfaces.

Option-click the Wifi icon in the menu bar to find out more information about the connected wireless network.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	12.6 Use of Secure Network Management and Communication Protocols Use secure network management and communication protocols (e.g., 802.1X, Wi-Fi Protected Access 2 (WPA2) Enterprise or greater).		●	●
v7	15.4 Disable Wireless Access on Devices if Not Required Disable wireless access on devices that do not have a business purpose for wireless access.			●
v7	15.5 Limit Wireless Access on Client Devices Configure wireless access on client machines that do have an essential wireless business purpose, to allow access only to authorized wireless networks and to restrict access to other wireless networks.			●

4.3 Ensure HTTP Server Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

macOS used to have a graphical front-end to the embedded Apache web server in the Operating System. Personal web sharing could be enabled to allow someone on another computer to download files or information from the user's computer. Personal web sharing from a user endpoint has long been considered questionable, and Apple has removed that capability from the GUI. Apache, however, is still part of the Operating System and can be easily turned on to share files and provide remote connectivity to an end-user computer. Web sharing should only be done through hardened web servers and appropriate cloud services.

Rationale:

Web serving should not be done from a user desktop. Dedicated webservers or appropriate cloud storage should be used. Open ports make it easier to exploit the computer.

Impact:

The web server is both a point of attack for the system and a means for unauthorized file transfers.

Audit:

Terminal Method:
Run the following command to verify that the HTTP server services are not currently enabled. This check does not reflect any auto-start settings, only whether the web server is currently enabled:

```
$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep -c "org.apache.httpd"

0
```

Remediation:

Terminal Method:
Run the following command to disable the HTTP server services:

```
$ sudo /usr/bin/sudo /bin/launchctl unload -w
/System/Library/LaunchDaemons/org.apache.httpd.plist
```

References:

1. https://www.stigviewer.com/stig/apple_macos_11_big_sur/2021-06-16/finding/V-230793

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

4.4 Ensure NFS Server Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

macOS can act as an NFS fileserver. NFS sharing could be enabled to allow someone on another computer to mount shares and gain access to information from the user's computer. File sharing from a user endpoint has long been considered questionable, and Apple has removed that capability from the GUI. NFSD is still part of the Operating System and can be easily turned on to export shares and provide remote connectivity to an end-user computer.

The etc/exports file contains the list of NFS shared directories. If the file exists, it is likely that NFS sharing has been enabled in the past or may be available periodically. As an additional check, the audit verifies that there is no /etc/exports file.

Rationale:

File serving should not be done from a user desktop. Dedicated servers should be used. Open ports make it easier to exploit the computer.

Impact:

The nfs server is both a point of attack for the system and a means for unauthorized file transfers.

Audit:

Terminal Method:
Run the following commands to verify that the NFS fileserver service is not enabled:

```
$ /usr/bin/sudo /bin/launchctl list | /usr/bin/grep -c com.apple.nfsd

0

$ /usr/bin/sudo /bin/cat /etc/exports

cat: /etc/exports: No such file or directory
```

Remediation:

Terminal Method:
Run the following command to disable the nfsd fileserver services:

```
$ /usr/bin/sudo /bin/launchctl disable system/com.apple.nfsd
```

Remove the exported Directory listing.

```
$ /usr/bin/sudo /bin/rm /etc/exports
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

5 System Access, Authentication and Authorization

The controls in this section are a combination of hardening controls that are not specifically in a System Preference pane. Many of these controls are only accessible using the Command Line or a Device Profile and not available in the Graphical User Interface. The Benchmark does contain simple, easy to follow instructions for technical staff to audit and implement recommended controls.

5.1 File System Permissions and Access Controls

File system permissions have always been part of computer security. There are several principles that are part of best practices for a POSIX-based system which are contained in this section. This section does not contain a complete list of every permission on a macOS System that might be problematic. Developers and use cases differ, and what some administrators who are long in the profession might consider a travesty a risk assessor steeped in BYOD trends may not give a second glance at. Here we document controls that should point out truly bad practices or anomalies which should be looked at and considered closely. Many of the controls are to mitigate the risk of privilege escalation attacks and data exposure to unauthorized parties.

5.1.1 Ensure Home Folders Are Secure (Automated)

Profile Applicability:

- Level 1

Description:

By default, macOS allows all valid users into the top level of every other user's home folder and restricts access to the Apple default folders within. Another user on the same system can see you have a "Documents" folder but cannot see inside it. This configuration does work for personal file sharing but can expose user files to standard accounts on the system.

The best parallel for Enterprise environments is that everyone who has a Dropbox account can see everything that is at the top level but can't see your pictures. Similarly with macOS, users can see into every new Directory that is created because of the default permissions.

Home folders should be restricted to access only by the user. Sharing should be used on dedicated servers or cloud instances that are managing access controls. Some environments may encounter problems if execute rights are removed as well as read and write. Either no access or execute only for group or others is acceptable.

Rationale:

Allowing all users to view the top level of all networked users' home folder may not be desirable since it may lead to the revelation of sensitive information.

Impact:

If implemented, users will not be able to use the "Public" folders in other users' home folders. "Public" folders with appropriate permissions would need to be set up in the /Shared folder.

Audit:

Terminal Method:
Run the following command to ensure that all home folders are secure:

```
$ /usr/bin/sudo /bin/ls -l /Users/ | grep -v Shared
```

The output for each home folder should be either `drwx------` or `drwx--x--x`

example:

```
$ /usr/bin/sudo /bin/ls -l /Users/

total 0
drwxr-xr-x+ 12 Guest       _guest  384 24 Jul 13:42 Guest
drwx--x--x+ 18 firstuser   staff   576 10 Aug 14:36 firstuser
drwx--x--x+ 15 seconduser  staff   480 10 Aug 09:16 seconduser
drwxrwxrwx+ 11 thirduser   staff   352 10 Aug 14:53 thirduser
drwxrw-rw-+ 11 fourthuser  staff   352 10 Aug 14:53 fourthuser
```

Remediation:

Terminal Method:
For each user, run the following command to secure all home folders:

```
$ /usr/bin/sudo /bin/chmod -R og-rwx /Users/<username>
```

Alternately, run the following command if there needs to be executable access for a home folder:

```
$ /usr/bin/sudo /bin/chmod -R og-rw /Users/<username>
```

example:

```
$ /usr/bin/sudo /bin/chmod -R og-rw /Users/thirduser/

$ /usr/bin/sudo /bin/chmod -R og-rwx /Users/fourthuser/

# /bin/ls -l /Users/

total 0
drwxr-xr-x+ 12 Guest       _guest  384 24 Jul 13:42 Guest
drwxrwxrwt   4 root        wheel   128 22 Jul 11:00 Shared
drwx--x--x+ 18 firstuser   staff   576 10 Aug 14:36 firstuser
drwx--x--x+ 15 seconduser  staff   480 10 Aug 09:16 seconduser
drwx--x--x+ 11 thirduser   staff   352 10 Aug 14:53 thirduser
drwx------+ 11 fourthuser  staff   352 10 Aug 14:53 fourthuser
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

5.1.2 Ensure System Integrity Protection Status (SIP) Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

System Integrity Protection is a security feature introduced in OS X 10.11 El Capitan. System Integrity Protection restricts access to System domain locations and restricts runtime attachment to system processes. Any attempt to inspect or attach to a system process will fail. Kernel Extensions are now restricted to /Library/Extensions and are required to be signed with a Developer ID.

Rationale:

Running without System Integrity Protection on a production system runs the risk of the modification of system binaries or code injection of system processes that would otherwise be protected by SIP.

Impact:

System binaries and processes could become compromised.

Audit:

Terminal Method:
Run the following command to verify that System Integrity Protection is enabled:

```
$ /usr/bin/sudo /usr/bin/csrutil status

`System Integrity Protection status: enabled.`
```

Remediation:

Terminal Method:
Perform the following steps to enable System Integrity Protection:

1. Reboot into the `Recovery Partition` (reboot and hold down `Command (⌘) + R`)
2. Select `Utilities`
3. Select `Terminal`
4. Run the following command:

```
$ /usr/bin/sudo /usr/bin/csrutil enable

Successfully enabled System Integrity Protection. Please restart the machine
for the changes to take effect.
```

5. Reboot the computer

Note: If SIP has been disabled, the admin should research why it was disabled. You might also want to assume that the operating system has been compromised. If you believe that, back up any files, and do a clean install to a known good Operating System It might be a better option to erase the Mac and reinstall the operating system. That is at your discretion.
Note: You cannot enable System Integrity Protection from the booted operating system. If the remediation is attempted in the booted OS and not the Recovery Partition the output will give the error `csrutil: failed to modify system integrity configuration. This tool needs to be executed from the Recovery OS.`

References:

1. https://developer.apple.com/documentation/security/disabling_and_enabling_system_integrity_protection
2. https://support.apple.com/en-us/HT204899

Additional Information:

Related to SIP controls, Library Validation is a security feature introduced in macOS 10.10 Yosemite. Library Validation protects processes from loading arbitrary libraries. This stops root from loading arbitrary libraries into any process (depending on SIP status), and keeps root from becoming more powerful. Security is strengthened, because some user processes can no longer be fooled to run additional code without root's explicit request, which may grant access to daemons that depend on Library Validation for secure validation of code identity.

With SIP enabled, Library Validation cannot be disabled. To test against a non-validated library, you will need to disabled SIP AND disable Library Validation.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.3 Address Unauthorized Software Ensure that unauthorized software is either removed from use on enterprise assets or receives a documented exception. Review monthly, or more frequently.	●	●	●
v8	2.6 Allowlist Authorized Libraries Use technical controls to ensure that only authorized software libraries, such as specific .dll, .ocx, .so, etc., files, are allowed to load into a system process. Block unauthorized libraries from loading into a system process. Reassess bi-annually, or more frequently.		●	●
v8	10.5 Enable Anti-Exploitation Features Enable anti-exploitation features on enterprise assets and software, where possible, such as Microsoft® Data Execution Prevention (DEP), Windows® Defender Exploit Guard (WDEG), or Apple® System Integrity Protection (SIP) and Gatekeeper™.		●	●
v7	2.6 Address unapproved software Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●

5.1.3 Ensure Apple Mobile File Integrity (AMFI) Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Apple Mobile File Integrity (AMFI) was first released in macOS 10.12. The daemon and service block attempts to run unsigned code. AMFI uses lanchd, code signatures, certificates, entitlements, and provisioning profiles to create a filtered entitlement dictionary for an app. AMFI is the macOS kernel module that enforces code-signing and library validation.

Rationale:

Apple Mobile File Integrity validates that application code is validated.

Impact:

Applications could be compromised with malicious code.

Audit:

Terminal Method:
Run the following command to verify that Apple Mobile File Integrity is enabled:

```
$ /usr/bin/sudo /usr/sbin/nvram -p | /usr/bin/grep -c
"amfi_get_out_of_my_way=1"

0
```

Note: AMFI cannot be disabled with SIP enabled, but a change attempt can be made that will appear successful, and report incorrectly as successful. If the AMFI audit fails, and the SIP audit passes, this is still an issue the admin should research.

Remediation:

Terminal Method:
Run the following command to enable the Apple Mobile File Integrity service:

```
$ /usr/bin/sudo /usr/sbin/nvram boot-args=""
```

References:

1. https://eclecticlight.co/2018/12/29/amfi-checking-file-integrity-on-your-mac/
2. https://github.com/usnistgov/macos_security/issues/39
3. https://github.com/usnistgov/macos_security/issues/40
4. https://www.naut.ca/blog/2020/11/13/forbidden-commands-to-liberate-macos/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.3 Address Unauthorized Software Ensure that unauthorized software is either removed from use on enterprise assets or receives a documented exception. Review monthly, or more frequently.	●	●	●
v8	2.6 Allowlist Authorized Libraries Use technical controls to ensure that only authorized software libraries, such as specific .dll, .ocx, .so, etc., files, are allowed to load into a system process. Block unauthorized libraries from loading into a system process. Reassess bi-annually, or more frequently.		●	●
v7	2.6 Address unapproved software Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●

5.1.4 Ensure Appropriate Permissions Are Enabled for System Wide Applications (Automated)

Profile Applicability:

- Level 1

Description:

Applications in the System Applications Directory (/Applications) should be world-executable since that is their reason to be on the system. They should not be world-writable and allow any process or user to alter them for other processes or users to then execute modified versions.

Rationale:

Unauthorized modifications of applications could lead to the execution of malicious code.

Impact:

Applications changed will no longer be world-writable. Depending on the environment, there will be different risk tolerances on each non-conforming application. Global changes should not be performed where mission-critical applications are misconfigured.

Audit:

Terminal Method:
Run the following command to verify that all applications have the correct permissions:

```
$ /usr/bin/sudo /usr/bin/find /Applications -iname "*\.app" -type d -perm -2
-ls | /usr/bin/wc -l | /usr/bin/xargs

0
```

Remediation:

Terminal Method:
Run the following command to change the permissions for each application that does not meet the requirements:

```
$ /usr/bin/sudo IFS=$'\n'
for apps in $( /usr/bin/find /Applications -iname "*\.app" -type d -perm -2
); do
  /bin/chmod -R o-w "$apps"
done
```

Note: Global changes should not be performed where mission-critical applications are part of the improperly permissioned applications.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

5.1.5 Ensure No World Writable Files Exist in the System Folder (Automated)

Profile Applicability:

- Level 1

Description:

Software sometimes insists on being installed in the `/System/Volumes/Data/System` Directory and has inappropriate world-writable permissions.

Macs with writable files in System should be investigated forensically. A file with open writable permissions is a sign of at best a rogue application. It could also be a sign of a computer compromise and a persistent presence on the system.

Rationale:

Folders in `/System/Volumes/Data/System` should not be world-writable. The audit check excludes the "Drop Box" folder that is part of Apple's default user template.

Impact:

Changing file permissions could disrupt the use of applications that rely on files in the System Folder with vulnerable permissions.

Audit:

Terminal Method:
Run the following command to check for directories in the /System folder that are world-writable:

```
$ /usr/bin/sudo /usr/bin/find /System/Volumes/Data/System -type d -perm -2 -
ls | /usr/bin/grep -v "Drop Box" | /usr/bin/wc -l | /usr/bin/xargs

0
```

Remediation:

Terminal Method:
Run the following command to set permissions so that folders are not world-writable in the /System folder:

```
$ /usr/bin/sudo IFS=$'\n'
for sysPermissions in $( /usr/bin/find /System/Volumes/Data/System -type d -
perm -2 | /usr/bin/grep -v "Drop Box" ); do
  /bin/chmod -R o-w "$sysPermissions"
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

5.1.6 Ensure No World Writable Files Exist in the Library Folder (Automated)

Profile Applicability:

- Level 2

Description:

Software sometimes insists on being installed in the `/System/Volumes/Data/Library Directory` and has inappropriate world-writable permissions.

Rationale:

Folders in `/System/Volumes/Data/Library` should not be world-writable. The audit check excludes the `/System/Volumes/Data/Library/Caches` and `/System/Volumes/Data/Library/Preferences/Audio/Data` folders where the sticky bit is set.

Audit:

Terminal Method:
Run the following to verify that no directories in the `/System/Volumes/Data/Library` folder are world-writable:

```
$ /usr/bin/sudo /usr/bin/find /System/Volumes/Data/Library -type d -perm -2 -
ls | /usr/bin/grep -v Caches | /usr/bin/grep -v /Preferences/Audio/Data |
/usr/bin/wc -l | /usr/bin/xargs

0
```

Remediation:

Terminal Method:
Run the following command to set permissions so that folders are not world-writable in the `/System/Volumes/Data/Library` folder:

```
$ /usr/bin/sudo IFS=$'\n'
for libPermissions in $( /usr/bin/find /System/Volumes/Data/Library -type d -
perm -2 | /usr/bin/grep -v Caches | /usr/bin/grep -v /Preferences/Audio/Data
); do
  /bin/chmod -R o-w "$libPermissions"
done
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

5.2 Password Management

Password security is an important part of general IT security where passwords are in use. For macOS, passwords are still much more widely used than other methods for account access. While there are other authentication and authorization methods for access from a macOS computer to organizational services, console access to the Mac is probably done using a password. This section contains password controls.

Apple has provided sufficient security controls to resist password attacks against the locked console, thus the CIS benchmark no longer recommends locking the keychain in addition to locking the console or display.

Recent updates based on research by NIST in SP800-63 call into question traditional password complexity and rotation requirements. Sticky notes are not a password management program, and password vault APIs are under increasing attack. Ideally, the user will remember their important passwords. The new understanding has informed changes to the previous password recommendations.

Length, threshold, and a yearly rotation requirement are the only scored controls below. Other controls will remain as unscored options. Passwords used for macOS are likely to also function as encryption keys for FileVault. Depending on the information confidentiality on FileVault volumes, stronger passwords may be required than are necessary to pass the controls in this Benchmark.

Before applying your organizations password policy, the existing password policy should be cleared so there is no out-dated or conflicting legacy settings in the password policy. A pre-existing password policly could result in false results. To clear the password policy before applying the newest options, use the command `/usr/bin/sudo /usr/bin/pwpolicy -clearaccountpolicies`

Apple-supported solutions for managing local passwords on macOS are to use either an XML file that contains password rules that are imported with pwpolicy or through the use of a profile. In either case, the controls in this section can be implemented with an organizationally-approved password policy.

Content is available where security hardening content is available and is native to Management suites and MDM tools.

Content also available here: https://github.com/ronc-LAemigre/macos-sec-config

NIST guidance on passwords starting at 5.1.1.1

https://pages.nist.gov/800-63-3/sp800-63b.html

Additional references:

- https://developer.apple.com/documentation/devicemanagement/passcode
- https://krypted.com/mac-security/programatically-setting-password-policies/
- https://www.macworld.co.uk/news/flaw-mac-t2-chip-passwords-3813616/

Note: The current method of creating and setting password policy is using the `pwpolicy -setglobalpolicy` command. That command has been deprecated by Apple, but is still in use in the current version of macOS. The Benchmark will continue to use this command line method for passwords until Apple removes it from the OS. Setting password policy with mobile configuration profiles is the preferred method going forward.

5.2.1 Ensure Password Account Lockout Threshold Is Configured (Automated)

Profile Applicability:

- Level 1

Description:

The account lockout threshold specifies the amount of times a user can enter an incorrect password before a lockout will occur.

Ensure that a lockout threshold is part of the password policy on the computer.

Rationale:

The account lockout feature mitigates brute-force password attacks on the system.

Impact:

The number of incorrect log on attempts should be reasonably small to minimize the possibility of a successful password attack, while allowing for honest errors made during a normal user log on.

The locked account will auto-unlock after a few minutes when bad password attempts stop. The computer will accept the still-valid password if remembered or recovered.

Audit:

Graphical Method:
Perform the following steps to ensure that the Password Account Threshold is set to less than or equal to 5:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Max Failed Attempts` set to ≤ `5`

Terminal Method:
Run the following command to verify that the number of failed attempts is less than or equal to 5:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -getaccountpolicies 2> /dev/null |
/usr/bin/tail +2 | /usr/bin/xmllint --xpath
'//dict/key[text()="policyAttributeMaximumFailedAuthentications"]/following-
sibling::integer[1]/text()' -
```

The output should be ≤ `5`

Remediation:

Terminal Method:
Run the following command to set the maximum number of failed login attempts to less than or equal to 5:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"maxFailedLoginAttempts=<value≤5>"
```

Note: When the account lockout threshold is set with `pwpolicy`, it will also set a reset value to `policyAttributeMinutesUntilFailedAuthenticationReset` that defaults to 1 minute. You can change this value with the command:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"policyAttributeMinutesUntilFailedAuthenticationReset=<value in minutes>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"maxFailedLoginAttempts=5"

/usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"policyAttributeMinutesUntilFailedAuthenticationReset=10"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mobiledevice.passwordpolicy`
2. The key to include is `maxFailedAttempts`
3. The key must be set to `<integer><value≤5></integer>`

Note: When setting the lockout threshold with a mobile configuration profile there is no default reset to the lockout. To set the reset value use the key `autoEnableInSeconds` and set the key to `<integer><value in seconds></integer>`.
Note: The profile method is the preferred method for setting password policy since `-setglobalpolicy` in `pwpolicy` is deprecated and will likely be removed in a future macOS release.

References:

1. CIS Password Policy - https://workbench.cisecurity.org/communities/113

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.	●	●	●
v7	16.7 Establish Process for Revoking Access Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor . Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.		●	●

5.2.2 Ensure Password Minimum Length Is Configured (Automated)

Profile Applicability:

- Level 1

Description:

A minimum password length is the fewest number of characters a password can contain to meet a system's requirements.

Ensure that a minimum of a 15-character password is part of the password policy on the computer.

Where the confidentiality of encrypted information in FileVault is more of a concern, requiring a longer password or passphrase may be sufficient rather than imposing additional complexity requirements that may be self-defeating.

Rationale:

Information systems that are not protected with strong password schemes including passwords of minimum length provide a greater opportunity for attackers to crack the password and gain access to the system.

Impact:

Short passwords can be easily attacked.

Audit:

Graphical Method:
Perform the following steps to ensure that the Password Account Threshold is set to greater than or equal to 15:

1. Open System Preferences
2. Select Profiles
3. Verify that an installed profile has `Min Password Length` set to ≥ `15`

Terminal Method:
Run the following command to verify that the password length is greater than or equal to 15:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep -e
"policyAttributePassword matches" | /usr/bin/cut -b 46-53 | /usr/bin/cut -
d',' -f1 | /usr/bin/cut -d'{' -f2
```

The output value should be ≥ `15`

Remediation:

Terminal Method:
Run the following command to set the password length to greater than or equal to 15:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"minChars=<value≥15>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"minChars=15"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mobiledevice.passwordpolicy`
2. The key to include is `minLength`
3. The key must be set to `<integer><value≥15></integer>`

Note: The profile method is the preferred method for setting password policy since `-setglobalpolicy` in `pwpolicy` is deprecated and will likely be removed in a future macOS release.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

5.2.3 Ensure Complex Password Must Contain Alphabetic Characters Is Configured (Manual)

Profile Applicability:

- Level 2

Description:

Complex passwords contain one character from each of the following classes: English uppercase letters, English lowercase letters, Westernized Arabic numerals, and non-alphanumeric characters.

Ensure that an Alphabetic character is part of the password policy on the computer.

Rationale:

The more complex a password, the more resistant it will be against persons seeking unauthorized access to a system.

Impact:

Password policy should be in effect to reduce the risk of exposed services being compromised easily through dictionary attacks or other social engineering attempts.

Audit:

Graphical Method:
Perform the following steps to ensure that the passwords must contain at least 1 alphabetic characters:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Requires Alphanumeric` set to `True`

Terminal Method:
Run the following command to verify that the password requires at least one letter:

```
$ /usr/bin/sudo pref1=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep
-e "Contain at least one number and one alphabetic character." | cut -b 13-
68) && pref2=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep -A1
minimumLetters | /usr/bin/tail -1 | /usr/bin/cut -d'>' -f2 | /usr/bin/cut -d
'<' -f1) && if [ "$pref1" = "Contain at least one number and one alphabetic
character" ]; then echo "true"; elif [[ "$pref2" != "" && pref2 -ge 1 ]];
then echo "true"; else echo "false"; fi

true
```

Remediation:

Terminal Method:
Run the following command to set the that passwords must contain at least one letter:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy -
setaccountpolicies "requiresAlpha=<value≥1>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"requiresAlpha=1"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mobiledevice.passwordpolicy`
2. The key to include is `requireAlphanumeric`
3. The key must be set to `<true/>`

Note: This profile sets a requirement of both an alphabetical and a numeric character.
Note: The profile method is the preferred method for setting password policy since `-setglobalpolicy` in `pwpolicy` is deprecated and will likely be removed in a future macOS release.

Additional Information:

Note: The CIS macOS community has decided to not require the additional password complexity settings (Recommendations `5.3` - `5.6`). Because of that, we have left the complexity recommendations as a manual assessment. Since there are a large amount of admins in the greater macOS world that do need these settings, we include both the guidance for the proper setting as well as probes for CIS-CAT to test.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

5.2.4 Ensure Complex Password Must Contain Numeric Character Is Configured (Manual)

Profile Applicability:

- Level 2

Description:

Complex passwords contain one character from each of the following classes: English uppercase letters, English lowercase letters, Westernized Arabic numerals, and non-alphanumeric characters.

Ensure that a number or numeric value is part of the password policy on the computer.

Rationale:

The more complex a password, the more resistant it will be against persons seeking unauthorized access to a system.

Impact:

Password policy should be in effect to reduce the risk of exposed services being compromised easily through dictionary attacks or other social engineering attempts.

Audit:

Graphical Method:
Perform the following steps to ensure that the passwords must contain at least 1 numeric characters:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Requires Alphanumeric` set to `True`

Terminal Method:
Run the following command to verify that passwords require at least one number:

```
$ /usr/bin/sudo pref1=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep
-e "Contain at least one number and one alphabetic character." | cut -b 13-
68) && pref2=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep -A1
minimumNumericCharacters | /usr/bin/tail -1 | /usr/bin/cut -d'>' -f2 |
/usr/bin/cut -d '<' -f1) && if [ "$pref1" = "Contain at least one number and
one alphabetic character" ]; then echo "true"; elif [[ "$pref2" != "" &&
pref2 -ge 1 ]]; then echo "true"; else echo "false"; fi

true
```

Remediation:

Terminal Method:
Run the following command to set passwords to require at least one number:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy -
setaccountpolicies "requiresNumeric=<value≥1>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"requiresNumeric=2"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mobiledevice.passwordpolicy`
2. The key to include is `requireAlphanumeric`
3. The key must be set to `<true/>`

Note: This profile sets a requirement of both an alphabetical and a numeric character.
Note: The profile method is the preferred method for setting password policy since `-setglobalpolicy` in `pwpolicy` is deprecated and will likely be removed in a future macOS release.

Additional Information:

Note: The CIS macOS community has decided to not require the additional password complexity settings (Recommendations `5.3` - `5.6`). Because of that, we have left the complexity recommendations as a manual assessment. Since there are a large amount of admins in the greater macOS world that do need these settings, we include both the guidance for the proper setting as well as probes for CIS-CAT to test.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

5.2.5 Ensure Complex Password Must Contain Special Character Is Configured (Manual)

Profile Applicability:

- Level 2

Description:

Complex passwords contain one character from each of the following classes: English uppercase letters, English lowercase letters, Westernized Arabic numerals, and non-alphanumeric characters. Ensure that a special character is part of the password policy on the computer.

Rationale:

The more complex a password, the more resistant it will be against persons seeking unauthorized access to a system.

Impact:

Password policy should be in effect to reduce the risk of exposed services being compromised easily through dictionary attacks or other social engineering attempts.

Audit:

Graphical Method:
Perform the following steps to ensure that the passwords must contain at least 1 special characters:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Min Complex Length` set to ≥ `1`

Terminal Method:
Run the following command to set verify that the password requires at least one special character:

```
$ /usr/bin/sudo pref1=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep
-e "policyAttributePassword matches '(.*[^a-zA-Z0-9].*){1,}'" | cut -b 12-67)
&& pref2=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep -A1
minimumSymbols | /usr/bin/tail -1 | /usr/bin/cut -d'>' -f2 | /usr/bin/cut -d
'<' -f1) && if [ "$pref1" = "policyAttributePassword matches '(.*[^a-zA-Z0-
9].*){1,}'" ]; then echo "true"; elif [[ "$pref2" != "" && pref2 -ge 1 ]];
then echo "true"; else echo "false"; fi

true
```

Remediation:

Terminal Method:
Run the following command to set passwords to require at least one special character:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy -
setaccountpolicies "requiresSymbol=<value≥1>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"requiresSymbol=1"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mobiledevice.passwordpolicy`
2. The key to include is `minComplexChars`
3. The key must be set to `<integer><value≥1></integer>`

Note: The profile method is the preferred method for setting password policy since `-setglobalpolicy` in `pwpolicy` is deprecated and will likely be removed in a future macOS release.

Additional Information:

Note: The CIS macOS community has decided to not require the additional password complexity settings (Recommendations `5.3` - `5.6`). Because of that, we have left the complexity recommendations as a manual assessment. Since there are a large amount of admins in the greater macOS world that do need these settings, we include both the guidance for the proper setting as well as probes for CIS-CAT to test.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

5.2.6 Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured (Manual)

Profile Applicability:

- Level 2

Description:

Complex passwords contain one character from each of the following classes: English uppercase letters, English lowercase letters, Westernized Arabic numerals, and non-alphanumeric characters.

Ensure that both uppercase and lowercase letters are part of the password policy on the computer.

Rationale:

The more complex a password, the more resistant it will be against persons seeking unauthorized access to a system.

Impact:

Password policy should be in effect to reduce the risk of exposed services being compromised easily through dictionary attacks or other social engineering attempts.

Audit:

Terminal Method:
Run the following command to verify that the password requires an upper and lower case letter:

```
$ /usr/bin/sudo pref1=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep
-A1 minimumMixedCaseCharacters | /usr/bin/tail -1 | /usr/bin/cut -d'>' -f2 |
/usr/bin/cut -d '<' -f1) && if [[ "$pref2" != "" && pref2 -ge 1 ]]; then echo
"true"; else echo "false"; fi

true
```

Remediation:

Terminal Method:
Run the following command to set passwords to require at upper and lower case letter:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"requiresMixedCase=<value≥1>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"requiresMixedCase=1"
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

5.2.7 Ensure Password Age Is Configured (Automated)

Profile Applicability:

- Level 1

Description:

Over time, passwords can be captured by third parties through mistakes, phishing attacks, third-party breaches, or merely brute-force attacks. To reduce the risk of exposure and to decrease the incentives of password reuse (passwords that are not forced to be changed periodically generally are not ever changed), users should reset passwords periodically. This control uses 365 days as the acceptable value. Some organizations may be more or less restrictive. This control mainly exists to mitigate against password reuse of the macOS account password in other realms that may be more prone to compromise. Attackers take advantage of exposed information to attack other accounts.

Rationale:

Passwords should be changed periodically to reduce exposure.

Impact:

Required password changes will lead to some locked computers requiring admin assistance.

Audit:

Graphical Method:
Perform the following steps to ensure that the passwords expire after at most 365 days:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Max Age (days)` set to ≤ `365`

Terminal Method:
Run the following command to verify that the password expires after at most 365 days:

```
$ /usr/bin/sudo pref1=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep
-A1 policyAttributeExpiresEveryNDays | /usr/bin/tail -1 | /usr/bin/cut -d'>'
-f2 | /usr/bin/cut -d '<' -f1) && pref2=$(/usr/bin/pwpolicy -
getaccountpolicies | /usr/bin/grep -A1 policyAttributeDaysUntilExpiration |
/usr/bin/tail -1 | /usr/bin/cut -d'>' -f2 | /usr/bin/cut -d '<' -f1) && if [[
"$pref1" != "" && pref1 -le 365 ]]; then echo "true"; elif [[ "$pref2" != ""
&& pref2 -le 365 ]]; then echo "true"; else echo "false"; fi

true
```

Remediation:

Terminal Method:
Run the following command to require that passwords expire after at most 365 days:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"maxMinutesUntilChangePassword=<value≤525600>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"maxMinutesUntilChangePassword=43200"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mobiledevice.passwordpolicy`
2. The key to include is `maxPINAgeInDays`
3. The key must be set to `<integer><value≥365></integer>`

Note: The profile method is the preferred method for setting password policy since `-setglobalpolicy` in `pwpolicy` is deprecated and will likely be removed in a future macOS release.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.3 Disable Dormant Accounts Delete or disable any dormant accounts after a period of 45 days of inactivity, where supported.	●	●	●
v7	16.9 Disable Dormant Accounts Automatically disable dormant accounts after a set period of inactivity.	●	●	●

5.2.8 Ensure Password History Is Configured (Automated)

Profile Applicability:

- Level 1

Description:

Over time, passwords can be captured by third parties through mistakes, phishing attacks, third-party breaches, or merely brute-force attacks. To reduce the risk of exposure and to decrease the incentives of password reuse (passwords that are not forced to be changed periodically generally are not ever changed), users must reset passwords periodically. This control ensures that previous passwords are not reused immediately by keeping a history of previous password hashes. Ensure that password history checks are part of the password policy on the computer. This control checks whether a new password is different than the previous 15. The latest NIST guidance based on exploit research referenced in this section details how one of the greatest risks is password exposure rather than password cracking. Passwords should be changed to a new unique value whenever a password might have been exposed to anyone other than the account holder. Attackers have maintained persistent control based on predictable password change patterns and substantially different patterns should be used in case of a leak.

Rationale:

Old passwords should not be reused.

Impact:

Required password changes will lead to some locked computers requiring admin assistance.

Audit:

Graphical Method:
Perform the following steps to ensure that the password is not the same as at least the last 15 passwords:

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Max History Kept` set to ≥ `15`

Terminal Method:
Run the following command to verify that the password is required to be different from at least the last 15 passwords:

```
$ /usr/bin/sudo pref1=$(/usr/bin/pwpolicy -getaccountpolicies | /usr/bin/grep
-A1 policyAttributePasswordHistoryDepth | /usr/bin/tail -1 | /usr/bin/cut -
d'>' -f2 | /usr/bin/cut -d '<' -f1) && pref2=$(/usr/bin/pwpolicy -
getaccountpolicies | /usr/bin/grep -A1 policyAttributePasswordHistoryDepth |
/usr/bin/tail -1 | /usr/bin/cut -d'>' -f2 | /usr/bin/cut -d '<' -f1) && if [[
"$pref1" != "" && pref1 -ge 15 ]]; then echo "true"; elif [[ "$pref2" != ""
&& pref2 -ge 15 ]]; then echo "true"; else echo "false"; fi

true
```

Remediation:

Terminal Method:
Run the following command to require that the password must to be different from at least the last 15 passwords:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"usingHistory=<value≥15>"
```

example:

```
$ /usr/bin/sudo /usr/bin/pwpolicy -n /Local/Default -setglobalpolicy
"usingHistory=15"
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.mobiledevice.passwordpolicy`
2. The key to include is `pinHistory`
3. The key must be set to `<integer><value≥15></integer>`

Note: The profile method is the preferred method for setting password policy since `-setglobalpolicy` in `pwpolicy` is deprecated and will likely be removed in a future macOS release.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

5.3 Ensure the Sudo Timeout Period Is Set to Zero (Automated)

Profile Applicability:

- Level 1

Description:

The sudo command allows the user to run programs as the root user. Working as the root user allows the user an extremely high level of configurability within the system. This control, along with the control to use a separate timestamp for each tty, limits the window where an unauthorized user, process, or attacker could utilize legitimate credentials that are valid for longer than required.

Rationale:

The `sudo` command stays logged in as the root user for five minutes before timing out and re-requesting a password. This five-minute window should be eliminated since it leaves the system extremely vulnerable. This is especially true if an exploit were to gain access to the system, since they would be able to make changes as a root user.

Impact:

This control has a serious impact where users often have to use sudo. It is even more of an impact where users have to use sudo multiple times in quick succession as part of normal work processes. Organizations with that common use case will likely find this control too onerous and are better to accept the risk of not requiring a 0 grace period.

In some ways the use of sudo -s, which is undesirable, is better than a long grace period since that use does change the hash to show that it is a root shell rather than a normal shell where sudo commands will be implemented without a password.

Audit:

Terminal Method:
Perform the following to verify the sudo timeout period:

```
$ /usr/bin/sudo /usr/bin/sudo -V | /usr/bin/grep -c "Authentication timestamp
timeout: 0.0 minutes"

1
```

Run the following commands to verify that the root is the owner of the `/etc/sudoers.d` folder, and that wheel is the group

```
$ /usr/bin/stat /etc/sudoers.d

16777229 19662948 drwxr-xr-x 2 root wheel 0 64 "Jun  7 23:12:24 2022" "May  9
17:30:48 2022" "Jun  7 23:12:24 2022" "May  9 17:30:48 2022" 4096 0 0
/etc/sudoers.d
```

Remediation:

Terminal Method:
Run the following command to edit the sudo settings:

```
$ /usr/bin/sudo /usr/sbin/visudo -f /etc/sudoers.d/<configuration file name>
```

example: `$ /usr/bin/sudo /usr/sbin/visudo -f /etc/sudoers.d/10_cissudoconfiguration`
Note: Unlike other Unix and/or Linux distros, macOS will ignore configuration files in the sudoers.d folder that contain a `.` so do not add a file extension to the configuration file.
Add the line `Defaults timestamp_timeout=0` to the configuration file.
If /etc/sudoers.d/ is not owned by root or in the wheel group, run the following to change ownership and group:

```
$ /usr/bin/sudo /usr/sbin/chown -R root:wheel /etc/security/sudoers.d/
```

Additional Information:

In previous iterations and OS versions of the macOS Benchmark, the guidance was to edit the sudoers file directly. While this would properly configure the OS, any update would change the settings back to the default configuration. Creating a configuration file in the `/etc/sudoers.d/` folder will not be modified on an OS update and will keep the proper configuration.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.3 Configure Automatic Session Locking on Enterprise Assets Configure automatic session locking on enterprise assets after a defined period of inactivity. For general purpose operating systems, the period must not exceed 15 minutes. For mobile end-user devices, the period must not exceed 2 minutes.	●	●	●
v7	16.11 Lock Workstation Sessions After Inactivity Automatically lock workstation sessions after a standard period of inactivity.	●	●	●

5.4 Ensure a Separate Timestamp Is Enabled for Each User/tty Combo (Automated)

Profile Applicability:

- Level 1

Description:

Using tty tickets ensures that a user must enter the sudo password in each Terminal session.

With sudo versions 1.8 and higher, introduced in 10.12, the default value is to have tty tickets for each interface so that root access is limited to a specific terminal. The default configuration can be overwritten or not configured correctly on earlier versions of macOS.

Rationale:

In combination with removing the sudo timeout grace period, a further mitigation should be in place to reduce the possibility of a background process using elevated rights when a user elevates to root in an explicit context or tty.

Additional mitigation should be in place to reduce the risk of privilege escalation of background processes.

Impact:

This control should have no user impact. Developers or installers may have issues if background processes are spawned with different interfaces than where sudo was executed.

Audit:

Terminal Method:
Run the following commands to verify that the default sudoers controls are in place with explicit tickets per tty:

```
$ /usr/bin/sudo /usr/bin/sudo -V | /usr/bin/grep -c "Type of authentication
timestamp record: tty"

1
```

Remediation:

Terminal Method:
Run the following command to edit the sudo settings:

```
$ /usr/bin/sudo /usr/sbin/visudo -f /etc/sudoers.d/<configuration file name>
```

example: `$ /usr/bin/sudo /usr/sbin/visudo -f /etc/sudoers.d/10_cissudoconfiguration`
Note: Unlike other Unix and/or Linux distros, macOS will ignore configuration files in the sudoers.d folder that contain a `.` so do not add a file extension to the configuration file.
Add the line `Defaults timestamp_type=tty` to the configuration file.
Note: The `Defaults timestamp_type=tty` line can be added to an existing configuration file or a new one. That will depend on your organization's preference and works either way.

Default Value:

If no value is set, the default value of tty_tickets enabled will be used.

Additional Information:

In previous iterations and OS versions of the macOS Benchmark, the guidance was to edit the sudoers file directly. While this would properly configure the OS, any update would change the settings back to the default configuration. Creating a configuration file in the `/etc/sudoers.d/` folder will not be modified on an OS update and will keep the proper configuration.

With the configuration file, there is no need to remove the `Defaults !tty_tickets` line from the `visudo` settings. The configuration file will take precedent.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.3 Configure Automatic Session Locking on Enterprise Assets Configure automatic session locking on enterprise assets after a defined period of inactivity. For general purpose operating systems, the period must not exceed 15 minutes. For mobile end-user devices, the period must not exceed 2 minutes.	●	●	●
v7	16.11 Lock Workstation Sessions After Inactivity Automatically lock workstation sessions after a standard period of inactivity.	●	●	●

5.5 Ensure the "root" Account Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

The root account is a superuser account that has access privileges to perform any actions and read/write to any file on the computer. With some versions of Linux, the system administrator may commonly use the root account to perform administrative functions.

Rationale:

Enabling and using the root account puts the system at risk since any successful exploit or mistake while the root account is in use could have unlimited access privileges within the system. Using the `sudo` command allows users to perform functions as a root user while limiting and password protecting the access privileges. By default the root account is not enabled on a macOS computer. An administrator can escalate privileges using the `sudo` command (use `-s` or `-i` to get a root shell).

Impact:

Some legacy POSIX software might expect an available root account.

Audit:

Graphical Method:
Perform the following steps to ensure that the root user is not enabled:

1. Open `/System/Library/CoreServices/Applications/Directory Utility`
2. Click the lock icon to unlock the service
3. Click `Edit` in the `menu bar`
4. Verify that the menu shows `Enable Root User`, not `Disable Root User`

Terminal Method:
Run the following command to verify the the root user has not been enabled:

```
$ /usr/bin/sudo /usr/bin/dscl . -read /Users/root AuthenticationAuthority


No such key: AuthenticationAuthority
```

Remediation:

Graphical Method:
Perform the following steps to ensure that the root user is disabled:

1. Open `/System/Library/CoreServices/Applications/Directory Utility`
2. Click the lock icon to unlock the service
3. Click `Edit` in the `menu bar`
4. Click `Disable Root User`

Terminal Method:
Run the following command to disable the root user:

```
$ /usr/bin/sudo /usr/sbin/dsenableroot -d

username = root
user password:
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●

5.6 Ensure Automatic Login Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

The automatic login feature saves a user's system access credentials and bypasses the login screen. Instead, the system automatically loads to the user's desktop screen.

Rationale:

Disabling automatic login decreases the likelihood of an unauthorized person gaining access to a system.

Impact:

If automatic login is not disabled, an unauthorized user could gain access to the system without supplying any credentials.

Audit:

Graphical Method:
Perform the following steps to ensure that automatic login is not enabled:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Click the lock to authenticate
4. Select `Login Options`
5. Verify that `Automatic login` is set to `Off`

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `DisableAutoLoginClient = 1` set

Terminal Method:
Run the following command to verify that automatic login has not been enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.loginwindow')
\
  .objectForKey('com.apple.login.mcx.DisableAutoLoginClient'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.loginwindow')
\
  .objectForKey('autoLoginUser'))
  if ( pref1 == 1 || pref2 == null ) {
    return("true")
  } else {
    return("false")
  }
}
EOS
```

Remediation:

Graphical Method:
Perform the following steps to set automatic login to off:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Click the lock to authenticate
4. Select `Login Options`
5. Select `Automatic login` and set it to `Off`

Terminal Method:
Run the following command to disable automatic login:

```
$ /usr/bin/sudo /usr/bin/defaults delete
/Library/Preferences/com.apple.loginwindow autoLoginUser
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.loginwindow`
2. The key to include is `com.apple.login.mcx.DisableAutoLoginClient`
3. The key must be set to `<true/>`

Note: If both the profile is enabled and a user is set to autologin, the profile will take precedent. In this case, the graphical or terminal remediation method should also be applied in case the profile is ever removed.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.7 Manage Default Accounts on Enterprise Assets and Software Manage default accounts on enterprise assets and software, such as root, administrator, and other pre-configured vendor accounts. Example implementations can include: disabling default accounts or making them unusable.	●	●	●
v7	4.2 Change Default Passwords Before deploying any new asset, change all default passwords to have values consistent with administrative level accounts.	●	●	●

5.7 Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session (Automated)

Profile Applicability:

- Level 1

Description:

macOS has a privilege that can be granted to any user that will allow that user to unlock active user's sessions.

Rationale:

Disabling the administrator's and/or user's ability to log into another user's active and locked session prevents unauthorized persons from viewing potentially sensitive and/or personal information.

Impact:

While Fast user switching is a workaround for some lab environments, especially where there is even less of an expectation of privacy, this setting change may impact some maintenance workflows.

Audit:

Terminal Method:
Run the following command to verify that a user cannot log into another user's active and/or locked session:

```
$ /usr/bin/sudo /usr/bin/security authorizationdb read
system.login.screensaver 2>&1 | /usr/bin/grep -c 'use-login-window-ui'

1
```

Remediation:

Terminal Method:
Run the following command to disable a user logging into another user's active and/or locked session:

```
$ /usr/bin/sudo /usr/bin/security authorizationdb write
system.login.screensaver use-login-window-ui

YES (0)
```

References:

1. https://derflounder.wordpress.com/2014/02/16/managing-the-authorization-database-in-os-x-mavericks/
2. https://www.jamf.com/jamf-nation/discussions/18195/system-login-screensaver

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.3 Configure Automatic Session Locking on Enterprise Assets Configure automatic session locking on enterprise assets after a defined period of inactivity. For general purpose operating systems, the period must not exceed 15 minutes. For mobile end-user devices, the period must not exceed 2 minutes.	●	●	●
v7	16.11 Lock Workstation Sessions After Inactivity Automatically lock workstation sessions after a standard period of inactivity.	●	●	●

5.8 Ensure a Login Window Banner Exists (Automated)

Profile Applicability:

- Level 2

Description:

A Login window banner warning informs the user that the system is reserved for authorized use only. It enforces an acknowledgment by the user that they have been informed of the use policy in the banner if required. The system recognizes either the `.txt` and the `.rtf` formats.

Rationale:

An access warning may reduce a casual attacker's tendency to target the system. Access warnings may also aid in the prosecution of an attacker by evincing the attacker's knowledge of the system's private status, acceptable use policy, and authorization requirements.

Impact:

Users will have to click on the window with the Login text before logging into the computer.

Audit:

Terminal Method:
Run the following command to verify the login window text:

```
$ /usr/bin/sudo /bin/cat /Library/Security/PolicyBanner.*
```

If the output includes `no matches found: /Library/Security/PolicyBanner.*` the system is not compliant.
Run the following to verify permissions of the policy banner file:

```
$ /usr/bin/stat -f %A /Library/Security/PolicyBanner.*
```

The output should have `4` as the 3rd digit.
If there is an output, then the policy banner will not display.

example:

```
$ /usr/bin/sudo /bin/cat /Library/Security/PolicyBanner.txt

Center for Internet Security Test Message

$ /usr/bin/sudo /bin/cat /Library/Security/PolicyBanner.rtf

{\rtf1\ansi\ansicpg1252\cocoartf1561\cocoasubrtf610
{\fonttbl\f0\fswiss\fcharset0 Helvetica;}
{\colortbl;\red255\green255\blue255;}
{\*\expandedcolortbl;;}
\margl1440\margr1440\vieww10800\viewh8400\viewkind0
\pard\tx566\tx1133\tx1700\tx2267\tx2834\tx3401\tx3968\tx4535\tx5102\tx5669\tx
6236\tx6803\pardirnatural\partightenfactor0

\f0\fs24 \cf0 Center for Internet Security Test Message}

$ /usr/bin/sudo /bin/cat /Library/Security/PolicyBanner.*

{\rtf1\ansi\ansicpg1252\cocoartf1561\cocoasubrtf610
{\fonttbl\f0\fswiss\fcharset0 Helvetica;}
{\colortbl;\red255\green255\blue255;}
{\*\expandedcolortbl;;}
\margl1440\margr1440\vieww10800\viewh8400\viewkind0
\pard\tx566\tx1133\tx1700\tx2267\tx2834\tx3401\tx3968\tx4535\tx5102\tx5669\tx
6236\tx6803\pardirnatural\partightenfactor0

\f0\fs24 \cf0 Center for Internet Security Test Message}Center for Internet
Security Test Message

$ /usr/bin/sudo stat -f %A /Library/Security/PolicyBanner.*

644
```

Remediation:

Terminal Method:
Run the following commands to create or edit the login window text and set the proper permissions:
Edit (or create) a `PolicyBanner.txt` or `PolicyBanner.rtf` file, in the `/Library/Security/` folder, to include the required login window banner text.
Perform the following to set permissions on the policy banner file:

```
$ /usr/bin/sudo /usr/sbin/chown o+r /Library/Security/PolicyBanner.txt

$ /usr/bin/sudo /usr/sbin/chown o+r /Library/Security/PolicyBanner.rtf
```

Note: If your organization uses an `.rtfd` file to set the policy banner, run `$ /usr/bin/sudo /usr/sbin/chown -R o+rx /Library/Security/PolicyBanner.rtfd` to update the permissions.

References:

1. https://support.apple.com/en-au/HT202277

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

5.9 Ensure Users' Accounts Do Not Have a Password Hint (Automated)

Profile Applicability:

- Level 1

Description:

Password hints help the user recall their passwords for various systems and/or accounts. In most cases, password hints are simple and closely related to the user's password.

Rationale:

Password hints that are closely related to the user's password are a security vulnerability, especially in the social media age. Unauthorized users are more likely to guess a user's password if there is a password hint. The password hint is very susceptible to social engineering attacks and information exposure on social media networks.

Audit:

Terminal Method:
Run the following command to verify that no users have a password hint:

```
$ /usr/bin/sudo /usr/bin/dscl . -list /Users hint
```

The output will list all users. If there are any text listed with the user, then the machine is not compliant.
example:

```
$ /usr/bin/sudo /usr/bin/dscl . -list /Users hint . -list /Users hint

firstuser    passwordhint
seconduser   passwordhint2
thirduser
fourthuser
Guest
```

Remediation:

Graphical Method:
Perform the following steps to remove a user's password hint:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Select the Current User
4. Select `Change Password`
5. Change the password and ensure that no text is entered in the Password hint box

Note: This will only change the currently logged-in user's password, and not any others that are not compliant on the Mac. Use the terminal method if multiple users are not in compliance.
Terminal Method:
Run the following command to remove a user's password hint:

```
$ /usr/bin/sudo /usr/bin/dscl . -list /Users hint . -delete /Users/<username>
hint
```

example:

```
$ /usr/bin/sudo /usr/bin/dscl . -list /Users hint . -delete /Users/firstuser
hint

$ /usr/bin/sudo /usr/bin/dscl . -list /Users hint . -delete /Users/seconduser
hint
```

Additional Information:

Organizations might consider entering an organizational help desk phone number or other text (such as a warning to the user). A help desk number is only appropriate for organizations with trained help desk personnel that are validating user identities for password resets.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

5.10 Ensure Secure Keyboard Entry Terminal.app Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Secure Keyboard Entry prevents other applications on the system and/or network from detecting and recording what is typed into Terminal. Unauthorized applications and malicious code could intercept keystrokes entered in the Terminal.

Rationale:

Enabling Secure Keyboard Entry minimizes the risk of a key logger from detecting what is entered in Terminal.

Impact:

Enabling this in Terminal would prevent an application that is otherwise validly intercepting keyboard input from intercepting that input in Terminal.app. This could impact productivity tools.

Audit:

Graphical Method:
Perform the following steps to ensure that keyboard entries are secure in Terminal:

1. Open the `Applications` folder
2. Open the `Utilities` folder
3. Open `Terminal`
4. Select `Terminal` in the Menu Bar
5. Verify that `Secure Keyboard Entry` is enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `SecureKeyboardEntry` is set to `1`

Terminal Method:
For each user, run the following command to verify that keyboard entries in Terminal are secured:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read -app Terminal
SecureKeyboardEntry

1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read -app Terminal
SecureKeyboardEntry

0

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read -app Terminal
SecureKeyboardEntry

1
```

In the above example the user seconduser is compliant, and the user firstuser is not compliant.
or
Run the following command to verify that a profile is installed that enables secure keyboard entry in Terminal:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.Terminal')\
.objectForKey('SecureKeyboardEntry').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable secure keyboard entries in Terminal:

1. Open the `Applications` folder
2. Open the `Utilities` folder
3. Open `Terminal`
4. Select `Terminal` in the Menu Bar
5. Set `Secure Keyboard Entry` to enabled

Terminal Method:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write -app Terminal
SecureKeyboardEntry -bool true
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write -app Terminal
SecureKeyboardEntry -bool true
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.Terminal`
2. The key to include is `SecureKeyboardEntry`
3. The key must be set to `<true/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://support.apple.com/en-ca/guide/terminal/trml109/mac
2. https://developer.apple.com/library/archive/technotes/tn2150/_index.html
3. https://krypted.com/mac-os-x/secure-keyboard-entry-on-macos/

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v7	4.1 Maintain Inventory of Administrative Accounts Use automated tools to inventory all administrative accounts, including domain and local accounts, to ensure that only authorized individuals have elevated privileges.		●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

6 User Accounts and Environment

Account management is a central part of security for any computer system, including macOS. General practices should be followed to ensure that all accounts on a system are still needed, and that default accounts have been removed. Users with administrator roles should have distinct accounts for both administrator functions as well as day-to-day work where the passwords are different and known only by the user assigned to the account. Accounts with elevated privileges should not be easily discerned from the account name from standard accounts.

When any computer system is added to a directory system there are additional controls available, including user account management, that are not available in a standalone computer. One of the drawbacks is the local computer is no longer in control of the accounts that can access or manage it if given permission. For macOS, if the computer is connected to a directory, any standard user can now log into the computer at console, which by default may be desirable or not depending on the use case. If an administrator group is allowed to administer the local computer, the membership of that group is controlled completely in the directory.

macOS computers connected to a directory should be configured so that the risk is appropriate for the mission use of the computer. Only those accounts that require local authentication should be allowed, and only required administrator accounts should be in the local administrator group. Authenticated users for console access and domain admins for administration may be too broad or too limited.

6.1 Accounts Preferences Action Items

Proper account management is critical to computer security. Many options and settings in the Account System Preference Pane can be used to increase the security of the Mac.

6.1.1 Ensure Login Window Displays as Name and Password Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

The login window prompts a user for his/her credentials, verifies their authorization level, and then allows or denies the user access to the system.

Rationale:

Prompting the user to enter both their username and password makes it twice as hard for unauthorized users to gain access to the system since they must discover two attributes.

Audit:

Graphical Method:
Perform the following steps to verify that the login window displays name and password:

1. Open `System Preferences
2. Select `Users and Groups
3. Select `Login Options`
4. Verify that `Display login window as:` is set to `Name and password`

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Show Full Name` set to `True`

Terminal Method:
Run the following command to verify the login window displays name and password:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.loginwindow')\
.objectForKey('SHOWFULLNAME').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to ensure the login window display name and password:

1. Open `System Preferences`
2. Select `Users and Groups`
3. Select `Login Options`
4. Set `Display login window as:` to `Name and password`

Terminal Method:
Run the following command to enable the login window to display name and password:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.loginwindow SHOWFULLNAME -bool true
```

Note: The GUI will not display the updated setting until the current user(s) logs out.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.loginwindow`
2. The key to include is `SHOWFULLNAME`
3. The key must be set to `<true/>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

6.1.2 Ensure Show Password Hints Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Password hints are user-created text displayed when an incorrect password is used for an account.

Rationale:

Password hints make it easier for unauthorized persons to gain access to systems by displaying information provided by the user to assist in remembering the password. This info could include the password itself or other information that might be readily discerned with basic knowledge of the end user.

Impact:

The user can set the hint to any value, including the password itself or clues that allow trivial social engineering attacks.

Audit:

Graphical Method:
Perform the following steps to verify if password hints are shown:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Select `Login Options`
4. Verify that `Show password hints` is disabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Retires Before Hint Shown` is set to `0`

Terminal Method:
Run the following command to verify that password hints are not displayed:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.loginwindow')\
.objectForKey('RetriesUntilHint').js
EOS

0
```

Remediation:

Graphical Method:
Perform the following steps to disable password hints from being shown:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Select `Login Options`
4. Set `Show password hints` to disabled

Terminal Method:
Run the following command to disable password hints:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.loginwindow RetriesUntilHint -int 0
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.loginwindow`
2. The key to include is `RetriesUntilHint`
3. The key must be set to `<integer>0</integer>`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

6.1.3 Ensure Guest Account Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

The guest account allows users access to the system without having to create an account or password. Guest users are unable to make setting changes and cannot remotely login to the system. All files, caches, and passwords created by the guest user are deleted upon logging out.

Rationale:

Disabling the guest account mitigates the risk of an untrusted user doing basic reconnaissance and possibly using privilege escalation attacks to take control of the system.

Impact:

A guest user can use that access to find out additional information about the system and might be able to use privilege escalation vulnerabilities to establish greater access.

Audit:

Graphical Method:
Perform the following steps to ensure that the guest account is not available:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Select `Guest User`
4. Verify that `Allow guests to log in to this computer` is not enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `Disable Guest Account` set to `True`

Terminal Method:
Run the following command to verify if the guest account is enabled:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
function run() {
  let pref1 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.MCX')\
  .objectForKey('DisableGuestAccount'))
  let pref2 =
ObjC.unwrap($.NSUserDefaults.alloc.initWithSuiteName('com.apple.loginwindow')
\
  .objectForKey('GuestEnabled'))
  if ( pref1 == 1 || pref2 == 0 ) {
    return("true")
  } else {
    return("false")
  }
}
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to disable guest account availability:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Select `Guest User`
4. Set `Allow guests to log in to this computer` to disabled

Terminal Method:
Run the following command to disable the guest account:

```
$ /usr/bin/sudo /usr/bin/defaults write
/Library/Preferences/com.apple.loginwindow GuestEnabled -bool false
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.MCX`
2. The key to include is `DisableGuestAccount`
3. The key must be set to `<true/>`

Additional Information:

By default, the guest account is enabled for access to sharing services but is not allowed to log into the computer.

The guest account does not need a password when it is enabled to log into the computer.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.	●	●	●
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

6.1.4 Ensure Guest Access to Shared Folders Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Allowing guests to connect to shared folders enables users to access selected shared folders and their contents from different computers on a network.

Rationale:

Not allowing guests to connect to shared folders mitigates the risk of an untrusted user doing basic reconnaissance and possibly use privilege escalation attacks to take control of the system.

Impact:

Unauthorized users could access shared files on the system.

Audit:

Graphical Method:
Perform the following steps to ensure that guests cannot connect to shared folders:

1. Open `System Preferences`
2. Select `Users & Groups`
3. Select `Guest User`
4. Verify that `Allow guest users to connect to shared folders` is not enabled

Terminal Method:
Run the following commands to verify that shared folders are not accessible to guest users:

```
$ /usr/bin/sudo /usr/sbin/sysadminctl -smbGuestAccess status
```

The output should include `SMB guest access disabled`.

Remediation:

Graphical Method:
Perform the following steps to no longer allow guest user access to shared folders:

1. Open `System Preferences
2. Select `Users & Groups
3. Select `Guest User
4. Set `Allow guest users to connect to shared folders` to disabled

Terminal Method:
Run the following commands to verify that shared folders are not accessible to guest users:

```
$ /usr/bin/sudo /usr/sbin/sysadminctl -smbGuestAccess off
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

6.1.5 Ensure the Guest Home Folder Does Not Exist (Automated)

Profile Applicability:

- Level 1

Description:

In the previous two controls, the guest account login has been disabled and sharing to guests has been disabled, as well. There is no need for the legacy Guest home folder to remain in the file system. When normal user accounts are removed, you have the option to archive it, leave it in place, or delete. In the case of the guest folder, the folder remains in place without a GUI option to remove it. If at some point in the future a Guest account is needed, it will be re-created. The presence of the Guest home folder can cause automated audits to fail when looking for compliant settings within all User folders, as well. Rather than ignoring the folder's continued existence, it is best removed.

Rationale:

The Guest home folders are unneeded after the Guest account is disabled and could be used inappropriately.

Impact:

The Guest account should not be necessary after it is disabled, and it will be automatically re-created if the Guest account is re-enabled

Audit:

Terminal Method:
Run the following command to verify if the Guest user home folder exists:

```
$ /usr/bin/sudo /bin/ls /Users/ | /usr/bin/grep Guest
```

Remediation:

Terminal Method:
Run the following command to remove the Guest user home folder:

```
$ /usr/bin/sudo /bin/rm -R /Users/Guest
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.	●	●	●
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.	●	●	●

6.2 Ensure Show All Filename Extensions Setting is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

A filename extension is a suffix added to a base filename that indicates the base filename's file format.

Rationale:

Visible filename extensions allow the user to identify the file type and the application it is associated with which leads to quick identification of misrepresented malicious files.

Impact:

The user of the system can open files of unknown or unexpected filetypes if the extension is not visible.

Audit:

Graphical Method:
Perform the following steps to ensure that keyboard entries are secure in Terminal:

1. Open the `Applications` folder
2. Open the `Utilities` folder
3. Open `Terminal`
4. Select `Terminal` in the Menu Bar
5. Verify that `Secure Keyboard Entry` is enabled

or

1. Open `System Settings`
2. Select `Profiles`
3. Verify that an installed profile has `SecureKeyboardEntry` is set to `1`

Terminal Method:
For each user, run the following command to verify that keyboard entries in Terminal are secured:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read -app Terminal
SecureKeyboardEntry

1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read -app Terminal
SecureKeyboardEntry

0

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read -app Terminal
SecureKeyboardEntry

1
```

In the above example the user seconduser is compliant, and the user firstuser is not compliant.
or
Run the following command to verify that a profile is installed that enables secure keyboard entry in Terminal:

```
$ /usr/bin/sudo /usr/bin/osascript -l JavaScript << EOS
$.NSUserDefaults.alloc.initWithSuiteName('com.apple.Terminal')\
.objectForKey('SecureKeyboardEntry').js
EOS

true
```

Remediation:

Graphical Method:
Perform the following steps to enable secure keyboard entries in Terminal:

1. Open the `Applications` folder
2. Open the `Utilities` folder
3. Open `Terminal`
4. Select `Terminal` in the Menu Bar
5. Set `Secure Keyboard Entry` to enabled

Terminal Method:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write -app Terminal
SecureKeyboardEntry -bool true
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write -app Terminal
SecureKeyboardEntry -bool true
```

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.Terminal`
2. The key to include is `SecureKeyboardEntry`
3. The key must be set to `<true/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

Default Value:

Filename extensions are turned off by default.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.3 Address Unauthorized Software Ensure that unauthorized software is either removed from use on enterprise assets or receives a documented exception. Review monthly, or more frequently.	●	●	●
v7	2.6 Address unapproved software Ensure that unauthorized software is either removed or the inventory is updated in a timely manner	●	●	●

7 Applications

All default Operating System default installs include OS vendor applications, or vendor endorsed channels. Most of these applications do not require explicit security controls as part of an OS Benchmark or a separate Benchmark. When the application is OS specific it is much more efficient to include security guidance as part of the OS Benchmark where security controls are available. This section contains a small list of macOS applications where security controls exist and should be reviewed.

There is no insistence that any of the built-in application must be used, three are many alternative third party applications that may be used instead. Included applications are often core to the Operating System and a lack of security controls will likely make the OS itself more vulnerable.

7.1 Mail

Mail is Apple's OS included email client on both macOS and iOS. It supports a range of email server services including iCloud, Exchange, Gmail and standard IMAP and POP accounts. With the vast barrage of phishing attacks any Internet email client is an exploitable risk on a user system. Any email client should be hardened with controls that assist the user against social engineering attacks to reduce the risk of unwanted emails. This benchmark is not advocating for all users to use Apple Mail, just providing guidance on controls for the built-in email client. Other email clients will have their own set of security controls.

Apple Mail fully supports S/MIME without the use of any third party plugin. While there are very few remaining trusted CAs issuing free S/MIME certificates the use of a trusted CA and digital signing enables others to interact with you using end-to-end encryption through encrypted email. An internal CA may also be used but partner trust of the CA will have to be coordinated.

More S/MIME info

Sign or encrypt emails in Mail on Mac

Installing an S/MIME Certificate and Sending Secure Email in macOS

Obtaining and using an S/MIME certificate on Apple MacOS

Sources of Free S/MIME Certificates

7.1.1 Ensure Protect Mail Activity in Mail Is Enabled (Manual)

Profile Applicability:

- Level 2

Description:

Apple provides privacy protection that should be enabled for the mail app on macOS to reduce information collection from a user that receives email.

Rationale:

Email is routinely abused by attackers, spammers and marketers. The "Protect Mail Activity" control reduces risk by hiring the current IP address of your Mac and privately downloading remote content.

The Protect Mail Activity function of privately downloading remote content is not applicable for those users that do not download any remote content. Typical Internet email is no longer plain text and will not render properly without remote content, personal email or mailing list email may function without complaint by blocking remote content.

Impact:

Some remote content may be access controlled and refuse to download with this setting enabled.

Audit:

Graphical Method:
Perform the following steps to verify that protect mail activity is enabled:

1. Open `Mail`
2. Select `Mail` in the `menu bar`
3. Select `Preferences...`
4. Select `Privacy`
5. Verify that `Protect Mail Activity` is enabled

Remediation:

Graphical Method:
Perform the following steps to enabled protect mail activity:

1. Open `Mail`
2. Select `Mail` in the `menu bar`
3. Select `Preferences...`
4. Select `Privacy`
5. Set `Protect Mail Activity` to enabled

References:

1. https://support.apple.com/guide/mail/use-mail-privacy-protection-mlhl03be2866/mac

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v8	9.3 Maintain and Enforce Network-Based URL Filters Enforce and update network-based URL filters to limit an enterprise asset from connecting to potentially malicious or unapproved websites. Example implementations include category-based filtering, reputation-based filtering, or through the use of block lists. Enforce filters for all enterprise assets.		●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●
v7	7.4 Maintain and Enforce Network-Based URL Filters Enforce network-based URL filters that limit a system's ability to connect to websites not approved by the organization. This filtering shall be enforced for each of the organization's systems, whether they are physically at an organization's facilities or not.		●	●

7.2 Safari

Safari is Apple's included web browser. Many macOS services only operate, or operate more efficiently through the use of Safari.

https://www.apple.com/safari/

7.2.1 Ensure Automatic Opening of Safe Files in Safari Is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Safari will automatically run or execute what it considers safe files. This can include installers and other files that execute on the operating system. Safari evaluates file safety by using a list of filetypes maintained by Apple. The list of files include text, image, video and archive formats that would be run in the context of the OS rather than the browser.

Rationale:

Hackers have taken advantage of this setting via drive-by attacks. These attacks occur when a user visits a legitimate website that has been corrupted. The user unknowingly downloads a malicious file either by closing an infected pop-up or hovering over a malicious banner. An attacker can create a malicious file that will fall within Safari's safe file list that will download and execute without user input.

Impact:

Apple considers many files that the operating system itself auto-executes as "safe files." Many of these files could be malicious and could execute locally without the user even knowing that a file of a specific type had been downloaded.

Audit:

Graphical Method:
Perform the following to verify that safe files are not opened when download in Safari:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `General`
5. Verify that `Open "safe" files after downloading` is disabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `AutoOpenSafeDownloads` set `0`

Terminal Method:
Run the following command to verify that opening safe files after download in Safari is disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari AutoOpenSafeDownloads

0
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari AutoOpenSafeDownloads

0
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.
or
Run the following command to verify that a profile is installed that disables safe files from opening in Safari:

```
$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep
AutoOpenSafeDownloads

                                          AutoOpenSafeDownloads = 0;
```

Remediation:

Graphical Method:
Perform the following steps to set safe files to not open after downloading in Safari:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `General`
5. Set `Open "safe" files after downloading` to disabled

Terminal Method:
Run the following command to disable safe files from not opening when downloaded in Safari:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari AutoOpenSafeDownloads -bool false
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari AutoOpenSafeDownloads -bool false
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.Safari`
2. The key to include is `AutoOpenSafeDownloads`
3. The key must be set to: `<false/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v8	9.6 Block Unnecessary File Types Block unnecessary file types attempting to enter the enterprise's email gateway.		●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●
v7	7.9 Block Unnecessary File Types Block all e-mail attachments entering the organization's e-mail gateway if the file types are unnecessary for the organization's business.		●	●
v7	8.5 Configure Devices Not To Auto-run Content Configure devices to not auto-run content from removable media.	●	●	●

7.2.2 Audit History and Remove History Items (Manual)

Profile Applicability:

- Level 2

Description:

Organizational management of user web browsing history is a challenge effected by multiple facets. Organizations should decide whether to manage browser history and how much history should be maintained.

Rationale:

There are conflicting concerns in the retention of browser history. Unlimited retention:

- Consumes disk space
- Preferred by on disk forensics teams
- User searchable for old visited pages
- User privacy concerns
- Security concerns to retain old links that may be stale or lead to compromised pages or pages with changes or inappropriate content

Old browser history becomes stale and the use or misuse of the data can lead to unwanted outcomes. Search engine results are maintained and often provide much more relevant current information than old website visit information.

Impact:

If old browsing history is not available it will not be available to authorized or unauthorized users. Some users may find old and even stale information useful.

Audit:

Graphical Method:
Perform the following steps to verify how long the history in Safari is kept:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `General`
5. Verify that `Remove history items` is set to your organization's requirements

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `HistoryAgeInDaysLimit` set to your organization's requirements

Terminal Method:
Run the following command to verify how long Safari keeps history:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari HistoryAgeInDaysLimit
```

The output will be:
`1` - After one day
`7` - After one week
`14` - After two weeks
`31` - After one month
`365` - After one year
`36500` - Manually
Note: Setting the plist key to a value that is not represented by the GUI could cause issues.

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit

1

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read
/Users/seconduser/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari HistoryAgeInDaysLimit

7

$ /usr/bin/sudo -u thirduser /usr/bin/defaults read
/Users/thirduser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit

14

$ /usr/bin/sudo -u fourthuser /usr/bin/defaults read
/Users/fourthuser/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari HistoryAgeInDaysLimit

31

$ /usr/bin/sudo -u fifthuser /usr/bin/defaults read
/Users/fifthuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit

365

$ /usr/bin/sudo -u sixthuser /usr/bin/defaults read
/Users/sixthuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit

36500
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.

or

Run the following command to verify that a profile is installed that sets how long the history is kept in Safari:

```
$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep
HistoryAgeInDaysLimit
```

The output will be `HistoryAgeInDaysLimit =` followed by your organizations requirements.

Remediation:

Graphical Method:
Perform the following steps to set Safari to remove history after a set amount of days:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `General`
5. Set `Remove history items` to your organization's requirements

Terminal Method:
Run the following command to set when Safari will remove history items:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari HistoryAgeInDaysLimit -int <1,7,14,31,365,36500>
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit -int 36500

$ /usr/bin/sudo -u seconduser /usr/bin/defaults write
/Users/seconduser/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari HistoryAgeInDaysLimit -int 365

$ /usr/bin/sudo -u thirduser /usr/bin/defaults write
/Users/thirduser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit -int 31

$ /usr/bin/sudo -u fourthuser /usr/bin/defaults write
/Users/fourthuser/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari HistoryAgeInDaysLimit -int 14

$ /usr/bin/sudo -u fifthuser /usr/bin/defaults write
/Users/fifthuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit -int 7

$ /usr/bin/sudo -u sixthuser /usr/bin/defaults write
/Users/sixthuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari HistoryAgeInDaysLimit -int 1
```

Note: Setting the plist key to a value that is not represented by the GUI could cause issues.
Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.

Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.Safari`
2. The key to include is `HistoryAgeInDaysLimit`
3. The key must be set to: `<integer><1,7,14,31,365,36500></integer>`

Note: Setting the plist key to a value that is not represented by the GUI could cause issues.
Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●

7.2.3 Audit Passwords System Preference Setting (Manual)

Profile Applicability:

- Level 1

Description:

Apple has provided a new interface in macOS Monterey for managing passwords that mirrors the interfaced capability already available in iOS. Password management in macOS was previously available in both Safari Preferences and in Keychain Access. Apple is attempting to simplify password management for macOS and make the user experience more similar to iOS. Organizations are justifiably concerned about the risk of password managers, particularly as a possible backdoor to improved credential management regimes and greater use of Multi-Factor-Authentication (MFA).

Apple has information posted on this system preference with additional information.

Change Passwords preferences on Mac

A warning icon is shown next to a website for any of the following reasons:

- Easily guessed
- Appeared in a data leak
- Reused on another website

Rationale:

Organizations should remove what passwords can be saved on user computes and the ability of attackers to potentially steal organizational credentials. Limits on password storage must be evaluated based on both user risk and Enterprise risk.

Impact:

Organizations using passwords are constantly reported as having their password databases leaked to the Internet so every password a user has should be unique. Locking down secure password management solutions so that it cannot be used pushes users to password reuse, sticky notes or always open text files with long lists of credentials.

Audit:

Graphical Method:
Perform the following steps to audit the Password system settings:

1. Open `System Preferences`
2. Select `Passwords`
3. Enter the user password
4. Verify that `Detect compromised passwords` is set to match your organization's settings
5. Review applications with stored passwords to ensure that Enterprise managed passwords are not stored inappropriately. Application interfaces may need to be considered as well, as they allow the opportunity to store passwords that should not be saved.

Remediation:

Graphical Method:
Perform the following steps to set Password system settings to your organization's settings:

1. Open `System Preferences`
2. Select `Passwords`
3. Enter the user password
4. Select the `Detect compromised passwords` setting to match your organization's settings
5. Remove stored passwords that should not be saved.

References:

1. https://support.apple.com/guide/security/password-monitoring-sec78e79fc3b/1/web/1

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v8	5.6 Centralize Account Management Centralize account management through a directory or identity service.		●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

7.2.4 Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Apple uses the Google Safe Browsing API to check for fraudulent websites and report them to the user attempting visit one.

Rationale:

Attackers use crafted web pages to social engineer users to load unwanted content. Warning users prior to loading the content enables better security.

Impact:

Once compromised websites serving malware could be sanitized and remain in the database, though there is no widespread reporting of that risk.

Audit:

Graphical Method:
Perform the following to verify that warn when visiting a fraudulent site in Safari is enabled:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Security`
5. Verify that `Warn when visiting a fraudulent site` is enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `WarnAboutFraudulentWebsites` set to `1`

Terminal Method:
Run the following command to verify that warn when visiting a fraudulent site in Safari is not disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WarnAboutFraudulentWebsites

1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WarnAboutFraudulentWebsites

1
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.
or
Run the following command to verify that a profile is installed that warns when visiting fraudulent sites in Safari:

```
$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep
WarnAboutFraudulentWebsites

                                          AutoOpenSafeDownloads = 1;
```

Remediation:

Graphical Method:
Perform the following steps to set Safari to warn when visiting a fraudulent site:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Security`
5. Set `Warn when visiting a fraudulent site` to enabled

Terminal Method:
Run the following command to enable warn when visiting a fraudulent site in Safari:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WarnAboutFraudulentWebsites -bool false
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WarnAboutFraudulentWebsites -bool false
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.Safari`
2. The key to include is `WarnAboutFraudulentWebsites`
3. The key must be set to: `<true/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://support.apple.com/guide/safari/security-ibrw1074/16.0/mac/12.0

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v8	9.3 Maintain and Enforce Network-Based URL Filters Enforce and update network-based URL filters to limit an enterprise asset from connecting to potentially malicious or unapproved websites. Example implementations include category-based filtering, reputation-based filtering, or through the use of block lists. Enforce filters for all enterprise assets.		●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●
v7	7.4 Maintain and Enforce Network-Based URL Filters Enforce network-based URL filters that limit a system's ability to connect to websites not approved by the organization. This filtering shall be enforced for each of the organization's systems, whether they are physically at an organization's facilities or not.		●	●

7.2.5 Ensure Prevent Cross-site Tracking in Safari Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

There is a vast network of groups that collect, use and sell user data. One method used to collect user data is pay and provide contented and services for website owners, along with that "assistance" the site owners push tracking cookies on visitors. In many cases the help allows a content owner to keep the site up. The tracking cookies allow information brokers to track web users across visited sites. For better privacy and to provide some resistance to data brokers prevent cross-tracking.

Rationale:

Cross-tracking allows data-brokers to follow you across the Internet to enable their business model of selling personal data. Users should protect their data and not volunteer it to marketing companies.

Impact:

Marketing companies will be unable to target you as effectively.

Audit:

Graphical Method:
Perform the following to verify that preventing cross-site tracking in Safari is enabled:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Privacy`
5. Verify that `Prevent cross-site tracking` is enabled

or

1. Open `System Settings`
2. Select `Profiles`
3. Verify that an installed profile has `BlockStoragePolicy` set to `2`
4. Verify that an installed profile also has `WebKitPreferences.storageBlockingPolicy` set to `1`
5. Verify that an installed profile also has `WebKitStorageBlockingPolicy` set to `1`

Terminal Method:
Run the following command to verify that preventing cross-site tracking in Safari is not disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari BlockStoragePolicy

2

$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WebKitPreferences.storageBlockingPolicy

1

$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WebKitStorageBlockingPolicy

1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari BlockStoragePolicy

2

$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WebKitPreferences.storageBlockingPolicy

1

$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WebKitStorageBlockingPolicy

1
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.

or
Run the following command to verify that a profile is installed that prevents cross-site tracking in Safari:

```
$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep
BlockStoragePolicy

                                            BlockStoragePolicy = 2;

$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep
WebKitPreferences.storageBlockingPolicy


WebKitPreferences.storageBlockingPolicy = 1;

$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep
WebKitStorageBlockingPolicy

                                            WebKitStorageBlockingPolicy = 1;
```

Remediation:

Graphical Method:
Perform the following steps to set prevent cross-site tracking in Safari to enabled:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Privacy`
5. Set `Prevent cross-site tracking` is enable

Terminal Method:
Run the following command to enable Safari to prevent cross-site tracking:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari BlockStoragePolicy -int 2

$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WebKitPreferences.storageBlockingPolicy -int 1

$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WebKitStorageBlockingPolicy -int 1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari BlockStoragePolicy -int 2

$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WebKitPreferences.storageBlockingPolicy -int 1

$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WebKitStorageBlockingPolicy -int 1
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.Safari`
2. The key to include is `BlockStoragePolicy`
3. The key must be set to: 2
4. The key to also include is `WebKitPreferences.storageBlockingPolicy`
5. The key must be set to: 1
6. The key to also include is `WebKitStorageBlockingPolicy`
7. The key must be set to: 1

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://support.apple.com/guide/safari/prevent-cross-site-tracking-sfri40732/mac

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●

7.2.6 Audit Hide IP Address in Safari Setting (Manual)

Profile Applicability:

- Level 2

Description:

Public (Routable) IP addresses can be used to track people to their current location, including home and business addresses. While a valid IP addess is necessary to load the site the valid address does not need to be provided to known trackers and should be hidden.

Rationale:

Trackers can correlate your visits through various applications including websites and is a threat to your privacy.

Impact:

Website address blocking through iCloud Private Relay made prevent some wanted pages to load that use IP geolocation access controls.

Audit:

Graphical Method:
Perform the following steps to verify the setting for hiding IP addresses from trackers in Safari:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Privacy`
5. Verify that `Hide IP address from trackers` is set to your organization's requirement

Terminal Method:
Run the following command to verify if IP addresses are hidden from trackers in Safari:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WBSPrivacyProxyAvailabilityTraffic
```

The output will be either `3296` if hide IP address from trackers is disabled, or `3300`, if it is enabled.
example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WBSPrivacyProxyAvailabilityTraffic

3296

$ /usr/bin/sudo -u seconduser /usr/bin/defaults read
/Users/seconduser/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WBSPrivacyProxyAvailabilityTraffic

3300
```

In the above example the firstuser has hide ip address from trackers disabled. Seconduser has it enabled.
Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.

Remediation:

Graphical Method:
Perform the following steps to set Safari whether or not to hide IP addresses from trackers:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Privacy`
5. Set `Hide IP address from trackers` to your organization's requirements

Terminal Method:
Run the following command to enable or disable hiding IP addresses from trackers in Safari:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WBSPrivacyProxyAvailabilityTraffic -int <3296/3300>
```

`3296` will set hide IP address from trackers to disabled. `3300` will enable it.
example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WBSPrivacyProxyAvailabilityTraffic -int 3300

$ /usr/bin/sudo -u seconduser /usr/bin/defaults write
/Users/seconduser/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WBSPrivacyProxyAvailabilityTraffic -int 3296
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.

References:

1. https://support.apple.com/en-bn/guide/safari/sfri35610/16.0/mac/12.0

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v8	9.3 Maintain and Enforce Network-Based URL Filters Enforce and update network-based URL filters to limit an enterprise asset from connecting to potentially malicious or unapproved websites. Example implementations include category-based filtering, reputation-based filtering, or through the use of block lists. Enforce filters for all enterprise assets.		●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●
v7	7.4 Maintain and Enforce Network-Based URL Filters Enforce network-based URL filters that limit a system's ability to connect to websites not approved by the organization. This filtering shall be enforced for each of the organization's systems, whether they are physically at an organization's facilities or not.		●	●

7.2.7 Ensure Advertising Privacy Protection in Safari Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Apple provides a framework that allows advertisers to target Apple users and end-users with advertisements. While many people prefer that when they see advertising it is relevant to them and their interests, the detailed information that is data mining collected, correlated, and available to advertisers in repositories is often disconcerting. This information is valuable to both advertisers and attackers and has been used with other metadata to reveal users' identities.

Organizations should manage advertising settings on computers rather than allow users to configure the settings.

Apple Information

Ad tracking should be limited on 10.15 and prior.

Rationale:

Organizations should manage user privacy settings on managed devices to align with organizational policies and user data protection requirements.

Impact:

Uses will see generic advertising rather than targeted advertising. Apple warns that this will reduce the number of relevant ads.

Audit:

Graphical Method:
Perform the following steps to verify that allow privacy-preserving measurement of ad effectiveness in Safari is enabled:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Prefernces`
4. Select `Privacy`
5. Verify that `Allow privacy-preserving measurement of ad effectiveness` is enabled

Terminal Method:
Run the following command to verify that allow privacy-preserving measurement of ad effectiveness in Safari is not disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WebKitPreferences.privateClickMeasurementEnabled

1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WebKitPreferences.privateClickMeasurementEnabled

1
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.

Remediation:

Graphical Method:
Perform the following steps to set Safari to allow privacy-preserving measurement of ad effectiveness:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Prefernces`
4. Select `Privacy`
5. Set `Warn when visiting a fraudulent site` to enabled

Terminal Method:
Run the following command to enable allow privacy-preserving measurement of ad effectiveness in Safari:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari WebKitPreferences.privateClickMeasurementEnabled -bool
true
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari WebKitPreferences.privateClickMeasurementEnabled -bool true
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●

7.2.8 Ensure Show Full Website Address in Safari Is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Attackers use websites with malicious or unwanted content to exploit the user or the computer. Part of the attack chain is to lure someone to load their content rather than the desired content. In order to reduce the risk in interacting with unwanted content the full website address should always be displayed in Safari.

Rationale:

Full visibility into what site is being visited is important for privacy and security.

Impact:

Many URLs are very long and complicated, particularly for internal content management systems. Some complete URLS in the Smart Search Field may be difficult to parse.

Audit:

Graphical Method:
Perform the following steps to verify that showing full website addresses in Safari is enabled:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Advanced`
5. Verify that `Show full website address` is enabled

or

1. Open `System Preferences`
2. Select `Profiles`
3. Verify that an installed profile has `ShowFullURLInSmartSearchField` set `1`

Terminal Method:
Run the following command to verify that showing full website addresses in Safari is not disabled:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults read
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari ShowFullURLInSmartSearchField

1
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults read
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari ShowFullURLInSmartSearchField

1
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.
or
Run the following command to verify that a profile is installed that disables safe files from opening in Safari:

```
$ /usr/bin/sudo /usr/bin/profiles -P -o stdout | /usr/bin/grep
ShowFullURLInSmartSearchField

                                            ShowFullURLInSmartSearchField =
1;
```

Remediation:

Graphical Method:
Perform the following steps to set Safari to show full website addresses:

1. Open `Safari`
2. Select `Safari` from the `menu bar`
3. Select `Preferences`
4. Select `Security`
5. Set `Show full website address` to enabled

Terminal Method:
Run the following command to enable showing full website addresses in Safari:

```
$ /usr/bin/sudo -u <username> /usr/bin/defaults write
/Users/<username>/Library/Containers/com.apple.Safari/Data/Library/Preference
s/com.apple.Safari ShowFullURLInSmartSearchField -bool true
```

example:

```
$ /usr/bin/sudo -u firstuser /usr/bin/defaults write
/Users/firstuser/Library/Containers/com.apple.Safari/Data/Library/Preferences
/com.apple.Safari ShowFullURLInSmartSearchField -bool true
```

Note: To run the Terminal commands, Terminal must be granted Full Disk Access in the Security & Privacy pane in System Preferences.
Profile Method:
Create or edit a configuration profile with the following information:

1. The PayloadType string is `com.apple.Safari`
2. The key to include is `ShowFullURLInSmartSearchField`
3. The key must be set to: `<true/>`

Note: Since the profile method sets a system-wide setting and not a user-level one, the profile method is the preferred method. It is always better to set system-wide than per user.

References:

1. https://apple.stackexchange.com/questions/371473/always-show-full-url-in-safari-address-bar

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	9.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure only fully supported browsers and email clients are allowed to execute in the enterprise, only using the latest version of browsers and email clients provided through the vendor.	●	●	●
v7	7.1 Ensure Use of Only Fully Supported Browsers and Email Clients Ensure that only fully supported web browsers and email clients are allowed to execute in the organization, ideally only using the latest version of the browsers and email clients provided by the vendor.	●	●	●

Appendix: Summary Table

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
1	**Install Updates, Patches and Additional Security Software**		
1.1	Ensure All Apple-provided Software Is Current (Automated)	☐	☐
1.2	Ensure Auto Update Is Enabled (Automated)	☐	☐
1.3	Ensure Download New Updates When Available Is Enabled (Automated)	☐	☐
1.4	Ensure Installation of App Update Is Enabled (Automated)	☐	☐
1.5	Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled (Automated)	☐	☐
1.6	Ensure Install of macOS Updates Is Enabled (Automated)	☐	☐
1.7	Ensure Software Update Deferment Is Less Than or Equal to 30 Days (Automated)	☐	☐
1.8	Ensure Computer Name Does Not Contain PII or Protected Organizational Information (Manual)	☐	☐
2	**System Preferences**		
2.1	**Bluetooth**		
2.1.1	Ensure Show Bluetooth Status in Menu Bar Is Enabled (Automated)	☐	☐
2.2	**Date & Time**		
2.2.1	Ensure "Set time and date automatically" Is Enabled (Automated)	☐	☐
2.2.2	Ensure Time Is Set Within Appropriate Limits (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
2.3	**Desktop & Screen Saver**		
2.3.1	Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled (Automated)	☐	☐
2.3.2	Ensure Screen Saver Corners Are Secure (Automated)	☐	☐
2.4	**Sharing**		
2.4.1	Ensure Remote Apple Events Is Disabled (Automated)	☐	☐
2.4.2	Ensure Internet Sharing Is Disabled (Automated)	☐	☐
2.4.3	Ensure Screen Sharing Is Disabled (Automated)	☐	☐
2.4.4	Ensure Printer Sharing Is Disabled (Automated)	☐	☐
2.4.5	Ensure Remote Login Is Disabled (Automated)	☐	☐
2.4.6	Ensure DVD or CD Sharing Is Disabled (Automated)	☐	☐
2.4.7	Ensure Bluetooth Sharing Is Disabled (Automated)	☐	☐
2.4.8	Ensure File Sharing Is Disabled (Automated)	☐	☐
2.4.9	Ensure Remote Management Is Disabled (Automated)	☐	☐
2.4.10	Ensure Content Caching Is Disabled (Automated)	☐	☐
2.4.11	Ensure AirDrop Is Disabled (Automated)	☐	☐
2.4.12	Ensure Media Sharing Is Disabled (Automated)	☐	☐
2.5	**Security & Privacy**		
2.5.1	**Encryption**		
2.5.1.1	Ensure FileVault Is Enabled (Automated)	☐	☐
2.5.1.2	Ensure all user storage APFS volumes are encrypted (Manual)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
2.5.1.3	Ensure all user storage CoreStorage volumes are encrypted (Manual)	☐	☐
2.5.2	**Firewall**		
2.5.2.1	Ensure Firewall Is Enabled (Automated)	☐	☐
2.5.2.2	Ensure Firewall Stealth Mode Is Enabled (Automated)	☐	☐
2.5.3	Ensure Location Services Is Enabled (Automated)	☐	☐
2.5.4	Audit Location Services Access (Manual)	☐	☐
2.5.5	Ensure Sending Diagnostic and Usage Data to Apple Is Disabled (Automated)	☐	☐
2.5.6	Ensure Limit Ad Tracking Is Enabled (Automated)	☐	☐
2.5.7	Ensure Gatekeeper Is Enabled (Automated)	☐	☐
2.5.8	Ensure a Custom Message for the Login Screen Is Enabled (Automated)	☐	☐
2.5.9	Ensure an Administrator Password Is Required to Access System-Wide Preferences (Automated)	☐	☐
2.5.10	Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled (Automated)	☐	☐
2.6	**Apple ID**		
2.6.1	**iCloud**		
2.6.1.1	Audit iCloud Keychain (Manual)	☐	☐
2.6.1.2	Audit iCloud Drive (Manual)	☐	☐
2.6.1.3	Ensure iCloud Drive Document and Desktop Sync Is Disabled (Automated)	☐	☐
2.6.2	Audit App Store Password Settings (Manual)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
2.7	**Time Machine**		
2.7.1	Ensure Backup Automatically is Enabled If Time Machine Is Enabled (Automated)	☐	☐
2.7.2	Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled (Automated)	☐	☐
2.8	**Energy Saver**		
2.8.1	Ensure Wake for Network Access Is Disabled (Automated)	☐	☐
2.8.2	Ensure Power Nap Is Disabled for Intel Macs (Automated)	☐	☐
2.8.3	Ensure the OS is not Activate When Resuming from Sleep (Automated)	☐	☐
2.9	Ensure Legacy EFI Is Valid and Updating (Automated)	☐	☐
2.10	Audit Siri Settings (Manual)	☐	☐
2.11	Audit Sidecar Settings (Manual)	☐	☐
2.12	Audit Touch ID and Wallet & Apple Pay Settings (Manual)	☐	☐
3	**Logging and Auditing**		
3.1	Ensure Security Auditing Is Enabled (Automated)	☐	☐
3.2	Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements (Automated)	☐	☐
3.3	Ensure install.log Is Retained for 365 or More Days and No Maximum Size (Automated)	☐	☐
3.4	Ensure Security Auditing Retention Is Enabled (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
3.5	Ensure Access to Audit Records Is Controlled (Automated)	☐	☐
3.6	Ensure Firewall Logging Is Enabled and Configured (Automated)	☐	☐
3.7	Audit Software Inventory (Manual)	☐	☐
4	**Network Configurations**		
4.1	Ensure Bonjour Advertising Services Is Disabled (Automated)	☐	☐
4.2	Ensure Show Wi-Fi status in Menu Bar Is Enabled (Automated)	☐	☐
4.3	Ensure HTTP Server Is Disabled (Automated)	☐	☐
4.4	Ensure NFS Server Is Disabled (Automated)	☐	☐
5	**System Access, Authentication and Authorization**		
5.1	**File System Permissions and Access Controls**		
5.1.1	Ensure Home Folders Are Secure (Automated)	☐	☐
5.1.2	Ensure System Integrity Protection Status (SIP) Is Enabled (Automated)	☐	☐
5.1.3	Ensure Apple Mobile File Integrity (AMFI) Is Enabled (Automated)	☐	☐
5.1.4	Ensure Appropriate Permissions Are Enabled for System Wide Applications (Automated)	☐	☐
5.1.5	Ensure No World Writable Files Exist in the System Folder (Automated)	☐	☐
5.1.6	Ensure No World Writable Files Exist in the Library Folder (Automated)	☐	☐
5.2	**Password Management**		

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
5.2.1	Ensure Password Account Lockout Threshold Is Configured (Automated)	☐	☐
5.2.2	Ensure Password Minimum Length Is Configured (Automated)	☐	☐
5.2.3	Ensure Complex Password Must Contain Alphabetic Characters Is Configured (Manual)	☐	☐
5.2.4	Ensure Complex Password Must Contain Numeric Character Is Configured (Manual)	☐	☐
5.2.5	Ensure Complex Password Must Contain Special Character Is Configured (Manual)	☐	☐
5.2.6	Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured (Manual)	☐	☐
5.2.7	Ensure Password Age Is Configured (Automated)	☐	☐
5.2.8	Ensure Password History Is Configured (Automated)	☐	☐
5.3	Ensure the Sudo Timeout Period Is Set to Zero (Automated)	☐	☐
5.4	Ensure a Separate Timestamp Is Enabled for Each User/tty Combo (Automated)	☐	☐
5.5	Ensure the "root" Account Is Disabled (Automated)	☐	☐
5.6	Ensure Automatic Login Is Disabled (Automated)	☐	☐
5.7	Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session (Automated)	☐	☐
5.8	Ensure a Login Window Banner Exists (Automated)	☐	☐
5.9	Ensure Users' Accounts Do Not Have a Password Hint (Automated)	☐	☐
5.10	Ensure Secure Keyboard Entry Terminal.app Is Enabled (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
6	**User Accounts and Environment**		
6.1	**Accounts Preferences Action Items**		
6.1.1	Ensure Login Window Displays as Name and Password Is Enabled (Automated)	☐	☐
6.1.2	Ensure Show Password Hints Is Disabled (Automated)	☐	☐
6.1.3	Ensure Guest Account Is Disabled (Automated)	☐	☐
6.1.4	Ensure Guest Access to Shared Folders Is Disabled (Automated)	☐	☐
6.1.5	Ensure the Guest Home Folder Does Not Exist (Automated)	☐	☐
6.2	Ensure Show All Filename Extensions Setting is Enabled (Automated)	☐	☐
7	**Applications**		
7.1	**Mail**		
7.1.1	Ensure Protect Mail Activity in Mail Is Enabled (Manual)	☐	☐
7.2	**Safari**		
7.2.1	Ensure Automatic Opening of Safe Files in Safari Is Disabled (Automated)	☐	☐
7.2.2	Audit History and Remove History Items (Manual)	☐	☐
7.2.3	Audit Passwords System Preference Setting (Manual)	☐	☐
7.2.4	Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled (Automated)	☐	☐
7.2.5	Ensure Prevent Cross-site Tracking in Safari Is Enabled (Automated)	☐	☐
7.2.6	Audit Hide IP Address in Safari Setting (Manual)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
7.2.7	Ensure Advertising Privacy Protection in Safari Is Enabled (Automated)	☐	☐
7.2.8	Ensure Show Full Website Address in Safari Is Enabled (Automated)	☐	☐

Appendix: CIS Controls v7 IG 1 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure All Apple-provided Software Is Current	☐	☐
1.2	Ensure Auto Update Is Enabled	☐	☐
1.3	Ensure Download New Updates When Available Is Enabled	☐	☐
1.4	Ensure Installation of App Update Is Enabled	☐	☐
1.5	Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled	☐	☐
1.6	Ensure Install of macOS Updates Is Enabled	☐	☐
1.7	Ensure Software Update Deferment Is Less Than or Equal to 30 Days	☐	☐
2.3.1	Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled	☐	☐
2.3.2	Ensure Screen Saver Corners Are Secure	☐	☐
2.4.1	Ensure Remote Apple Events Is Disabled	☐	☐
2.4.2	Ensure Internet Sharing Is Disabled	☐	☐
2.4.3	Ensure Screen Sharing Is Disabled	☐	☐
2.4.4	Ensure Printer Sharing Is Disabled	☐	☐
2.4.5	Ensure Remote Login Is Disabled	☐	☐
2.4.6	Ensure DVD or CD Sharing Is Disabled	☐	☐
2.4.7	Ensure Bluetooth Sharing Is Disabled	☐	☐
2.4.8	Ensure File Sharing Is Disabled	☐	☐
2.4.9	Ensure Remote Management Is Disabled	☐	☐
2.4.11	Ensure AirDrop Is Disabled	☐	☐
2.4.12	Ensure Media Sharing Is Disabled	☐	☐
2.5.1.1	Ensure FileVault Is Enabled	☐	☐
2.5.1.2	Ensure all user storage APFS volumes are encrypted	☐	☐
2.5.1.3	Ensure all user storage CoreStorage volumes are encrypted	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.5.2.1	Ensure Firewall Is Enabled	☐	☐
2.5.2.2	Ensure Firewall Stealth Mode Is Enabled	☐	☐
2.5.3	Ensure Location Services Is Enabled	☐	☐
2.5.4	Audit Location Services Access	☐	☐
2.5.5	Ensure Sending Diagnostic and Usage Data to Apple Is Disabled	☐	☐
2.5.7	Ensure Gatekeeper Is Enabled	☐	☐
2.5.8	Ensure a Custom Message for the Login Screen Is Enabled	☐	☐
2.5.9	Ensure an Administrator Password Is Required to Access System-Wide Preferences	☐	☐
2.5.10	Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled	☐	☐
2.6.1.1	Audit iCloud Keychain	☐	☐
2.6.1.2	Audit iCloud Drive	☐	☐
2.6.1.3	Ensure iCloud Drive Document and Desktop Sync Is Disabled	☐	☐
2.6.2	Audit App Store Password Settings	☐	☐
2.7.1	Ensure Backup Automatically is Enabled If Time Machine Is Enabled	☐	☐
2.7.2	Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled	☐	☐
2.8.2	Ensure Power Nap Is Disabled for Intel Macs	☐	☐
2.8.3	Ensure the OS is not Activate When Resuming from Sleep	☐	☐
2.9	Ensure Legacy EFI Is Valid and Updating	☐	☐
2.10	Audit Siri Settings	☐	☐
2.11	Audit Sidecar Settings	☐	☐
2.12	Audit Touch ID and Wallet & Apple Pay Settings	☐	☐
3.1	Ensure Security Auditing Is Enabled	☐	☐
3.2	Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements	☐	☐
3.5	Ensure Access to Audit Records Is Controlled	☐	☐

Recommendation		Set Correctly	
		Yes	No
3.6	Ensure Firewall Logging Is Enabled and Configured	☐	☐
3.7	Audit Software Inventory	☐	☐
4.1	Ensure Bonjour Advertising Services Is Disabled	☐	☐
4.3	Ensure HTTP Server Is Disabled	☐	☐
4.4	Ensure NFS Server Is Disabled	☐	☐
5.1.1	Ensure Home Folders Are Secure	☐	☐
5.1.2	Ensure System Integrity Protection Status (SIP) Is Enabled	☐	☐
5.1.3	Ensure Apple Mobile File Integrity (AMFI) Is Enabled	☐	☐
5.1.4	Ensure Appropriate Permissions Are Enabled for System Wide Applications	☐	☐
5.1.5	Ensure No World Writable Files Exist in the System Folder	☐	☐
5.1.6	Ensure No World Writable Files Exist in the Library Folder	☐	☐
5.2.7	Ensure Password Age Is Configured	☐	☐
5.3	Ensure the Sudo Timeout Period Is Set to Zero	☐	☐
5.4	Ensure a Separate Timestamp Is Enabled for Each User/tty Combo	☐	☐
5.5	Ensure the "root" Account Is Disabled	☐	☐
5.6	Ensure Automatic Login Is Disabled	☐	☐
5.7	Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session	☐	☐
5.8	Ensure a Login Window Banner Exists	☐	☐
5.10	Ensure Secure Keyboard Entry Terminal.app Is Enabled	☐	☐
6.1.1	Ensure Login Window Displays as Name and Password Is Enabled	☐	☐
6.1.2	Ensure Show Password Hints Is Disabled	☐	☐
6.1.4	Ensure Guest Access to Shared Folders Is Disabled	☐	☐
6.1.5	Ensure the Guest Home Folder Does Not Exist	☐	☐
6.2	Ensure Show All Filename Extensions Setting is Enabled	☐	☐
7.1.1	Ensure Protect Mail Activity in Mail Is Enabled	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.2.1	Ensure Automatic Opening of Safe Files in Safari Is Disabled	☐	☐
7.2.2	Audit History and Remove History Items	☐	☐
7.2.4	Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled	☐	☐
7.2.5	Ensure Prevent Cross-site Tracking in Safari Is Enabled	☐	☐
7.2.6	Audit Hide IP Address in Safari Setting	☐	☐
7.2.7	Ensure Advertising Privacy Protection in Safari Is Enabled	☐	☐
7.2.8	Ensure Show Full Website Address in Safari Is Enabled	☐	☐

Appendix: CIS Controls v7 IG 2 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure All Apple-provided Software Is Current	☐	☐
1.2	Ensure Auto Update Is Enabled	☐	☐
1.3	Ensure Download New Updates When Available Is Enabled	☐	☐
1.4	Ensure Installation of App Update Is Enabled	☐	☐
1.5	Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled	☐	☐
1.6	Ensure Install of macOS Updates Is Enabled	☐	☐
1.7	Ensure Software Update Deferment Is Less Than or Equal to 30 Days	☐	☐
1.8	Ensure Computer Name Does Not Contain PII or Protected Organizational Information	☐	☐
2.1.1	Ensure Show Bluetooth Status in Menu Bar Is Enabled	☐	☐
2.2.1	Ensure "Set time and date automatically" Is Enabled	☐	☐
2.2.2	Ensure Time Is Set Within Appropriate Limits	☐	☐
2.3.1	Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled	☐	☐
2.3.2	Ensure Screen Saver Corners Are Secure	☐	☐
2.4.1	Ensure Remote Apple Events Is Disabled	☐	☐
2.4.2	Ensure Internet Sharing Is Disabled	☐	☐
2.4.3	Ensure Screen Sharing Is Disabled	☐	☐
2.4.4	Ensure Printer Sharing Is Disabled	☐	☐
2.4.5	Ensure Remote Login Is Disabled	☐	☐
2.4.6	Ensure DVD or CD Sharing Is Disabled	☐	☐
2.4.7	Ensure Bluetooth Sharing Is Disabled	☐	☐
2.4.8	Ensure File Sharing Is Disabled	☐	☐
2.4.9	Ensure Remote Management Is Disabled	☐	☐
2.4.10	Ensure Content Caching Is Disabled	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.4.11	Ensure AirDrop Is Disabled	☐	☐
2.4.12	Ensure Media Sharing Is Disabled	☐	☐
2.5.1.1	Ensure FileVault Is Enabled	☐	☐
2.5.1.2	Ensure all user storage APFS volumes are encrypted	☐	☐
2.5.1.3	Ensure all user storage CoreStorage volumes are encrypted	☐	☐
2.5.2.1	Ensure Firewall Is Enabled	☐	☐
2.5.2.2	Ensure Firewall Stealth Mode Is Enabled	☐	☐
2.5.3	Ensure Location Services Is Enabled	☐	☐
2.5.4	Audit Location Services Access	☐	☐
2.5.5	Ensure Sending Diagnostic and Usage Data to Apple Is Disabled	☐	☐
2.5.6	Ensure Limit Ad Tracking Is Enabled	☐	☐
2.5.7	Ensure Gatekeeper Is Enabled	☐	☐
2.5.8	Ensure a Custom Message for the Login Screen Is Enabled	☐	☐
2.5.9	Ensure an Administrator Password Is Required to Access System-Wide Preferences	☐	☐
2.5.10	Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled	☐	☐
2.6.1.1	Audit iCloud Keychain	☐	☐
2.6.1.2	Audit iCloud Drive	☐	☐
2.6.1.3	Ensure iCloud Drive Document and Desktop Sync Is Disabled	☐	☐
2.6.2	Audit App Store Password Settings	☐	☐
2.7.1	Ensure Backup Automatically is Enabled If Time Machine Is Enabled	☐	☐
2.7.2	Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled	☐	☐
2.8.1	Ensure Wake for Network Access Is Disabled	☐	☐
2.8.2	Ensure Power Nap Is Disabled for Intel Macs	☐	☐
2.8.3	Ensure the OS is not Activate When Resuming from Sleep	☐	☐
2.9	Ensure Legacy EFI Is Valid and Updating	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.10	Audit Siri Settings	☐	☐
2.11	Audit Sidecar Settings	☐	☐
2.12	Audit Touch ID and Wallet & Apple Pay Settings	☐	☐
3.1	Ensure Security Auditing Is Enabled	☐	☐
3.2	Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements	☐	☐
3.3	Ensure install.log Is Retained for 365 or More Days and No Maximum Size	☐	☐
3.4	Ensure Security Auditing Retention Is Enabled	☐	☐
3.5	Ensure Access to Audit Records Is Controlled	☐	☐
3.6	Ensure Firewall Logging Is Enabled and Configured	☐	☐
3.7	Audit Software Inventory	☐	☐
4.1	Ensure Bonjour Advertising Services Is Disabled	☐	☐
4.3	Ensure HTTP Server Is Disabled	☐	☐
4.4	Ensure NFS Server Is Disabled	☐	☐
5.1.1	Ensure Home Folders Are Secure	☐	☐
5.1.2	Ensure System Integrity Protection Status (SIP) Is Enabled	☐	☐
5.1.3	Ensure Apple Mobile File Integrity (AMFI) Is Enabled	☐	☐
5.1.4	Ensure Appropriate Permissions Are Enabled for System Wide Applications	☐	☐
5.1.5	Ensure No World Writable Files Exist in the System Folder	☐	☐
5.1.6	Ensure No World Writable Files Exist in the Library Folder	☐	☐
5.2.1	Ensure Password Account Lockout Threshold Is Configured	☐	☐
5.2.2	Ensure Password Minimum Length Is Configured	☐	☐
5.2.3	Ensure Complex Password Must Contain Alphabetic Characters Is Configured	☐	☐
5.2.4	Ensure Complex Password Must Contain Numeric Character Is Configured	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.2.5	Ensure Complex Password Must Contain Special Character Is Configured	☐	☐
5.2.6	Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured	☐	☐
5.2.7	Ensure Password Age Is Configured	☐	☐
5.2.8	Ensure Password History Is Configured	☐	☐
5.3	Ensure the Sudo Timeout Period Is Set to Zero	☐	☐
5.4	Ensure a Separate Timestamp Is Enabled for Each User/tty Combo	☐	☐
5.5	Ensure the "root" Account Is Disabled	☐	☐
5.6	Ensure Automatic Login Is Disabled	☐	☐
5.7	Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session	☐	☐
5.8	Ensure a Login Window Banner Exists	☐	☐
5.9	Ensure Users' Accounts Do Not Have a Password Hint	☐	☐
5.10	Ensure Secure Keyboard Entry Terminal.app Is Enabled	☐	☐
6.1.1	Ensure Login Window Displays as Name and Password Is Enabled	☐	☐
6.1.2	Ensure Show Password Hints Is Disabled	☐	☐
6.1.3	Ensure Guest Account Is Disabled	☐	☐
6.1.4	Ensure Guest Access to Shared Folders Is Disabled	☐	☐
6.1.5	Ensure the Guest Home Folder Does Not Exist	☐	☐
6.2	Ensure Show All Filename Extensions Setting is Enabled	☐	☐
7.1.1	Ensure Protect Mail Activity in Mail Is Enabled	☐	☐
7.2.1	Ensure Automatic Opening of Safe Files in Safari Is Disabled	☐	☐
7.2.2	Audit History and Remove History Items	☐	☐
7.2.3	Audit Passwords System Preference Setting	☐	☐
7.2.4	Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled	☐	☐
7.2.5	Ensure Prevent Cross-site Tracking in Safari Is Enabled	☐	☐
7.2.6	Audit Hide IP Address in Safari Setting	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.2.7	Ensure Advertising Privacy Protection in Safari Is Enabled	☐	☐
7.2.8	Ensure Show Full Website Address in Safari Is Enabled	☐	☐

Appendix: CIS Controls v7 IG 3 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure All Apple-provided Software Is Current	☐	☐
1.2	Ensure Auto Update Is Enabled	☐	☐
1.3	Ensure Download New Updates When Available Is Enabled	☐	☐
1.4	Ensure Installation of App Update Is Enabled	☐	☐
1.5	Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled	☐	☐
1.6	Ensure Install of macOS Updates Is Enabled	☐	☐
1.7	Ensure Software Update Deferment Is Less Than or Equal to 30 Days	☐	☐
1.8	Ensure Computer Name Does Not Contain PII or Protected Organizational Information	☐	☐
2.1.1	Ensure Show Bluetooth Status in Menu Bar Is Enabled	☐	☐
2.2.1	Ensure "Set time and date automatically" Is Enabled	☐	☐
2.2.2	Ensure Time Is Set Within Appropriate Limits	☐	☐
2.3.1	Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled	☐	☐
2.3.2	Ensure Screen Saver Corners Are Secure	☐	☐
2.4.1	Ensure Remote Apple Events Is Disabled	☐	☐
2.4.2	Ensure Internet Sharing Is Disabled	☐	☐
2.4.3	Ensure Screen Sharing Is Disabled	☐	☐
2.4.4	Ensure Printer Sharing Is Disabled	☐	☐
2.4.5	Ensure Remote Login Is Disabled	☐	☐
2.4.6	Ensure DVD or CD Sharing Is Disabled	☐	☐
2.4.7	Ensure Bluetooth Sharing Is Disabled	☐	☐
2.4.8	Ensure File Sharing Is Disabled	☐	☐
2.4.9	Ensure Remote Management Is Disabled	☐	☐
2.4.10	Ensure Content Caching Is Disabled	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.4.11	Ensure AirDrop Is Disabled	☐	☐
2.4.12	Ensure Media Sharing Is Disabled	☐	☐
2.5.1.1	Ensure FileVault Is Enabled	☐	☐
2.5.1.2	Ensure all user storage APFS volumes are encrypted	☐	☐
2.5.1.3	Ensure all user storage CoreStorage volumes are encrypted	☐	☐
2.5.2.1	Ensure Firewall Is Enabled	☐	☐
2.5.2.2	Ensure Firewall Stealth Mode Is Enabled	☐	☐
2.5.3	Ensure Location Services Is Enabled	☐	☐
2.5.4	Audit Location Services Access	☐	☐
2.5.5	Ensure Sending Diagnostic and Usage Data to Apple Is Disabled	☐	☐
2.5.6	Ensure Limit Ad Tracking Is Enabled	☐	☐
2.5.7	Ensure Gatekeeper Is Enabled	☐	☐
2.5.8	Ensure a Custom Message for the Login Screen Is Enabled	☐	☐
2.5.9	Ensure an Administrator Password Is Required to Access System-Wide Preferences	☐	☐
2.5.10	Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled	☐	☐
2.6.1.1	Audit iCloud Keychain	☐	☐
2.6.1.2	Audit iCloud Drive	☐	☐
2.6.1.3	Ensure iCloud Drive Document and Desktop Sync Is Disabled	☐	☐
2.6.2	Audit App Store Password Settings	☐	☐
2.7.1	Ensure Backup Automatically is Enabled If Time Machine Is Enabled	☐	☐
2.7.2	Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled	☐	☐
2.8.1	Ensure Wake for Network Access Is Disabled	☐	☐
2.8.2	Ensure Power Nap Is Disabled for Intel Macs	☐	☐
2.8.3	Ensure the OS is not Activate When Resuming from Sleep	☐	☐
2.9	Ensure Legacy EFI Is Valid and Updating	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.10	Audit Siri Settings	☐	☐
2.11	Audit Sidecar Settings	☐	☐
2.12	Audit Touch ID and Wallet & Apple Pay Settings	☐	☐
3.1	Ensure Security Auditing Is Enabled	☐	☐
3.2	Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements	☐	☐
3.3	Ensure install.log Is Retained for 365 or More Days and No Maximum Size	☐	☐
3.4	Ensure Security Auditing Retention Is Enabled	☐	☐
3.5	Ensure Access to Audit Records Is Controlled	☐	☐
3.6	Ensure Firewall Logging Is Enabled and Configured	☐	☐
3.7	Audit Software Inventory	☐	☐
4.1	Ensure Bonjour Advertising Services Is Disabled	☐	☐
4.2	Ensure Show Wi-Fi status in Menu Bar Is Enabled	☐	☐
4.3	Ensure HTTP Server Is Disabled	☐	☐
4.4	Ensure NFS Server Is Disabled	☐	☐
5.1.1	Ensure Home Folders Are Secure	☐	☐
5.1.2	Ensure System Integrity Protection Status (SIP) Is Enabled	☐	☐
5.1.3	Ensure Apple Mobile File Integrity (AMFI) Is Enabled	☐	☐
5.1.4	Ensure Appropriate Permissions Are Enabled for System Wide Applications	☐	☐
5.1.5	Ensure No World Writable Files Exist in the System Folder	☐	☐
5.1.6	Ensure No World Writable Files Exist in the Library Folder	☐	☐
5.2.1	Ensure Password Account Lockout Threshold Is Configured	☐	☐
5.2.2	Ensure Password Minimum Length Is Configured	☐	☐
5.2.3	Ensure Complex Password Must Contain Alphabetic Characters Is Configured	☐	☐
5.2.4	Ensure Complex Password Must Contain Numeric Character Is Configured	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.2.5	Ensure Complex Password Must Contain Special Character Is Configured	☐	☐
5.2.6	Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured	☐	☐
5.2.7	Ensure Password Age Is Configured	☐	☐
5.2.8	Ensure Password History Is Configured	☐	☐
5.3	Ensure the Sudo Timeout Period Is Set to Zero	☐	☐
5.4	Ensure a Separate Timestamp Is Enabled for Each User/tty Combo	☐	☐
5.5	Ensure the "root" Account Is Disabled	☐	☐
5.6	Ensure Automatic Login Is Disabled	☐	☐
5.7	Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session	☐	☐
5.8	Ensure a Login Window Banner Exists	☐	☐
5.9	Ensure Users' Accounts Do Not Have a Password Hint	☐	☐
5.10	Ensure Secure Keyboard Entry Terminal.app Is Enabled	☐	☐
6.1.1	Ensure Login Window Displays as Name and Password Is Enabled	☐	☐
6.1.2	Ensure Show Password Hints Is Disabled	☐	☐
6.1.3	Ensure Guest Account Is Disabled	☐	☐
6.1.4	Ensure Guest Access to Shared Folders Is Disabled	☐	☐
6.1.5	Ensure the Guest Home Folder Does Not Exist	☐	☐
6.2	Ensure Show All Filename Extensions Setting is Enabled	☐	☐
7.1.1	Ensure Protect Mail Activity in Mail Is Enabled	☐	☐
7.2.1	Ensure Automatic Opening of Safe Files in Safari Is Disabled	☐	☐
7.2.2	Audit History and Remove History Items	☐	☐
7.2.3	Audit Passwords System Preference Setting	☐	☐
7.2.4	Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled	☐	☐
7.2.5	Ensure Prevent Cross-site Tracking in Safari Is Enabled	☐	☐
7.2.6	Audit Hide IP Address in Safari Setting	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.2.7	Ensure Advertising Privacy Protection in Safari Is Enabled	☐	☐
7.2.8	Ensure Show Full Website Address in Safari Is Enabled	☐	☐

Appendix: CIS Controls v7 Unmapped Recommendations

Recommendation		Set Correctly	
		Yes	No
	No unmapped recommendations to CIS Controls v7.0	☐	☐

Appendix: CIS Controls v8 IG 1 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure All Apple-provided Software Is Current	☐	☐
1.2	Ensure Auto Update Is Enabled	☐	☐
1.3	Ensure Download New Updates When Available Is Enabled	☐	☐
1.4	Ensure Installation of App Update Is Enabled	☐	☐
1.5	Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled	☐	☐
1.6	Ensure Install of macOS Updates Is Enabled	☐	☐
1.7	Ensure Software Update Deferment Is Less Than or Equal to 30 Days	☐	☐
1.8	Ensure Computer Name Does Not Contain PII or Protected Organizational Information	☐	☐
2.3.1	Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled	☐	☐
2.3.2	Ensure Screen Saver Corners Are Secure	☐	☐
2.4.1	Ensure Remote Apple Events Is Disabled	☐	☐
2.4.2	Ensure Internet Sharing Is Disabled	☐	☐
2.4.3	Ensure Screen Sharing Is Disabled	☐	☐
2.4.4	Ensure Printer Sharing Is Disabled	☐	☐
2.4.5	Ensure Remote Login Is Disabled	☐	☐
2.4.6	Ensure DVD or CD Sharing Is Disabled	☐	☐
2.4.7	Ensure Bluetooth Sharing Is Disabled	☐	☐
2.4.8	Ensure File Sharing Is Disabled	☐	☐
2.4.9	Ensure Remote Management Is Disabled	☐	☐
2.4.11	Ensure AirDrop Is Disabled	☐	☐
2.4.12	Ensure Media Sharing Is Disabled	☐	☐
2.5.1.1	Ensure FileVault Is Enabled	☐	☐
2.5.1.2	Ensure all user storage APFS volumes are encrypted	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.5.2.1	Ensure Firewall Is Enabled	☐	☐
2.5.2.2	Ensure Firewall Stealth Mode Is Enabled	☐	☐
2.5.3	Ensure Location Services Is Enabled	☐	☐
2.5.4	Audit Location Services Access	☐	☐
2.5.5	Ensure Sending Diagnostic and Usage Data to Apple Is Disabled	☐	☐
2.5.7	Ensure Gatekeeper Is Enabled	☐	☐
2.5.8	Ensure a Custom Message for the Login Screen Is Enabled	☐	☐
2.5.9	Ensure an Administrator Password Is Required to Access System-Wide Preferences	☐	☐
2.5.10	Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled	☐	☐
2.6.1.1	Audit iCloud Keychain	☐	☐
2.6.1.2	Audit iCloud Drive	☐	☐
2.6.1.3	Ensure iCloud Drive Document and Desktop Sync Is Disabled	☐	☐
2.6.2	Audit App Store Password Settings	☐	☐
2.7.1	Ensure Backup Automatically is Enabled If Time Machine Is Enabled	☐	☐
2.7.2	Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled	☐	☐
2.8.2	Ensure Power Nap Is Disabled for Intel Macs	☐	☐
2.8.3	Ensure the OS is not Activate When Resuming from Sleep	☐	☐
2.9	Ensure Legacy EFI Is Valid and Updating	☐	☐
2.10	Audit Siri Settings	☐	☐
2.11	Audit Sidecar Settings	☐	☐
2.12	Audit Touch ID and Wallet & Apple Pay Settings	☐	☐
3.1	Ensure Security Auditing Is Enabled	☐	☐
3.2	Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements	☐	☐

Recommendation		Set Correctly	
		Yes	No
3.3	Ensure install.log Is Retained for 365 or More Days and No Maximum Size	☐	☐
3.4	Ensure Security Auditing Retention Is Enabled	☐	☐
3.5	Ensure Access to Audit Records Is Controlled	☐	☐
3.6	Ensure Firewall Logging Is Enabled and Configured	☐	☐
3.7	Audit Software Inventory	☐	☐
4.1	Ensure Bonjour Advertising Services Is Disabled	☐	☐
4.3	Ensure HTTP Server Is Disabled	☐	☐
4.4	Ensure NFS Server Is Disabled	☐	☐
5.1.1	Ensure Home Folders Are Secure	☐	☐
5.1.2	Ensure System Integrity Protection Status (SIP) Is Enabled	☐	☐
5.1.3	Ensure Apple Mobile File Integrity (AMFI) Is Enabled	☐	☐
5.1.4	Ensure Appropriate Permissions Are Enabled for System Wide Applications	☐	☐
5.1.5	Ensure No World Writable Files Exist in the System Folder	☐	☐
5.1.6	Ensure No World Writable Files Exist in the Library Folder	☐	☐
5.2.1	Ensure Password Account Lockout Threshold Is Configured	☐	☐
5.2.2	Ensure Password Minimum Length Is Configured	☐	☐
5.2.3	Ensure Complex Password Must Contain Alphabetic Characters Is Configured	☐	☐
5.2.4	Ensure Complex Password Must Contain Numeric Character Is Configured	☐	☐
5.2.5	Ensure Complex Password Must Contain Special Character Is Configured	☐	☐
5.2.6	Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured	☐	☐
5.2.7	Ensure Password Age Is Configured	☐	☐
5.2.8	Ensure Password History Is Configured	☐	☐
5.3	Ensure the Sudo Timeout Period Is Set to Zero	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.4	Ensure a Separate Timestamp Is Enabled for Each User/tty Combo	☐	☐
5.5	Ensure the "root" Account Is Disabled	☐	☐
5.6	Ensure Automatic Login Is Disabled	☐	☐
5.7	Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session	☐	☐
5.8	Ensure a Login Window Banner Exists	☐	☐
5.9	Ensure Users' Accounts Do Not Have a Password Hint	☐	☐
6.1.1	Ensure Login Window Displays as Name and Password Is Enabled	☐	☐
6.1.2	Ensure Show Password Hints Is Disabled	☐	☐
6.1.3	Ensure Guest Account Is Disabled	☐	☐
6.1.4	Ensure Guest Access to Shared Folders Is Disabled	☐	☐
6.1.5	Ensure the Guest Home Folder Does Not Exist	☐	☐
6.2	Ensure Show All Filename Extensions Setting is Enabled	☐	☐
7.1.1	Ensure Protect Mail Activity in Mail Is Enabled	☐	☐
7.2.1	Ensure Automatic Opening of Safe Files in Safari Is Disabled	☐	☐
7.2.2	Audit History and Remove History Items	☐	☐
7.2.3	Audit Passwords System Preference Setting	☐	☐
7.2.4	Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled	☐	☐
7.2.5	Ensure Prevent Cross-site Tracking in Safari Is Enabled	☐	☐
7.2.6	Audit Hide IP Address in Safari Setting	☐	☐
7.2.7	Ensure Advertising Privacy Protection in Safari Is Enabled	☐	☐
7.2.8	Ensure Show Full Website Address in Safari Is Enabled	☐	☐

Appendix: CIS Controls v8 IG 2 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure All Apple-provided Software Is Current	☐	☐
1.2	Ensure Auto Update Is Enabled	☐	☐
1.3	Ensure Download New Updates When Available Is Enabled	☐	☐
1.4	Ensure Installation of App Update Is Enabled	☐	☐
1.5	Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled	☐	☐
1.6	Ensure Install of macOS Updates Is Enabled	☐	☐
1.7	Ensure Software Update Deferment Is Less Than or Equal to 30 Days	☐	☐
1.8	Ensure Computer Name Does Not Contain PII or Protected Organizational Information	☐	☐
2.1.1	Ensure Show Bluetooth Status in Menu Bar Is Enabled	☐	☐
2.2.1	Ensure "Set time and date automatically" Is Enabled	☐	☐
2.2.2	Ensure Time Is Set Within Appropriate Limits	☐	☐
2.3.1	Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled	☐	☐
2.3.2	Ensure Screen Saver Corners Are Secure	☐	☐
2.4.1	Ensure Remote Apple Events Is Disabled	☐	☐
2.4.2	Ensure Internet Sharing Is Disabled	☐	☐
2.4.3	Ensure Screen Sharing Is Disabled	☐	☐
2.4.4	Ensure Printer Sharing Is Disabled	☐	☐
2.4.5	Ensure Remote Login Is Disabled	☐	☐
2.4.6	Ensure DVD or CD Sharing Is Disabled	☐	☐
2.4.7	Ensure Bluetooth Sharing Is Disabled	☐	☐
2.4.8	Ensure File Sharing Is Disabled	☐	☐
2.4.9	Ensure Remote Management Is Disabled	☐	☐
2.4.10	Ensure Content Caching Is Disabled	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.4.11	Ensure AirDrop Is Disabled	☐	☐
2.4.12	Ensure Media Sharing Is Disabled	☐	☐
2.5.1.1	Ensure FileVault Is Enabled	☐	☐
2.5.1.2	Ensure all user storage APFS volumes are encrypted	☐	☐
2.5.1.3	Ensure all user storage CoreStorage volumes are encrypted	☐	☐
2.5.2.1	Ensure Firewall Is Enabled	☐	☐
2.5.2.2	Ensure Firewall Stealth Mode Is Enabled	☐	☐
2.5.3	Ensure Location Services Is Enabled	☐	☐
2.5.4	Audit Location Services Access	☐	☐
2.5.5	Ensure Sending Diagnostic and Usage Data to Apple Is Disabled	☐	☐
2.5.6	Ensure Limit Ad Tracking Is Enabled	☐	☐
2.5.7	Ensure Gatekeeper Is Enabled	☐	☐
2.5.8	Ensure a Custom Message for the Login Screen Is Enabled	☐	☐
2.5.9	Ensure an Administrator Password Is Required to Access System-Wide Preferences	☐	☐
2.5.10	Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled	☐	☐
2.6.1.1	Audit iCloud Keychain	☐	☐
2.6.1.2	Audit iCloud Drive	☐	☐
2.6.1.3	Ensure iCloud Drive Document and Desktop Sync Is Disabled	☐	☐
2.6.2	Audit App Store Password Settings	☐	☐
2.7.1	Ensure Backup Automatically is Enabled If Time Machine Is Enabled	☐	☐
2.7.2	Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled	☐	☐
2.8.1	Ensure Wake for Network Access Is Disabled	☐	☐
2.8.2	Ensure Power Nap Is Disabled for Intel Macs	☐	☐
2.8.3	Ensure the OS is not Activate When Resuming from Sleep	☐	☐
2.9	Ensure Legacy EFI Is Valid and Updating	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.10	Audit Siri Settings	☐	☐
2.11	Audit Sidecar Settings	☐	☐
2.12	Audit Touch ID and Wallet & Apple Pay Settings	☐	☐
3.1	Ensure Security Auditing Is Enabled	☐	☐
3.2	Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements	☐	☐
3.3	Ensure install.log Is Retained for 365 or More Days and No Maximum Size	☐	☐
3.4	Ensure Security Auditing Retention Is Enabled	☐	☐
3.5	Ensure Access to Audit Records Is Controlled	☐	☐
3.6	Ensure Firewall Logging Is Enabled and Configured	☐	☐
3.7	Audit Software Inventory	☐	☐
4.1	Ensure Bonjour Advertising Services Is Disabled	☐	☐
4.2	Ensure Show Wi-Fi status in Menu Bar Is Enabled	☐	☐
4.3	Ensure HTTP Server Is Disabled	☐	☐
4.4	Ensure NFS Server Is Disabled	☐	☐
5.1.1	Ensure Home Folders Are Secure	☐	☐
5.1.2	Ensure System Integrity Protection Status (SIP) Is Enabled	☐	☐
5.1.3	Ensure Apple Mobile File Integrity (AMFI) Is Enabled	☐	☐
5.1.4	Ensure Appropriate Permissions Are Enabled for System Wide Applications	☐	☐
5.1.5	Ensure No World Writable Files Exist in the System Folder	☐	☐
5.1.6	Ensure No World Writable Files Exist in the Library Folder	☐	☐
5.2.1	Ensure Password Account Lockout Threshold Is Configured	☐	☐
5.2.2	Ensure Password Minimum Length Is Configured	☐	☐
5.2.3	Ensure Complex Password Must Contain Alphabetic Characters Is Configured	☐	☐
5.2.4	Ensure Complex Password Must Contain Numeric Character Is Configured	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.2.5	Ensure Complex Password Must Contain Special Character Is Configured	☐	☐
5.2.6	Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured	☐	☐
5.2.7	Ensure Password Age Is Configured	☐	☐
5.2.8	Ensure Password History Is Configured	☐	☐
5.3	Ensure the Sudo Timeout Period Is Set to Zero	☐	☐
5.4	Ensure a Separate Timestamp Is Enabled for Each User/tty Combo	☐	☐
5.5	Ensure the "root" Account Is Disabled	☐	☐
5.6	Ensure Automatic Login Is Disabled	☐	☐
5.7	Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session	☐	☐
5.8	Ensure a Login Window Banner Exists	☐	☐
5.9	Ensure Users' Accounts Do Not Have a Password Hint	☐	☐
5.10	Ensure Secure Keyboard Entry Terminal.app Is Enabled	☐	☐
6.1.1	Ensure Login Window Displays as Name and Password Is Enabled	☐	☐
6.1.2	Ensure Show Password Hints Is Disabled	☐	☐
6.1.3	Ensure Guest Account Is Disabled	☐	☐
6.1.4	Ensure Guest Access to Shared Folders Is Disabled	☐	☐
6.1.5	Ensure the Guest Home Folder Does Not Exist	☐	☐
6.2	Ensure Show All Filename Extensions Setting is Enabled	☐	☐
7.1.1	Ensure Protect Mail Activity in Mail Is Enabled	☐	☐
7.2.1	Ensure Automatic Opening of Safe Files in Safari Is Disabled	☐	☐
7.2.2	Audit History and Remove History Items	☐	☐
7.2.3	Audit Passwords System Preference Setting	☐	☐
7.2.4	Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled	☐	☐
7.2.5	Ensure Prevent Cross-site Tracking in Safari Is Enabled	☐	☐
7.2.6	Audit Hide IP Address in Safari Setting	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.2.7	Ensure Advertising Privacy Protection in Safari Is Enabled	☐	☐
7.2.8	Ensure Show Full Website Address in Safari Is Enabled	☐	☐

Appendix: CIS Controls v8 IG 3 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure All Apple-provided Software Is Current	☐	☐
1.2	Ensure Auto Update Is Enabled	☐	☐
1.3	Ensure Download New Updates When Available Is Enabled	☐	☐
1.4	Ensure Installation of App Update Is Enabled	☐	☐
1.5	Ensure System Data Files and Security Updates Are Downloaded Automatically Is Enabled	☐	☐
1.6	Ensure Install of macOS Updates Is Enabled	☐	☐
1.7	Ensure Software Update Deferment Is Less Than or Equal to 30 Days	☐	☐
1.8	Ensure Computer Name Does Not Contain PII or Protected Organizational Information	☐	☐
2.1.1	Ensure Show Bluetooth Status in Menu Bar Is Enabled	☐	☐
2.2.1	Ensure "Set time and date automatically" Is Enabled	☐	☐
2.2.2	Ensure Time Is Set Within Appropriate Limits	☐	☐
2.3.1	Ensure an Inactivity Interval of 20 Minutes Or Less for the Screen Saver Is Enabled	☐	☐
2.3.2	Ensure Screen Saver Corners Are Secure	☐	☐
2.4.1	Ensure Remote Apple Events Is Disabled	☐	☐
2.4.2	Ensure Internet Sharing Is Disabled	☐	☐
2.4.3	Ensure Screen Sharing Is Disabled	☐	☐
2.4.4	Ensure Printer Sharing Is Disabled	☐	☐
2.4.5	Ensure Remote Login Is Disabled	☐	☐
2.4.6	Ensure DVD or CD Sharing Is Disabled	☐	☐
2.4.7	Ensure Bluetooth Sharing Is Disabled	☐	☐
2.4.8	Ensure File Sharing Is Disabled	☐	☐
2.4.9	Ensure Remote Management Is Disabled	☐	☐
2.4.10	Ensure Content Caching Is Disabled	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.4.11	Ensure AirDrop Is Disabled	☐	☐
2.4.12	Ensure Media Sharing Is Disabled	☐	☐
2.5.1.1	Ensure FileVault Is Enabled	☐	☐
2.5.1.2	Ensure all user storage APFS volumes are encrypted	☐	☐
2.5.1.3	Ensure all user storage CoreStorage volumes are encrypted	☐	☐
2.5.2.1	Ensure Firewall Is Enabled	☐	☐
2.5.2.2	Ensure Firewall Stealth Mode Is Enabled	☐	☐
2.5.3	Ensure Location Services Is Enabled	☐	☐
2.5.4	Audit Location Services Access	☐	☐
2.5.5	Ensure Sending Diagnostic and Usage Data to Apple Is Disabled	☐	☐
2.5.6	Ensure Limit Ad Tracking Is Enabled	☐	☐
2.5.7	Ensure Gatekeeper Is Enabled	☐	☐
2.5.8	Ensure a Custom Message for the Login Screen Is Enabled	☐	☐
2.5.9	Ensure an Administrator Password Is Required to Access System-Wide Preferences	☐	☐
2.5.10	Ensure a Password is Required to Wake the Computer From Sleep or Screen Saver Is Enabled	☐	☐
2.6.1.1	Audit iCloud Keychain	☐	☐
2.6.1.2	Audit iCloud Drive	☐	☐
2.6.1.3	Ensure iCloud Drive Document and Desktop Sync Is Disabled	☐	☐
2.6.2	Audit App Store Password Settings	☐	☐
2.7.1	Ensure Backup Automatically is Enabled If Time Machine Is Enabled	☐	☐
2.7.2	Ensure Time Machine Volumes Are Encrypted If Time Machine Is Enabled	☐	☐
2.8.1	Ensure Wake for Network Access Is Disabled	☐	☐
2.8.2	Ensure Power Nap Is Disabled for Intel Macs	☐	☐
2.8.3	Ensure the OS is not Activate When Resuming from Sleep	☐	☐
2.9	Ensure Legacy EFI Is Valid and Updating	☐	☐

Recommendation		Set Correctly	
		Yes	No
2.10	Audit Siri Settings	☐	☐
2.11	Audit Sidecar Settings	☐	☐
2.12	Audit Touch ID and Wallet & Apple Pay Settings	☐	☐
3.1	Ensure Security Auditing Is Enabled	☐	☐
3.2	Ensure Security Auditing Flags For User-Attributable Events Are Configured Per Local Organizational Requirements	☐	☐
3.3	Ensure install.log Is Retained for 365 or More Days and No Maximum Size	☐	☐
3.4	Ensure Security Auditing Retention Is Enabled	☐	☐
3.5	Ensure Access to Audit Records Is Controlled	☐	☐
3.6	Ensure Firewall Logging Is Enabled and Configured	☐	☐
3.7	Audit Software Inventory	☐	☐
4.1	Ensure Bonjour Advertising Services Is Disabled	☐	☐
4.2	Ensure Show Wi-Fi status in Menu Bar Is Enabled	☐	☐
4.3	Ensure HTTP Server Is Disabled	☐	☐
4.4	Ensure NFS Server Is Disabled	☐	☐
5.1.1	Ensure Home Folders Are Secure	☐	☐
5.1.2	Ensure System Integrity Protection Status (SIP) Is Enabled	☐	☐
5.1.3	Ensure Apple Mobile File Integrity (AMFI) Is Enabled	☐	☐
5.1.4	Ensure Appropriate Permissions Are Enabled for System Wide Applications	☐	☐
5.1.5	Ensure No World Writable Files Exist in the System Folder	☐	☐
5.1.6	Ensure No World Writable Files Exist in the Library Folder	☐	☐
5.2.1	Ensure Password Account Lockout Threshold Is Configured	☐	☐
5.2.2	Ensure Password Minimum Length Is Configured	☐	☐
5.2.3	Ensure Complex Password Must Contain Alphabetic Characters Is Configured	☐	☐
5.2.4	Ensure Complex Password Must Contain Numeric Character Is Configured	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.2.5	Ensure Complex Password Must Contain Special Character Is Configured	☐	☐
5.2.6	Ensure Complex Password Must Contain Uppercase and Lowercase Characters Is Configured	☐	☐
5.2.7	Ensure Password Age Is Configured	☐	☐
5.2.8	Ensure Password History Is Configured	☐	☐
5.3	Ensure the Sudo Timeout Period Is Set to Zero	☐	☐
5.4	Ensure a Separate Timestamp Is Enabled for Each User/tty Combo	☐	☐
5.5	Ensure the "root" Account Is Disabled	☐	☐
5.6	Ensure Automatic Login Is Disabled	☐	☐
5.7	Ensure an Administrator Account Cannot Login to Another User's Active and Locked Session	☐	☐
5.8	Ensure a Login Window Banner Exists	☐	☐
5.9	Ensure Users' Accounts Do Not Have a Password Hint	☐	☐
5.10	Ensure Secure Keyboard Entry Terminal.app Is Enabled	☐	☐
6.1.1	Ensure Login Window Displays as Name and Password Is Enabled	☐	☐
6.1.2	Ensure Show Password Hints Is Disabled	☐	☐
6.1.3	Ensure Guest Account Is Disabled	☐	☐
6.1.4	Ensure Guest Access to Shared Folders Is Disabled	☐	☐
6.1.5	Ensure the Guest Home Folder Does Not Exist	☐	☐
6.2	Ensure Show All Filename Extensions Setting is Enabled	☐	☐
7.1.1	Ensure Protect Mail Activity in Mail Is Enabled	☐	☐
7.2.1	Ensure Automatic Opening of Safe Files in Safari Is Disabled	☐	☐
7.2.2	Audit History and Remove History Items	☐	☐
7.2.3	Audit Passwords System Preference Setting	☐	☐
7.2.4	Ensure Warn When Visiting A Fradulent Website in Safari Is Enabled	☐	☐
7.2.5	Ensure Prevent Cross-site Tracking in Safari Is Enabled	☐	☐
7.2.6	Audit Hide IP Address in Safari Setting	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.2.7	Ensure Advertising Privacy Protection in Safari Is Enabled	☐	☐
7.2.8	Ensure Show Full Website Address in Safari Is Enabled	☐	☐

Appendix: CIS Controls v8 Unmapped Recommendations

Recommendation		Set Correctly	
		Yes	No
	No unmapped recommendations to CIS Controls v8.0	☐	☐

Appendix: Change History

Date	Version	Changes for this version
Apr 6, 2020	1.0.0	Initial Release
Oct 14, 2020	1.1.0	1.1 - Updated Description, Audit, and Remediation
Oct 14, 2020	1.1.0	1.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	1.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	1.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	1.5 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.1.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.1.2 – Removed Previous Recommendation; Formerly 2.1.3; Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.1.3 – Moved to 2.1.2
Oct 14, 2020	1.1.0	2.2.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.2.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.3.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.3.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.3.3 - Updated Audit and Remediation

Date	Version	Changes for this version
Oct 14, 2020	1.1.0	2.4.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.5 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.6 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.7 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.8 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.9 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.4.10 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.1.1 - Updated Description, Audit, and Remediation
Oct 14, 2020	1.1.0	2.5.1.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.1.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.3 - Updated Audit and Remediation

Date	Version	Changes for this version
Oct 14, 2020	1.1.0	2.5.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.5 - Updated Audit and Remediation; Switched to Manual
Oct 14, 2020	1.1.0	2.5.6 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.7 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.8 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.5.9 – Added Recommendation
Oct 14, 2020	1.1.0	2.6.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.6.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.6.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.6.4 - Updated Audit and Remediation; Switched to Manual
Oct 14, 2020	1.1.0	2.7.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.7.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	2.8 - Updated Title, Description, Audit, and Remediation; Formerly 2.12
Oct 14, 2020	1.1.0	2.9 – Added Recommendation

Date	Version	Changes for this version
Oct 14, 2020	1.1.0	2.10 - Updated Audit and Remediation; Formerly 2.9
Oct 14, 2020	1.1.0	2.11 - Updated Audit and Remediation; Formerly 2.10
Oct 14, 2020	1.1.0	2.12 - Formerly 2.11
Oct 14, 2020	1.1.0	3.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	3.2 - Updated Audit and Remediation; Switched to Manual
Oct 14, 2020	1.1.0	3.3 – Updated Audit and Remediation
Oct 14, 2020	1.1.0	3.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	3.5 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	3.6 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	4.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	4.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	4.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	4.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	4.5 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.1.1 - Updated Audit and Remediation

Date	Version	Changes for this version
Oct 14, 2020	1.1.0	5.1.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.1.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.1.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.5 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.6 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.7 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.2.8 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.3 - Updated Audit, Remediation, and Additional Information
Oct 14, 2020	1.1.0	5.4 – Formerly 5.5; Updated Audit and Remediation; Switched to Manual
Oct 14, 2020	1.1.0	5.5 - Formerly 5.4; Updated Audit and Remediation

Date	Version	Changes for this version
Oct 14, 2020	1.1.0	5.6 - Updated Audit and Remediation; Switched to Manual
Oct 14, 2020	1.1.0	5.7 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.8 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.9 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.10 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.11 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.12 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.13 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.14 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.15 - Updated Audit and Remediation; Switched to Manual
Oct 14, 2020	1.1.0	5.16 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.17 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.18 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	5.19 - Updated Audit and Remediation

Date	Version	Changes for this version
Oct 14, 2020	1.1.0	6.1.1 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	6.1.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	6.1.3 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	6.1.4 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	6.1.5 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	6.2 - Updated Audit and Remediation
Oct 14, 2020	1.1.0	6.3 - Updated Audit and Remediation; Switched to Manual
Oct 14, 2020	1.1.0	6.4 – Removed Previous Recommendation
Oct 14, 2020	1.1.0	7.4 – Updated Description
Oct 14, 2020	1.1.0	7.6 – Added Audit and Remediation
Oct 14, 2020	1.1.0	7.12 - Added Audit and Remediation
Oct 14, 2020	1.1.0	7.16 - Added Rationale, Audit, and Remediation
Nov 11, 2020	1.1.0	First Revision Released
Mar 1, 2021	1.2.0	1.1 – Description Updated
Mar 1, 2021	1.2.0	1.2 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	1.3 – Updated Audit and Remediation

Date	Version	Changes for this version
Mar 1, 2021	1.2.0	1.4 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	1.5 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	1.6 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	1.7 – Moved from 7.3, Updated Rationale, Description, Audit, and Remediation
Mar 1, 2021	1.2.0	2.3.3 – Updated Rationale
Mar 1, 2021	1.2.0	2.4.3 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	2.4.4 – Updated Audit
Mar 1, 2021	1.2.0	2.4.6 – Updated Impact Statement, Description, Audit, and Remediation
Mar 1, 2021	1.2.0	2.4.7 – Updated Audit
Mar 1, 2021	1.2.0	2.4.8 – Updated Audit
Mar 1, 2021	1.2.0	2.4.10 – Updated Description
Mar 1, 2021	1.2.0	2.5.2 – Subsection Created
Mar 1, 2021	1.2.0	2.5.2.1 – Moved from 2.5.2, Updated Audit and Remediation
Mar 1, 2021	1.2.0	2.5.2.2 – Moved from 2.5.3, Updated Notes, References
Mar 1, 2021	1.2.0	2.5.2.3 – Moved from 2.5.4

Date	Version	Changes for this version
Mar 1, 2021	1.2.0	2.5.3 – Moved from 2.5.6, Updated Description, Audit, and Remediation
Mar 1, 2021	1.2.0	2.5.3 – Moved from 2.5.6
Mar 1, 2021	1.2.0	2.5.4 – Moved from 2.5.7, Updated Description
Mar 1, 2021	1.2.0	2.5.5 – Moved from 2.5.8
Mar 1, 2021	1.2.0	2.5.6 – Moved from 2.5.9, Updated Title, Impact Statement, Description, Audit, and Remediation
Mar 1, 2021	1.2.0	2.5.7 – Moved from 7.2, Updated Title, Description, Rationale, Audit, and Remediation
Mar 1, 2021	1.2.0	2.6 – Updated Overview
Mar 1, 2021	1.2.0	2.6.1 – Updated Impact Statement and Description
Mar 1, 2021	1.2.0	2.6.4 – Updated Title, Audit, and Remediation
Mar 1, 2021	1.2.0	2.7.2 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	2.8 – Updated Description
Mar 1, 2021	1.2.0	2.9 - Updated, Description, Rationale, Audit, and Remediation
Mar 1, 2021	1.2.0	2.11 – Removed Previous Recommendation, Moved from 2.12, Updated Description
Mar 1, 2021	1.2.0	2.12 – Moved from 7.6, Updated Title

Date	Version	Changes for this version
Mar 1, 2021	1.2.0	2.13 – Moved from 7.12, Updated Title, Audit, and Remediation
Mar 1, 2021	1.2.0	3.2 – Updated Rationale, Audit, Remediation, and Notes
Mar 1, 2021	1.2.0	3.5 – Updated Rationale, Audit, and Remediation
Mar 1, 2021	1.2.0	3.6 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	3.7 – Moved from 7.3, Updated Description, Audit, Remediation, and Notes
Mar 1, 2021	1.2.0	4.1 – Updated Audit
Mar 1, 2021	1.2.0	4.3 –Updated Description and Impact Statement
Mar 1, 2021	1.2.0	4.4 – Updated Audit and Remediation
Mar 1, 2021	1.2.0	4.5 – Updated Description, Audit, and Remediation
Mar 1, 2021	1.2.0	4.6 – Moved from 7.1, Updated Description, Rationale, Audit, and Remediation
Mar 1, 2021	1.2.0	5.1.2 - Updated Impact Statement and Description
Mar 1, 2021	1.2.0	5.2.1 - Updated Impact Statement, Remediation, and References
Mar 1, 2021	1.2.0	5.2.2 – Updated Remediation

Date	Version	Changes for this version
Mar 1, 2021	1.2.0	5.2.3 – Updated Remediation
Mar 1, 2021	1.2.0	5.2.4 – Updated Remediation
Mar 1, 2021	1.2.0	5.2.5 – Updated Remediation
Mar 1, 2021	1.2.0	5.2.6 – Updated Remediation
Mar 1, 2021	1.2.0	5.2.7 – Updated Remediation
Mar 1, 2021	1.2.0	5.2.8 – Updated Remediation
Mar 1, 2021	1.2.0	5.3 - Updated Impact Statement and Description
Mar 1, 2021	1.2.0	5.5 – Updated Rationale Statement
Mar 1, 2021	1.2.0	5.6 – Updated Description
Mar 1, 2021	1.2.0	5.7 – Updated Description, Audit, and Remediation
Mar 1, 2021	1.2.0	5.8 – Updated Impact Statement, Audit, and Remediation
Mar 1, 2021	1.2.0	5.9 – Updated Description
Mar 1, 2021	1.2.0	5.10 - Updated Description, Audit, Remediation, and Notes
Mar 1, 2021	1.2.0	5.12 - Updated Tile, Audit, Remediation, and References
Mar 1, 2021	1.2.0	5.13 – Updated Impact Statement

Date	Version	Changes for this version
Mar 1, 2021	1.2.0	5.14 – Updated Audit
Mar 1, 2021	1.2.0	5.16 – Updated Impact Statement
Mar 1, 2021	1.2.0	5.17 –Description
Mar 1, 2021	1.2.0	5.18 – Removed Previous Recommendation, Moved from 5.19, Updated Description
Mar 1, 2021	1.2.0	5.19 – Added Recommendation
Mar 1, 2021	1.2.0	6.1.2 – Updated Rationale Statement
Mar 1, 2021	1.2.0	6.2 – Updated Rationale Statement
Mar 1, 2021	1.2.0	7.1 – Moved from 7.8
Mar 1, 2021	1.2.0	7.2 – Moved from 7.9
Mar 1, 2021	1.2.0	7.3 – Moved from 7.10
Mar 1, 2021	1.2.0	7.4 – Moved from 7.11
Mar 1, 2021	1.2.0	7.5 – Removed Previous Recommendation, Moved from 7.13
Mar 1, 2021	1.2.0	7.6 – Moved from 7.14
Mar 1, 2021	1.2.0	7.7 – Moved from 7.15
Mar 1, 2021	1.2.0	7.8 – Moved from 7.16
Mar 1, 2021	1.2.0	7.9 – Moved from 7.17
Mar 1, 2021	1.2.0	7.10 – Moved from 7.18
Mar 1, 2021	1.2.0	7.11 – Moved from 7.19

Date	Version	Changes for this version
Mar 1, 2021	1.2.0	Second Iteration Released
Mar 11, 2021	1.3.0	2.4.7 – Moved to Manual from Automated
Mar 12, 2021	1.3.0	Third Iteration Released
May 30, 2021	1.4.0	1.1 – Grammatical Updates
May 30, 2021	1.4.0	1.3 – Grammatical Updates
May 30, 2021	1.4.0	1.4 – Grammatical Updates
May 30, 2021	1.4.0	1.5 – Grammatical Updates
May 30, 2021	1.4.0	1.6 – Grammatical Updates
May 30, 2021	1.4.0	1.7 – Grammatical Updates
May 30, 2021	1.4.0	2.1.1 – Grammatical Updates
May 30, 2021	1.4.0	2.1.2 – Grammatical Updates
May 30, 2021	1.4.0	2.2.2 – Grammatical Updates
May 30, 2021	1.4.0	2.3.1 – Grammatical Updates
May 30, 2021	1.4.0	2.3.2 – Grammatical Updates
May 30, 2021	1.4.0	2.4.2 – Grammatical Updates, Updated Audit
May 30, 2021	1.4.0	2.4.3 – Grammatical Updates, Updated Audit
May 30, 2021	1.4.0	2.4.4 – Grammatical Updates
May 30, 2021	1.4.0	2.4.5 – Grammatical Updates

Date	Version	Changes for this version
May 30, 2021	1.4.0	2.4.7 – Grammatical Updates, Updated Audit, Remediation, and Switched from Manual to Automatic
May 30, 2021	1.4.0	2.4.8 – Grammatical Updates
May 30, 2021	1.4.0	2.4.9 – Grammatical Updates
May 30, 2021	1.4.0	2.4.10 – Grammatical Updates
May 30, 2021	1.4.0	2.4.11 – Grammatical Updates
May 30, 2021	1.4.0	2.4.12 – Moved from 7.8
May 30, 2021	1.4.0	2.5.1 – Grammatical Updates
May 30, 2021	1.4.0	2.5.1.1 – Grammatical Updates
May 30, 2021	1.4.0	2.5.1.2 – Grammatical Updates
May 30, 2021	1.4.0	2.5.2 – Grammatical Updates
May 30, 2021	1.4.0	2.5.2.1 – Grammatical Updates
May 30, 2021	1.4.0	2.5.2.2 – Grammatical Updates
May 30, 2021	1.4.0	2.5.3 – Grammatical Updates
May 30, 2021	1.4.0	2.5.4 – Grammatical Updates
May 30, 2021	1.4.0	2.5.5 – Grammatical Updates

Date	Version	Changes for this version
May 30, 2021	1.4.0	2.5.6 – Grammatical Updates
May 30, 2021	1.4.0	2.5.7 – Grammatical Updates
May 30, 2021	1.4.0	2.6.1 – Grammatical Updates
May 30, 2021	1.4.0	2.6.2 – Grammatical Updates
May 30, 2021	1.4.0	2.6.4 – Grammatical Updates
May 30, 2021	1.4.0	2.7 – Grammatical Updates
May 30, 2021	1.4.0	2.7.1 – Grammatical Updates
May 30, 2021	1.4.0	2.7.2 – Grammatical Updates
May 30, 2021	1.4.0	2.8 – Grammatical Updates
May 30, 2021	1.4.0	2.9 – Grammatical Updates
May 30, 2021	1.4.0	2.10 – Grammatical Updates
May 30, 2021	1.4.0	2.11 – Grammatical Updates
May 30, 2021	1.4.0	2.12 – Grammatical Updates
May 30, 2021	1.4.0	2.13 – Grammatical Updates
May 30, 2021	1.4.0	2.14 – Grammatical Updates, Moved from 7.10
May 30, 2021	1.4.0	3 – Grammatical Updates, Updated Overview

Date	Version	Changes for this version
May 30, 2021	1.4.0	3.2 – Grammatical Updates
May 30, 2021	1.4.0	3.3 – Grammatical Updates
May 30, 2021	1.4.0	3.4 – Grammatical Updates
May 30, 2021	1.4.0	3.5 – Grammatical Updates
May 30, 2021	1.4.0	3.6 – Grammatical Updates
May 30, 2021	1.4.0	3.7 – Grammatical Updates
May 30, 2021	1.4.0	4.1 – Grammatical Updates
May 30, 2021	1.4.0	4.2 – Grammatical Updates
May 30, 2021	1.4.0	4.3 – Grammatical Updates
May 30, 2021	1.4.0	4.4 – Grammatical Updates
May 30, 2021	1.4.0	4.5 – Grammatical Updates
May 30, 2021	1.4.0	4.6 – Grammatical Updates
May 30, 2021	1.4.0	5.1.1 – Grammatical Updates
May 30, 2021	1.4.0	5.1.2 – Grammatical Updates
May 30, 2021	1.4.0	5.1.3 – Grammatical Updates
May 30, 2021	1.4.0	5.1.4 – Grammatical Updates
May 30, 2021	1.4.0	5.2.1 – Grammatical Updates
May 30, 2021	1.4.0	5.2.2 – Grammatical Updates
May 30, 2021	1.4.0	5.2.5 – Grammatical Updates

Date	Version	Changes for this version
May 30, 2021	1.4.0	5.2.6 – Grammatical Updates
May 30, 2021	1.4.0	5.2.7 – Grammatical Updates
May 30, 2021	1.4.0	5.2.8 – Grammatical Updates
May 30, 2021	1.4.0	5.3 – Grammatical Updates
May 30, 2021	1.4.0	5.4 – Grammatical Updates
May 30, 2021	1.4.0	5.5 – Grammatical Updates
May 30, 2021	1.4.0	5.6 – Grammatical Updates
May 30, 2021	1.4.0	5.7 – Grammatical Updates
May 30, 2021	1.4.0	5.8 – Grammatical Updates
May 30, 2021	1.4.0	5.9 – Grammatical Updates
May 30, 2021	1.4.0	5.10 – Grammatical Updates
May 30, 2021	1.4.0	5.11 – Grammatical Updates
May 30, 2021	1.4.0	5.12 – Grammatical Updates
May 30, 2021	1.4.0	5.13 – Grammatical Updates
May 30, 2021	1.4.0	5.14 – Grammatical Updates
May 30, 2021	1.4.0	5.15 – Grammatical Updates
May 30, 2021	1.4.0	5.16 – Grammatical Updates

Date	Version	Changes for this version
May 30, 2021	1.4.0	5.17 – Grammatical Updates
May 30, 2021	1.4.0	5.18 – Grammatical Updates
May 30, 2021	1.4.0	5.19 – Grammatical Updates, Updated Audit and Remediation
May 30, 2021	1.4.0	6.1.1 – Grammatical Updates
May 30, 2021	1.4.0	6.1.2 – Grammatical Updates
May 30, 2021	1.4.0	6.1.3 – Grammatical Updates
May 30, 2021	1.4.0	6.1.5 – Grammatical Updates
May 30, 2021	1.4.0	6.3 - Grammatical Updates, Updated Audit, Remediation, and Switched from Manual to Automatic
May 30, 2021	1.4.0	7.3 – Previous Recommendation Removed, Moved from 7.4
May 30, 2021	1.4.0	7.4 – Moved from 7.5
May 30, 2021	1.4.0	7.5 – Moved from 7.6
May 30, 2021	1.4.0	7.6 – Previous Recommendation Removed, Moved from 7.9
May 30, 2021	1.4.0	7.7 – Previous Recommendation Removed, Added to Section 3 Overview
May 30, 2021	1.4.0	7.8 –Moved to 2.4.12

Date	Version	Changes for this version
May 30, 2021	1.4.0	7.9 –Moved to 7.6
May 30, 2021	1.4.0	7.10 –Moved to 2.14
May 30, 2021	1.4.0	7.11 – Previous Recommendation Removed
Oct 18, 2021	2.0.0	Initial Draft Release
Oct 28, 2021	2.0.0	6.2 – Updated Title
Nov 1, 2021	2.0.0	Initial Release
July 1, 2022	2.1.0	Initial Draft Release
July 19, 2022	2.1.0	1.7 – Moved Previous Recommendation to 1.8, Created New Recommendation
July 19, 2022	2.1.0	1.8 – Moved from 1.7, Update Title and Description
July 19, 2022	2.1.0	2.1.1 – Previous Recommendation Removed, Moved from 2.1.1
July 19, 2022	2.1.0	2.1.2 – Moved to 2.1.1
July 19, 2022	2.1.0	2.2.1 – Removed Profile Methodology
July 19, 2022	2.1.0	2.2.2 – Updated Description. Audit, and Remediation
July 19, 2022	2.1.0	2.3.3 – Removed Recommendation
July 19, 2022	2.1.0	2.4.2 – Updated Audit
July 19, 2022	2.1.0	2.4.3 – Updated Audit
July 19, 2022	2.1.0	2.4.4 – Updated Audit
July 19, 2022	2.1.0	2.4.5 – Updated Description

Date	Version	Changes for this version
July 19, 2022	2.1.0	2.4.6 – Updated Audit
July 19, 2022	2.1.0	2.4.7 – Removed Profile Methodology
July 19, 2022	2.1.0	2.4.8 – Updated Description and Audit
July 19, 2022	2.1.0	2.4.10 – Updated Audit and Remediation
July 19, 2022	2.1.0	2.4.11 – Updated Description, Audit and Remediation
July 19, 2022	2.1.0	2.5 – Updated Description
July 19, 2022	2.1.0	2.5.1 – Updated Audit and Remediation
July 19, 2022	2.1.0	2.5.2.1 – Moved Previous Recommendation to 2.5.7, Moved from 2.5.2.2
July 19, 2022	2.1.0	2.5.2.2 – Moved Previous Recommendation to 2.5.2.1, Moved from 2.5.2.3
July 19, 2022	2.1.0	2.5.2.3 – Moved to 2.5.2.2
July 19, 2022	2.1.0	2.5.3 – Updated Audit and Remediation
July 19, 2022	2.1.0	2.5.6 – Updated Audit and Remediation
July 19, 2022	2.1.0	2.5.7 – Removed Previous Recommendation, Moved from 2.5.2.1
July 19, 2022	2.1.0	2.5.8 – Moved from 5.12
July 19, 2022	2.1.0	2.5.9 – Moved from 5.10
July 19, 2022	2.1.0	2.5.10 – Moved from 5.8

Date	Version	Changes for this version
July 19, 2022	2.1.0	2.6.1 – Updated Description
July 19, 2022	2.1.0	2.6.1.1 – Removed Previous Recommendation, Moved from 2.6.1.2
July 19, 2022	2.1.0	2.6.1.2 – Moved Previous Recommendation to 2.6.1.1, Moved from 2.6.1.3
July 19, 2022	2.1.0	2.6.1.3 – Moved Previous Recommendation to 2.6.1.2, Moved from 2.6.1.4
July 19, 2022	2.1.0	2.7 – Updated Description
July 19, 2022	2.1.0	2.7.2 – Updated Audit
July 19, 2022	2.1.0	2.8 – Moved Previous Recommendation to 2.8.1, Sub-Section Added
July 19, 2022	2.1.0	2.8.1 – Moved from 2.8
July 19, 2022	2.1.0	2.8.2 – Moved from 2.9
July 19, 2022	2.1.0	2.8.3 – Moved from 5.9
July 19, 2022	2.1.0	2.9 – Moved Previous Recommendation to 2.8.2, Moved from 2.11
July 19, 2022	2.1.0	2.10 – Moved Previous Recommendation to 5.11, Moved from 2.13
July 19, 2022	2.1.0	2.11 – Moved Previous Recommendation to 2.9, Moved from 2.14
July 19, 2022	2.1.0	2.12 – Removed Previous Recommendation, Moved from 2.15

Date	Version	Changes for this version
July 19, 2022	2.1.0	2.13 - Moved Previous Recommendation to 2.9
July 19, 2022	2.1.0	2.14 - Moved Previous Recommendation to 2.11
July 19, 2022	2.1.0	2.15 - Moved Previous Recommendation to 2.12
July 19, 2022	2.1.0	3.2 – Updated Audit
July 19, 2022	2.1.0	3.4 – Updated Description, Audit, Remediation
July 19, 2022	2.1.0	3.5 – Updated Audit, Remediation
July 19, 2022	2.1.0	4.3 – Removed Previous Recommendation, Moved from 4.4
July 19, 2022	2.1.0	4.4 – Moved Previous Recommendation to 4.3, Moved from 4.5, Updated Audit
July 19, 2022	2.1.0	4.5 - Moved Previous Recommendation to 4.4
July 19, 2022	2.1.0	4.6 - Removed Previous Recommendation
July 19, 2022	2.1.0	5.1.4 - Removed Previous Recommendation, Moved from 5.1.5
July 19, 2022	2.1.0	5.1.5 - Moved Previous Recommendation to 5.1.4, Moved from 5.1.6
July 19, 2022	2.1.0	5.1.6 - Moved Previous Recommendation to 5.1.5, Moved from 5.1.7

Date	Version	Changes for this version
July 19, 2022	2.1.0	5.1.7 - Moved Previous Recommendation to 5.1.6
July 19, 2022	2.1.0	5.2 – Updated Description
July 19, 2022	2.1.0	5.3 – Updated Audit
July 19, 2022	2.1.0	5.4 – Updated Audit
July 19, 2022	2.1.0	5.5 – Removed Previous Recommendation, Moved from 5.6
July 19, 2022	2.1.0	5.6 – Moved Previous Recommendation to 5.5, Moved from 5.7
July 19, 2022	2.1.0	5.7 – Moved Previous Recommendation to 5.5, Moved from 5.11
July 19, 2022	2.1.0	5.8 – Moved Previous Recommendation to 2.5.8, Moved from 5.13
July 19, 2022	2.1.0	5.9 – Moved Previous Recommendation to 5.5, Moved from 5.14
July 19, 2022	2.1.0	5.10 – Moved Previous Recommendation to 2.5.9, Moved from 5.15
July 19, 2022	2.1.0	5.11 - Moved Previous Recommendation to 2.5.9, Moved from 2.10
July 19, 2022	2.1.0	5.12 - Moved Previous Recommendation to 2.5.8
July 19, 2022	2.1.0	5.13 - Moved Previous Recommendation to 5.8
July 19, 2022	2.1.0	5.14 - Moved Previous Recommendation to 5.9

Date	Version	Changes for this version
July 19, 2022	2.1.0	5.15 - Moved Previous Recommendation to 5.10
July 19, 2022	2.1.0	6.1.3 – Updated Remediation
July 19, 2022	2.1.0	6.1.4 – Updated Audit, Remediation
July 19, 2022	2.1.0	6.1.5 – Updated Title
July 19, 2022	2.1.0	6.2 – Updated Remediation
July 19, 2022	2.1.0	7 – Section Removed
July 19, 2022	2.1.0	7.1 – Removed Recommendation
July 19, 2022	2.1.0	7.2 – Removed Recommendation
July 19, 2022	2.1.0	First Iteration Released
October 27, 2022	3.0.0	Initial Draft Release
November 21, 2022	3.0.0	Formatting was changed across the entire document to better differentiate between the 3 methods (Graphical, Terminal, and Profile)
November 21, 2022	3.0.0	Terminal commands in every audit and remediation have been updated to include the full path name
November 21, 2022	3.0.0	The formatting for the profile method in all remediations have been updated to give more explicit directions on what the mobile configuration profile should include

Date	Version	Changes for this version
November 21, 2022	3.0.0	1.2 – Updated Audit
November 21, 2022	3.0.0	1.3 – Updated Audit
November 21, 2022	3.0.0	1.4 – Updated Audit
November 21, 2022	3.0.0	1.5 – Updated Audit
November 21, 2022	3.0.0	1.6 – Updated Audit
November 21, 2022	3.0.0	1.7 – Updated Audit
November 21, 2022	3.0.0	2.1.1 – Updated Audit
November 21, 2022	3.0.0	2.3.1 – Updated Audit
November 21, 2022	3.0.0	2.4.2 – Updated Audit
November 21, 2022	3.0.0	2.4.10 – Updated Audit
November 21, 2022	3.0.0	2.4.11 – Updated Audit
November 21, 2022	3.0.0	2.4.12 – Updated Audit
November 21, 2022	3.0.0	2.5.1.1 – Updated Audit
November 21, 2022	3.0.0	2.5.2.1 – Updated Audit
November 21, 2022	3.0.0	2.5.2.2 – Updated Audit
November 21, 2022	3.0.0	2.5.3 – Updated Audit
November 21, 2022	3.0.0	2.5.5 – Updated Audit
November 21, 2022	3.0.0	2.5.6 – Updated Audit
November 21, 2022	3.0.0	2.5.8 – Updated Audit
November 21, 2022	3.0.0	2.5.10 – Updated Audit
November 21, 2022	3.0.0	2.6.1.3 – Updated Audit
November 21, 2022	3.0.0	2.7.1 – Updated Audit
November 21, 2022	3.0.0	2.10 – Updated Audit

Date	Version	Changes for this version
November 21, 2022	3.0.0	2.11 – Updated Audit
November 21, 2022	3.0.0	3.2 – Updated Audit
November 21, 2022	3.0.0	3.6 – Updated Audit
November 21, 2022	3.0.0	4.1 – Updated Audit
November 21, 2022	3.0.0	5.2 – Updated Description
November 21, 2022	3.0.0	5.2.1 – Updated Audit
November 21, 2022	3.0.0	5.2.2 – Updated Audit
November 21, 2022	3.0.0	5.2.3 – Updated Audit, Added Additional Information
November 21, 2022	3.0.0	5.2.4 – Updated Audit, Added Additional Information
November 21, 2022	3.0.0	5.2.5 – Updated Audit, Added Additional Information
November 21, 2022	3.0.0	5.2.6 – Updated Audit, Added Additional Information
November 21, 2022	3.0.0	5.2.7 – Updated Audit
November 21, 2022	3.0.0	5.2.8 – Updated Audit
November 21, 2022	3.0.0	5.3 – Updated Audit, Remediation
November 21, 2022	3.0.0	5.4 – Updated Audit, Remediation
November 21, 2022	3.0.0	5.6 – Updated Audit
November 21, 2022	3.0.0	5.10 – Remediation Removed, Moved Previous 5.11

Date	Version	Changes for this version
November 21, 2022	3.0.0	5.11 – Moved Previous Recommendation to 5.10
November 21, 2022	3.0.0	6.1.1 – Updated Audit
November 21, 2022	3.0.0	6.1.2 – Updated Audit
November 21, 2022	3.0.0	6.1.3 – Updated Audit
November 21, 2022	3.0.0	6.2 – Updated Audit
November 21, 2022	3.0.0	6.3 – Moved Previous Recommendation to 7.2.1
November 21, 2022	3.0.0	7 – New Section Added
November 21, 2022	3.0.0	7.1 – New Sub-section Added
November 21, 2022	3.0.0	7.1.1 – Recommendation Added
November 21, 2022	3.0.0	7.2 – New Sub-section Added
November 21, 2022	3.0.0	7.2.1 – Moved From 6.3
November 21, 2022	3.0.0	7.2.2 – New Recommendation
November 21, 2022	3.0.0	7.2.3 – New Recommendation
November 21, 2022	3.0.0	7.2.4 – Added Recommendation
November 21, 2022	3.0.0	7.2.5 – Added Recommendation
November 21, 2022	3.0.0	7.2.6 – Added Recommendation
November 21, 2022	3.0.0	7.2.7 – Added Recommendation

Date	Version	Changes for this version
November 21, 2022	3.0.0	7.2.8 – Added Recommendation
November 21, 2022	3.0.0	Initial Release